Antenna Study and Design for Ultra Wideband Communication Applications

by

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TO MY FAMILY

Abstract

Since the release by the Federal Communications Commission (FCC) of a bandwidth of 7.5GHz (from 3.1GHz to 10.6GHz) for ultra wideband (UWB) wireless communications, UWB is rapidly advancing as a high data rate wireless communication technology.

As is the case in conventional wireless communication systems, an antenna also plays a very crucial role in UWB systems. However, there are more challenges in designing a UWB antenna than a narrow band one. A suitable UWB antenna should be capable of operating over an ultra wide bandwidth as allocated by the FCC. At the same time, satisfactory radiation properties over the entire frequency range are also necessary. Another primary requirement of the UWB antenna is a good time domain performance, i.e. a good impulse response with minimal distortion.

This thesis focuses on UWB antenna design and analysis. Studies have been undertaken covering the areas of UWB fundamentals and antenna theory. Extensive investigations were also carried out on two different types of UWB antennas.

The first type of antenna studied in this thesis is circular disc monopole antenna. The vertical disc monopole originates from conventional straight wire monopole by replacing the wire element with a disc plate to enhance the operating bandwidth substantially. Based on the understanding of vertical disc monopole, two more compact versions featuring low-profile and compatibility to printed circuit board are proposed and studied. Both of them are printed circular disc monopoles, one fed by a micro-strip line, while the other fed by a co-planar waveguide (CPW).

The second type of UWB antenna is elliptical/circular slot antenna, which can also be fed by either micro-strip line or CPW.

The performances and characteristics of UWB disc monopole and elliptical/circular slot

antenna are investigated in both frequency domain and time domain. The design parameters for achieving optimal operation of the antennas are also analyzed extensively in order to understand the antenna operations.

It has been demonstrated numerically and experimentally that both types of antennas are suitable for UWB applications.

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List of Abbreviations

1G	First-Generation
2D	Two-Dimensional
2G	Second-Generation
3D	Three-Dimensional
3G	Third-Generation
4G	Fourth-Generation
ABW	Absolute BandWidth
AWGN	Additive White Gaussian Noise
BPSK	Binary Phase-Shift Keying
BW	BandWidth
CEPT	Conference of European Posts and Telecommunications
CPW	CoPlanar Waveguide
DAA	Detect and Avoid
DC	Direct Current
DSSS	Direct Sequence Spread Spectrum
DS-UWB	Direct Sequence Ultra Wideband
DVD	Digital Video Disc
ECC	Electronic Communications Committee
EM	ElectroMagnetic

ETRI	Electronics and Telecommunications Research Institute
FBW	Fractional BandWidth
FCC	Federal Communications Commission
FDTD	Finite-Difference Time-Domain
FE	Finite Element
FIT	Finite Integration Technique
FR4	Flame Resistant 4
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HDTV	High-Definition TV
IDA	Infocomm Development Authority
IEEE	Institute of Electrical and Electronics Engineers
IFFT	Inverse Fast Fourier Transform
ISM	Industrial Scientific and Medicine
ITU	International Telecommunication Union
LCD	Liquid Crystal Display
MIC	Ministry of Internal Affairs & Communications
MBOA	MultiBand OFDM Alliance
MoM	Method of Moments
MPEG	Moving Picture Experts Group
MB-OFDM	Multiband Orthogonal Frequency Division Multiplexing
OFDM	Orthogonal Frequency Division Multiplexing
PAM	Pulse-Amplitude Modulation
PBA®	Perfect Boundary Approximation
PC	Personal Computer
PCB	Printed Circuit Board
PDA	Personal Digital Assistant
PN	Pseudo-Noise

PPM	Pulse-Position Modulation
PSD	Power Spectral Density
PVP	Personal Video Player
SMA	SubMiniature version A
SNR	Signal-to-Noise Ratio
TEM	Transverse Electromagnetic
TST	Thin Sheet Technology^{TM}
UFZ	UWB Friendly Zone
USB	Universal Serial Bus
UWB	Ultra Wideband
VSWR	Voltage Standing Wave Ratio
Wi-Fi	Wireless Fidelity
WLAN	Wireless Local Area Network
WPAN	Wireless Personal Area Network

Chapter 1

Introduction

1.1 Introduction

Wireless communication technology has changed our lives during the past two decades. In countless homes and offices, the cordless phones free us from the short leash of handset cords. Cell phones give us even more freedom such that we can communicate with each other at any time and in any place. Wireless local area network (WLAN) technology provides us access to the internet without suffering from managing yards of unsightly and expensive cable.

The technical improvements have also enabled a large number of new services to emerge. The first-generation (1G) mobile communication technology only allowed analogue voice communication while the second-generation (2G) technology realized digital voice communication. Currently, the third-generation (3G) technology can provide video telephony, internet access, video/music download services as well as digital voice services. In the near future, the fourth-generation (4G) technology will be able to provide on-demand high quality audio and video services, and other advanced services.

In recent years, more interests have been put into wireless personal area network

(WPAN) technology worldwide. The future WPAN aims to provide reliable wireless connections between computers, portable devices and consumer electronics within a short range. Furthermore, fast data storage and exchange between these devices will also be accomplished. This requires a data rate which is much higher than what can be achieved through currently existing wireless technologies.

The maximum achievable data rate or capacity for the ideal band-limited additive white Gaussian noise (AWGN) channel is related to the bandwidth and signal-to-noise ratio (SNR) by Shannon-Nyquist criterion [1, 2], as shown in Equation 1.1.

$$C = B \log_2(1 + SNR) \tag{1.1}$$

where C denotes the maximum transmit data rate, B stands for the channel bandwidth.

Equation 1.1 indicates that the transmit data rate can be increased by increasing the bandwidth occupation or transmission power. However, the transmission power can not be readily increased because many portable devices are battery powered and the potential interference should also be avoided. Thus, a large frequency bandwidth will be the solution to achieve high data rate.

On February 14, 2002, the Federal Communications Commission (FCC) of the United States adopted the First Report and Order that permitted the commercial operation of ultra wideband (UWB) technology [3]. Since then, UWB technology has been regarded as one of the most promising wireless technologies that promises to revolutionize high data rate transmission and enables the personal area networking industry leading to new innovations and greater quality of services to the end users.

1.2 Review of the State-of-Art

At present, the global regulations and standards for UWB technology are still under consideration and construction. However, research on UWB has advanced greatly.

Freescale Semiconductor was the first company to produce UWB chips in the world and its XS110 solution is the only commercially available UWB chipset to date [4]. It provides full wireless connectivity implementing direct sequence ultra wideband (DS-UWB). The chipset delivers more than 110 Mbps data transfer rate supporting applications such as streaming video, streaming audio, and high-rate data transfer at very low levels of power consumption.

At 2005 3GSM World Congress, Samsung and Freescale demonstrated the world's first UWB-enabled cell phone featuring its UWB wireless chipset [5].



Figure 1.1: Samsung's UWB-enabled cell phone [5]

The Samsung's UWB-enabled cell phone, as shown in Figure 1.1, can connect wirelessly to a laptop and download files from the Internet. Additionally, pictures, MP3 audio files or data from the phone's address book can be selected and transferred directly to the laptop at very high data rate owing to the UWB technology exploited. These functions underscore the changing role of the cellular phone as new applications, such as cameras and video, require the ability for consumers to wirelessly connect their cell phone to

other devices and transfer their data and files very fast.

In June, 2005, Haier Corporation and Freescale Semiconductor announced the first commercial UWB product, i.e. a UWB-enabled Liquid Crystal Display (LCD) digital television and digital media server [6].



Figure 1.2: Haier's UWB-enabled LCD digital television and digital media server [6]

As shown in Figure 1.2, the Haier television is a 37-inch, LCD High-Definition TV (HDTV). The Freescale UWB antenna, which is a flat planar design etched on a single metal layer of common FR4 circuit board material, is embedded inside the television and is not visible to the user. The digital media server is the size of a standard DVD player but includes personal video player (PVP) functionality, a DVD playback capability and a tuner, as well as the Freescale UWB solution to wirelessly stream media to the HDTV. The digital media server can be placed as far away as 20 meters from the actual HDTV, providing considerable freedom in home theatre configuration. The rated throughput between these two devices is up to 110 megabits per second at this distance, which will allow several MPEG-2 video streams to be piped over the UWB link.

At the 2006 International Consumer Electronics Show in Las Vegas in January, Belkin announces its new CableFree USB (Universal Serial Bus) Hub, the first UWB-enabled product to be introduced in the U.S. market [7, 8].



Figure 1.3: Belkin's four-port Cablefree USB Hub [8]

The Belkin's four-port hub, as show in Figure 1.3, enables immediate high-speed wireless connectivity for any USB device without requiring software. USB devices plug into the hub with cords, but the hub does not require a cable to connect to the computer. So it gives desktop computer users the freedom to place their USB devices where it is most convenient for the users. Laptop users also gain the freedom to roam wirelessly with their laptop around the room while still maintaining access to their stationary USB devices, such as printers, scanners, hard drives, and MP3 players.

The regulatory bodies around the world are currently working on the UWB regulations and the Institute of Electrical and Electronics Engineers (IEEE) is busy making UWB standards. It is believed that when these works are finished, a great variety of UWB products will be available to the customers in the market.

1.3 Motivation

The UWB technology has experienced many significant developments in recent years. However, there are still challengers in making this technology live up to its full potential. One particular challenge is the UWB antenna.

Among the classical broadband antenna configurations that are under consideration

for use in UWB systems, a straight wire monopole features a simple structure, but its bandwidth is only around 10%. A vivaldi antenna is a directional antenna [9] and hence unsuitable for indoor systems and portable devices. A biconical antenna has a big size which limits its application [10]. Log periodic and spiral antennas tend to be dispersive and suffer severe ringing effect, apart from big size [11]. There is a growing demand for small and low cost UWB antennas that can provide satisfactory performances in both frequency domain and time domain.

In recent years, the circular disc monopole antenna has attracted considerable research interest due to its simple structure and UWB characteristics with nearly omni-directional radiation patterns [12, 13]. However, it is still not clear why this type of antenna can achieve ultra wide bandwidth and how exactly it operates over the entire bandwidth. In this thesis, the circular disc monopole antenna is investigated in detail in order to understand its operation, find out the mechanism that leads to the UWB characteristic and also obtain some quantitative guidelines for designing of this type of antenna. Based on the understanding of vertical disc monopole, two more compact versions, i.e. coplanar waveguide (CPW) fed and microstrip line fed circular disc monopoles, are proposed.

In contrast to circular disc monopoles which have relatively large electric near-fields, slot antennas have relatively large magnetic near-fields. This feature makes slot antennas more suitable for applications wherein near-field coupling is not desirable. As such, elliptical/circular slot antennas are proposed and studied for UWB systems in this thesis.

1.4 Organization of the thesis

This thesis is organised in seven chapters as follows:

Chapter 2: A brief introduction to UWB technology is presented in this chapter. The history of UWB technology is described. Its advantages and applications are also discussed. Besides, current regulation state and standards activities are addressed.

Chapter 3: This chapter covers the fundamental antenna theory. The primary requirements for a suitable UWB antenna are discussed. Some general approaches to achieve wide operating bandwidth of antenna are also introduced.

Chapter 4: In this chapter, circular disc monopole antennas are studied in frequency time. The operation principle of the antenna is addressed based on the investigation of the antenna performances and characteristics. The antenna configuration also evolves from a vertical type to a fully planar version.

Chapter 5: The frequency domain performances of elliptical/circular slot antennas are detailed in this chapter. The important parameters which affect the antenna performances are investigated both numerically and experimentally to derive the design rules.

Chapter 6: The time domain characteristics of circular disc monopoles and elliptical/circular slot antennas are evaluated in this chapter. The performances of antenna systems are analysed. Antenna responses in both transmit and receive modes are investigated. Further, the received signal waveforms are assessed by the pulse fidelity.

Chapter 7: This chapter concludes the researches that have been done in this thesis. Suggestions for future work are also given in this chapter.

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Chapter 2

UWB Technology

UWB technology has been used in the areas of radar, sensing and military communications during the past 20 years. A substantial surge of research interest has occurred since February 2002, when the FCC issued a ruling that UWB could be used for data communications as well as for radar and safety applications [1]. Since then, UWB technology has been rapidly advancing as a promising high data rate wireless communication technology for various applications.

This chapter presents a brief overview of UWB technology and explores its fundamentals, including UWB definition, advantages, current regulation state and standard activities.

2.1 Introduction

2.1.1 Background

UWB systems have been historically based on impulse radio because it transmitted data at very high data rates by sending pulses of energy rather than using a narrowband frequency carrier. Normally, the pulses have very short durations, typically a few

nanoseconds (billionths of a second) that results in an ultra wideband frequency spectrum.

The concept of impulse radio initially originated with Marconi, in the 1900s, when spark gap transmitters induced pulsed signals having very wide bandwidths [2]. At that time, there was no way to effectively recover the wideband energy emitted by a spark gap transmitter or discriminate among many such wideband signals in a receiver. As a result, wideband signals caused too much interference with one another. So the communications world abandoned wideband communication in favour of narrowband radio transmitter that were easy to regulate and coordinate.

In 1942-1945, several patents were filed on impulse radio systems to reduce interference and enhance reliability [3]. However, many of them were frozen for a long time because of the concerns about its potential military usage by the U.S. government. It is in the 1960s that impulse radio technologies started being developed for radar and military applications.

In the mid 1980s, the FCC allocated the Industrial Scientific and Medicine (ISM) bands for unlicensed wideband communication use. Owing to this revolutionary spectrum allocation, WLAN and Wireless Fidelity (Wi-Fi) have gone through a tremendous growth. It also leads the communication industry to study the merits and implications of wider bandwidth communication.

Shannon-Nyquist criterion (Equation 1.1) indicates that channel capacity increases linearly with bandwidth and decreases logarithmically as the SNR decreases. This relationship suggests that channel capacity can be enhanced more rapidly by increasing the occupied bandwidth than the SNR. Thus, for WPAN that only transmit over short distances, where signal propagation loss is small and less variable, greater capacity can be achieved through broader bandwidth occupancy.

In February, 2002, the FCC amended the Part 15 rules which govern unlicensed radio devices to include the operation of UWB devices. The FCC also allocated a bandwidth

of 7.5GHz, i.e. from 3.1GHz to 10.6GHz to UWB applications [1], by far the largest spectrum allocation for unlicensed use the FCC has ever granted.

According to the FCC's ruling, any signal that occupies at least 500MHz spectrum can be used in UWB systems. That means UWB is not restricted to impulse radio any more, it also applies to any technology that uses 500MHz spectrum and complies with all other requirements for UWB.

2.1.2 Signal Modulation Scheme

Information can be encoded in a UWB signal in various methods. The most popular signal modulation schemes for UWB systems include pulse-amplitude modulation (PAM) [4], pulse-position modulation (PPM) [3], binary phase-shift keying (BPSK) [5], and so on.

2.1.2.1 PAM

The principle of classic PAM scheme is to encode information based on the amplitude of the pulses, as illustrated in Figure 2.1.

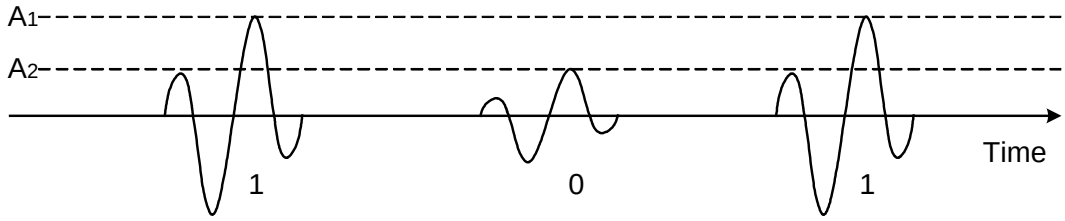


Figure 2.1: PAM modulation

The transmitted pulse amplitude modulated information signal $x(t)$ can be represented as:

$$x(t) = d_i \cdot w_{tr}(t) \tag{2.1}$$

where $w_{tr}(t)$ denotes the UWB pulse waveform, i is the bit transmitted (i.e. '1' or '0'),

and

$$d_i = \begin{cases} A_1, & i = 1 \\ A_2, & i = 0 \end{cases} \tag{2.2}$$

Figure 2.1 illustrates a two-level (A_1 and A_2) PAM scheme where one bit is encoded in one pulse. More amplitude levels can be used to encode more bits per symbol.

2.1.2.2 PPM

In PPM, the bit to be transmitted determines the position of the UWB pulse. As shown in Figure 2.2, the bit '0' is represented by a pulse which is transmitted at nominal position, while the bit '1' is delayed by a time of a from nominal position. The time delay a is normally much shorter than the time distance between nominal positions so as to avoid interference between pulses.

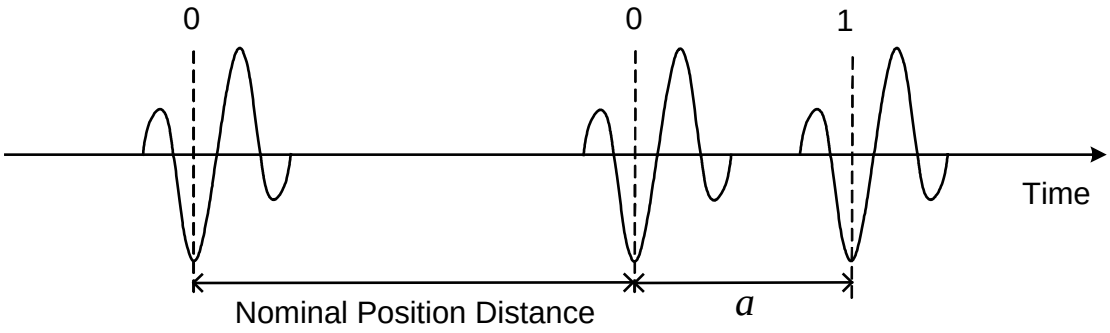


Figure 2.2: PPM modulation

The pulse position modulated signal $x(t)$ can be represented as:

$$x(t) = w_{tr}(t - a \cdot d_i) \tag{2.3}$$

where $w_{tr}(t)$ and i have been defined previously, and

$$d_i = \begin{cases} 1, & i = 1 \\ 0, & i = 0 \end{cases} \tag{2.4}$$

Figure 2.2 illustrates a two-position (0 and a) PPM scheme and additional positions can be used to achieve more bits per symbol.

2.1.2.3 BPSK

In BPSK modulation, the bit to be transmitted determines the phase of the UWB pulse. As shown in Figure 2.3, a pulse represents the bit '0'; when it is out of phase, it represents the bit '1'. In this case, only one bit is encoded per pulse because there are only two phases available. More bits per symbol may be obtained by using more phases.

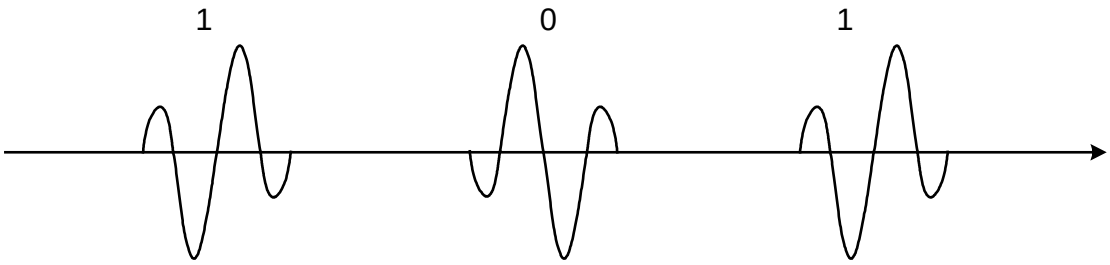


Figure 2.3: BPSK modulation

The BPSK modulated signal $x(t)$ can be represented as:

$$x(t) = w_{tr}(t)e^{-j(d_i \cdot \pi)} \tag{2.5}$$

where $w_{tr}(t)$ and i have been defined previously, and

$$d_i = \begin{cases} 1, & i = 1 \\ 0, & i = 0 \end{cases} \tag{2.6}$$

2.1.3 Band Assignment

The UWB band covers a frequency spectrum of 7.5GHz. Such a wide band can be utilized with two different approaches: single-band scheme and multiband scheme.

UWB systems based on impulse radio are single-band systems. They transmit short

pulses which are designed to have a spectrum covering the entire UWB band. Data is normally modulated using PPM method and multiple users can be supported using time hopping scheme.

Figure 2.4 presents an example of time hopping scheme. In each frame, there are eight time slots allocated to eight users; for each user, the UWB signal is transmitted at one specific slot which determined by a pseudo random sequence.

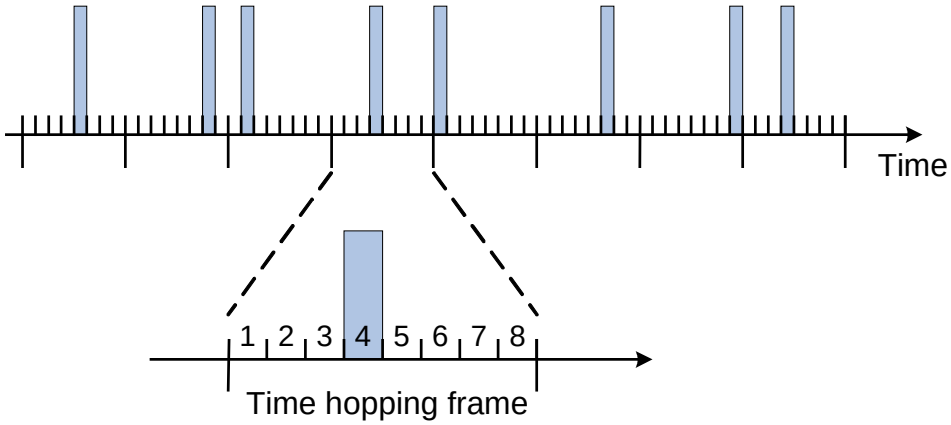


Figure 2.4: Time hopping concept

The other approach to UWB spectrum allocation is multiband scheme where the 7.5GHz UWB band is divided into several smaller sub-bands. Each sub-band has a bandwidth no less than 500MHz so as to conform to the FCC definition of UWB.

In multiband scheme, multiple access can be achieved by using frequency hopping. As exemplified in Figure 2.5, the UWB signal is transmitted over eight sub-bands in a sequence during the hopping period and it hops from frequency to frequency at fixed intervals. At any time, only one sub-band is active for transmission while the so-called time-frequency hopping codes are exploited to determine the sequence in which the sub-bands are used.

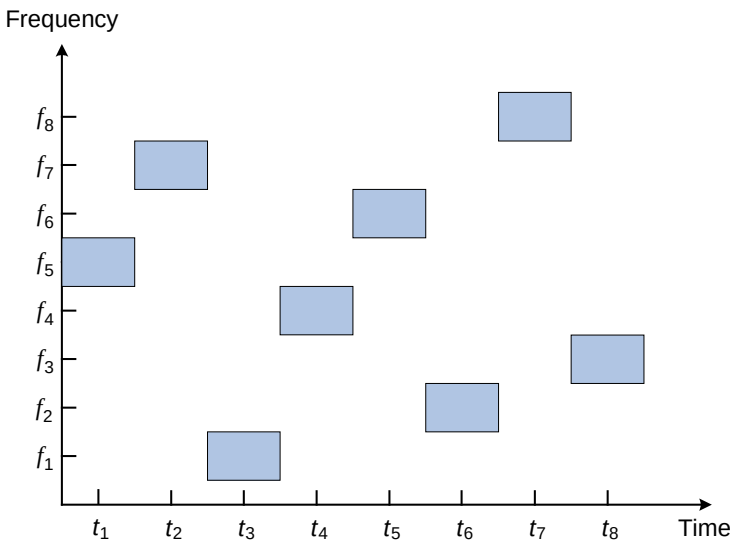


Figure 2.5: Frequency hopping concept

Single-band and multiband UWB systems present different features.

For single-band scheme, the transmitted pulse signal has extremely short duration, so very fast switching circuit is required. On the other hand, the multiband system needs a signal generator which is able to quickly switch between frequencies.

Single-band systems can achieve better multipath resolution compared to multiband systems because they employ discontinuous transmission of short pulses and normally the pulse duration is shorter than the multipath delay. While multiband systems may benefit from the frequency diversity across sub-bands to improve system performance.

Besides, multiband systems can provide good interference robustness and co-existence properties. For example, when the system detects the presence of other wireless systems, it can avoid the use of the sub-bands which share the spectrum with those systems.

To achieve the same result, a single-band system would need to exploit notch filters. However, this may increase the system complexity and distort the received signal waveform.

2.2 Advantages of UWB

UWB has a number of encouraging advantages that are the reasons why it presents a more eloquent solution to wireless broadband than other technologies.

Firstly, according to Shannon-Hartley theorem, channel capacity is in proportion to bandwidth. Since UWB has an ultra wide frequency bandwidth, it can achieve huge capacity as high as hundreds of Mbps or even several Gbps with distances of 1 to 10 meters [6].

Secondly, UWB systems operate at extremely low power transmission levels. By dividing the power of the signal across a huge frequency spectrum, the effect upon any frequency is below the acceptable noise floor [7], as illustrated in Figure 2.6.

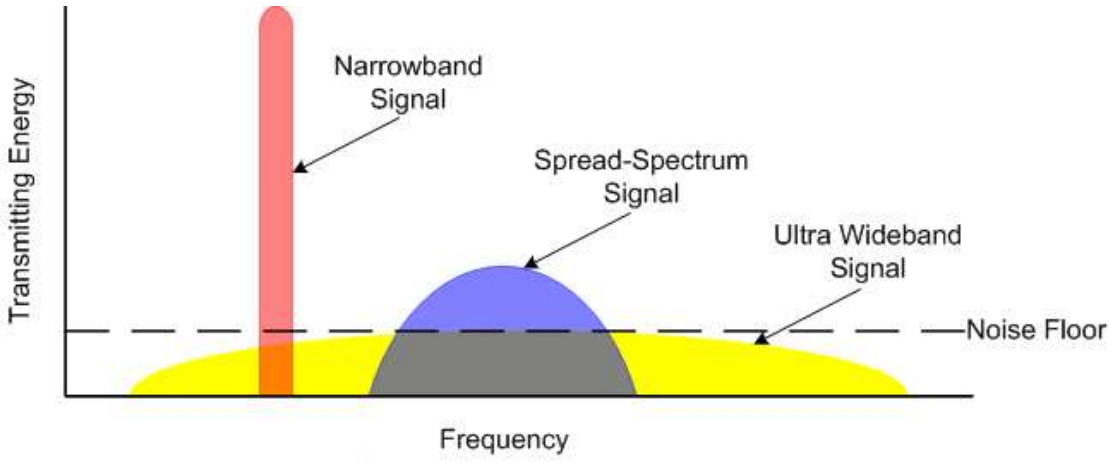


Figure 2.6: Ultra wideband communications spread transmitting energy across a wide spectrum of frequency (Reproduced from [7])

For example, 1 watt of power spread across 1GHz of spectrum results in only 1 nanowatt of power into each hertz band of frequency. Thus, UWB signals do not cause significant interference to other wireless systems.

Thirdly, UWB provides high secure and high reliable communication solutions. Due to the low energy density, the UWB signal is noise-like, which makes unintended detection quite difficult. Furthermore, the "noise-like" signal has a particular shape; in contrast,

real noise has no shape. For this reason, it is almost impossible for real noise to obliterate the pulse because interference would have to spread uniformly across the entire spectrum to obscure the pulse. Interference in only part of the spectrum reduces the amount of received signal, but the pulse still can be recovered to restore the signal. Hence UWB is perhaps the most secure means of wireless transmission ever previously available [8].

Lastly, UWB system based on impulse radio features low cost and low complexity which arise from the essentially baseband nature of the signal transmission. UWB does not modulate and demodulate a complex carrier waveform, so it does not require components such as mixers, filters, amplifiers and local oscillators.

2.3 Regulation Issues

Any technology has its own properties and constrains placed on it by physics as well as by regulations. Government regulators define the way that technologies operate so as to make coexistence more harmonious and also to ensure public safety [6].

Since UWB systems operate over an ultra wide frequency spectrum which will overlap with the existing wireless systems such as global positioning system (GPS), and the IEEE 802.11 WLAN, it is natural that regulations are an important issue.

The international regulations for UWB technology is still not available now and it will be mainly dependent on the findings and recommendations on the International Telecommunication Union (ITU).

Currently, United States, with the FCC approval, is the only country to have a complete ruling for UWB devices. While other regulatory bodies around the world have also been trying to build regulations for UWB.

2.3.1 The FCC's Rules in Unites States

After several years of debate, the FCC released its First Report and Order and adopted the rules for Part 15 operation of UWB devices on February 14th, 2002.

The FCC defines UWB operation as any transmission scheme that has a fractional bandwidth greater than or equal to 0.2 or an absolute bandwidth greater than or equal to 500MHz [1]. UWB bandwidth is the frequency band bounded by the points that are 10dB below the highest radiated emission, as based on the complete transmission system including the antenna. The upper boundary and the lower boundary are designated f_H and f_L, respectively. The fractional bandwidth FBW is then given in Equation 2.7.

$$FBW = 2\frac{f_H - f_L}{f_H + f_L} \tag{2.7}$$

Also, the frequency at which the highest radiated emission occurs is designated f_M and it must be contained with this bandwidth.

Although UWB systems have very low transmission power level, there is still serious concern about the potential interference they may cause to other wireless services. To avoid the harmful interference effectively, the FCC regulates emission mask which defines the maximum allowable radiated power for UWB devices.

In FCC's First Report and Order, the UWB devices are defined as imaging systems, vehicular radar systems, indoor systems and hand-held systems. The latter two categories are of primary interest to commercial UWB applications and will be discussed in this study.

The devices of indoor systems are intended solely for indoor operation and must operate with a fixed indoor infrastructure. It is prohibited to use outdoor antenna to direct the transmission outside of a building intentionally. The UWB bandwidth must be contained between 3.1GHz and 10.6GHz, and the radiated power spectral density

(PSD) should be compliant with the emission mask, as given in Table 2-A and Figure 2.7.

UWB hand-held devices do not employ a fixed infrastructure. They should transmit only when sending information to an associated receiver. Antennas should be mounted on the device itself and are not allowed to be placed on outdoor structures. UWB hand-held devices may operate indoors or outdoors. The outdoor emission mask is at the same level of -41.3dBm/MHz as the indoor mask within the UWB band from 3.1GHz to 10.6GHz, and it is 10dB lower outside this band to obtain better protection for other wireless services, as shown in Table 2-A and Figure 2.7.

Table 2-A: FCC emission limits for indoor and hand-held systems

Frequency range (MHz)	Indoor emission mask (dBm/MHz)	Outdoor emission mask (dBm/MHz)
960-1610	-75.3	-75.3
1610-1900	-53.3	-63.3
1900-3100	-51.3	-61.3
3100-10600	-41.3	-41.3
above 10600	-51.3	-61.3

As with all radio transmitters, the potential interference depends on many things, such as when and where the device is used, transmission power level, numbers of device operating, pulse repetition frequency, direction of the transmitted signal and so on. Although the FCC has allowed UWB devices to operate under mandatory emission masks, testing on the interference of UWB with other wireless systems will still continue.

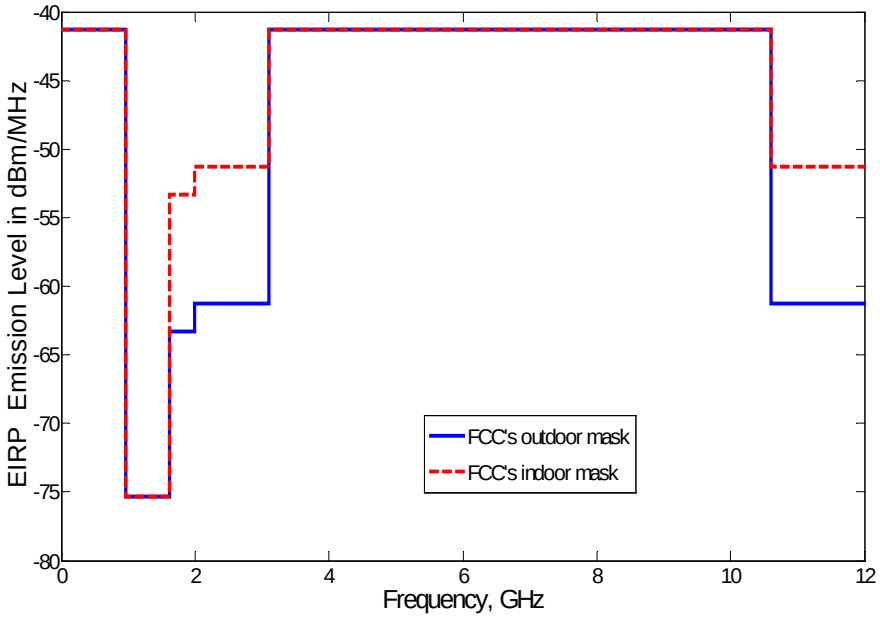


Figure 2.7: FCC's indoor and outdoor emission masks

2.3.2 Regulations Worldwide

The regulatory bodies outside United States are also actively conducting studies to reach a decision on the UWB regulations now. They are, of course, heavily influenced by the FCC's decision, but will not necessarily fully adopt the FCC's regulations.

In Europe, the Electronic Communications Committee (ECC) of the Conference of European Posts and Telecommunications (CEPT) completed the draft report on the protection requirement of radio communication systems from UWB applications [9]. In contrast to the FCC's single emission mask level over the entire UWB band, this report proposed two sub-bands with the low band ranging from 3.1GHz to 4.8GHz and the high band from 6GHz to 8.5GHz, respectively. The emission limit in the high band is -41.3dBm/MHz.

In order to ensure co-existence with other systems that may reside in the low band, the ECC's proposal includes the requirement of Detect and Avoid (DAA) which is an interference mitigation technique [10]. The emission level within the frequency range from 3.1GHz to 4.2GHz is -41.3dBm/MHz if the DAA protection mechanism is available.

Otherwise, it should be lower than -70dBm/MHz. Within the frequency range from 4.2GHz to 4.8GHz, there is no limitation until 2010 and the mask level is -41.3dBm/MHz.

The ECC proposed mask against the FCC one are plotted in Figure 2.8.

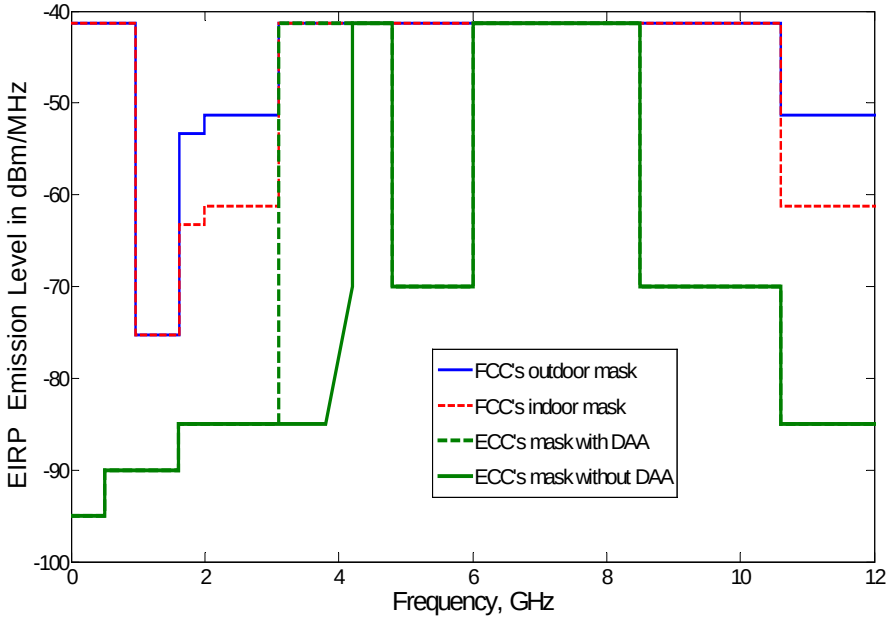


Figure 2.8: Proposed spectral mask of ECC

In Japan, the Ministry of Internal Affairs & Communications (MIC) completed the proposal draft in 2005 [11]. Similar to ECC, the MIC proposal has two sub-bands, but the low band is from 3.4GHz to 4.8GHz and the high band from 7.25GHz to 10.25GHz. DAA protection is also required for the low band.

In Korea, Electronics and Telecommunications Research Institute (ETRI) recommended an emission mask at a much lower level than the FCC spectral mask.

Compared to other countries, Singapore has a more tolerant attitude towards UWB. The Infocomm Development Authority (IDA) of Singapore has been conducting the studies on UWB regulations. Currently, while awaiting for the final regulations, IDA issued a UWB trial license to encourage experimentation and facilitate investigation [2, 12]. With this trial license, UWB systems are permitted to operate at an emission level 6dB higher than the FCC limit from 2.2GHz to 10.6GHz within the UWB Friendly

Zone (UFZ) which is located within Science Park II.

The UWB proposals in Japan, Korea and Singapore against the FCC one are illustrated in Figure 2.9.

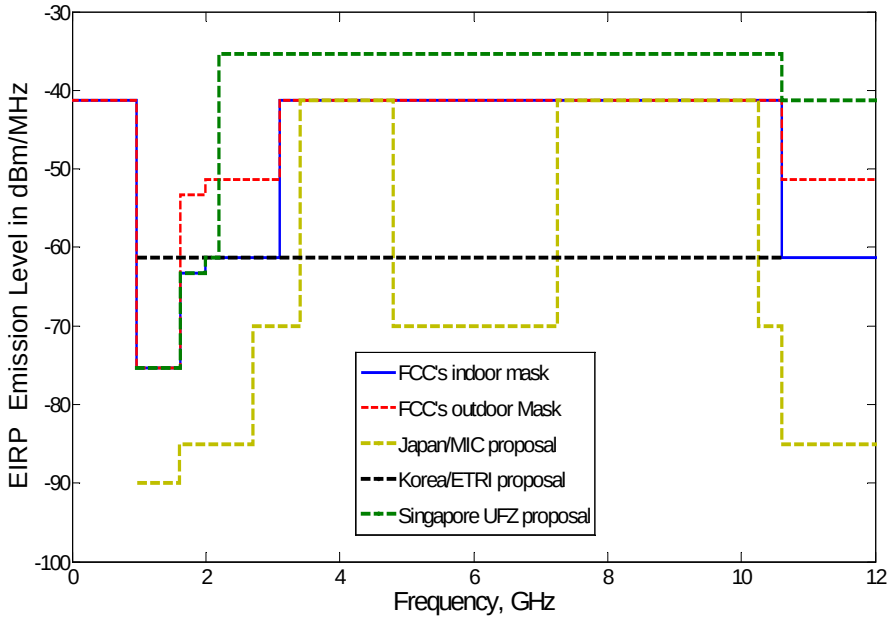


Figure 2.9: Proposed spectral masks in Asia

2.4 UWB Standards

A standard is the precondition for any technology to grow and develop because it makes possible the wide acceptance and dissemination of products from multiple manufacturers with an economy of scale that reduces costs to consumers. Conformance to standards makes it possible for different manufacturers to create products that are compatible or interchangeable with each other [2].

In UWB matters, the IEEE is active in making standards.

The IEEE 802.15.4a task group is focused on low rate alternative physical layer for WPANs. The technical requirements for 802.15.4a include low cost, low data rate (>250kbps), low complexity and low power consumption [13].

The IEEE 802.15.3a task group is aimed at developing high rate alternative physical layer for WPANs [14]. 802.15.3a is proposed to support a data rate of 110Mbps with a distance of 10 meters. When the distance is further reduced to 4 meters and 2 meters, the data rate will be increased to 200Mbps and 480Mbps, respectively. There are two competitive proposals for 802.15.3a, i.e. the Direct Sequence UWB (DS-UWB) and the Multiband Orthogonal Frequency Division Multiplexing (MB-OFDM).

DS-UWB proposal is the conventional impulse radio approach to UWB communication, i.e. it exploits short pulses which occupy a single band of several GHz for transmission. This proposal is mainly backed by Freescale and Japanese NICT and its proponents have established their own umbrella group, namely, the UWB Forum [15].

DS-UWB proposal employs direct sequence spreading of binary data sequences for transmission modulation.

The concept of direct sequence spread spectrum (DSSS) is illustrated in Figure 2.10. The input data is modulated by a pseudo-noise (PN) sequence which is a binary sequence that appears random but can be reproduced at the receiver. Each user is assigned a unique PN code which is approximately orthogonal to those of other users. The receiver can separate each user based on their PN code even if they share the same frequency band. Therefore, many users can simultaneously use the same bandwidth without significantly interfering one another [16].

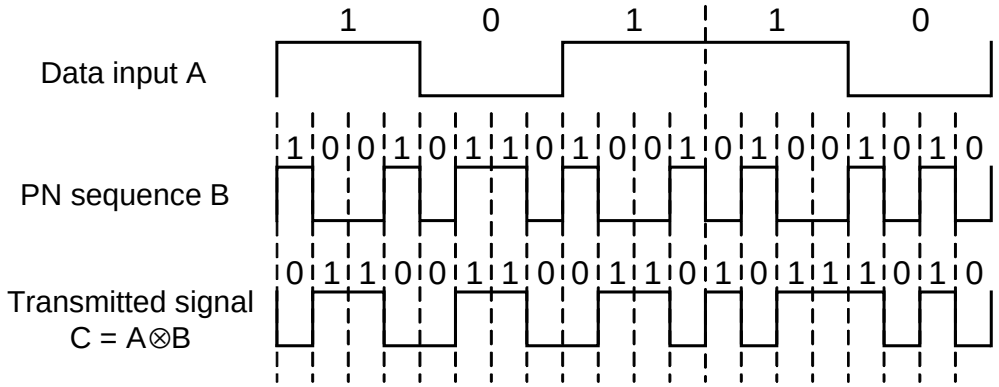


Figure 2.10: Example of direct sequence spread spectrum

The different achievable data rates are obtained by varying the convolutional code rate and the spreading code length. The code length determines the number of chips duration used to represent one symbol. Hence, a shorter code length will lead to a higher data rate for fixed error-correcting code rate [17].

The main advantage of DS-UWB is its immunity to the multipath fading due to the large frequency bandwidth. It is also flexible to adapt very high data rates in a very short distance.

However, there is also technical challenge to DS-UWB. As shown in Table 2-B, the FCC defined a single band of 7.5GHz for UWB communications, but this 3.1GHz-10.6GHz band is broken down into low and high sub-bands. Thus, an efficient pulse-shaping filter is required in order to comply with the various spectral masks proposed by different regulatory bodies.

Table 2-B: Proposed UWB band in the world

Region	UWB band
United States	Single band: 3.1GHz–10.6GHz
Europe	Low band: 3.1GHz–4.8GHz High band: 6GHz–8.5GHz
Japan	Low band: 3.4GHz–4.8GHz High band: 7.25GHz–10.25GHz

MB-OFDM proposal is supported by MultiBand OFDM Alliance (MBOA) which is comprised of more than 100 companies. MB-OFDM combines the multiband approach together with the orthogonal frequency division multiplexing (OFDM) techniques.

OFDM is a special case of multicarrier transmission, where a single data stream is transmitted over a number of lower rate sub-carriers. Because the sub-carriers are mathematically orthogonal, they can be arranged in a OFDM signal such that the sidebands of the individual sub-carriers overlap and the signals are still received without adjacent carrier interference. It is apparent that OFDM can achieve higher bandwidth efficiency compared with conventional multicarrier technique, as shown in Figure 2.11.

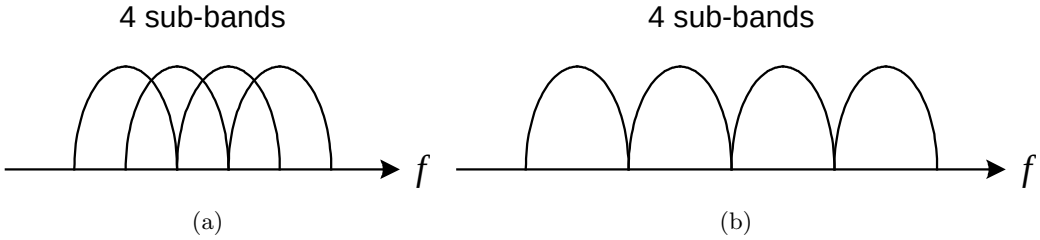


Figure 2.11: (a) OFDM technique versus (b) conventional multicarrier technique

In MB-OFDM proposal, the total UWB frequency band from 3.1GHz to 10.6GHz is divided into 14 sub-bands each of which has a bandwidth of 528MHz to conform to the FCC definition of UWB [18], as shown in Figure 2.12. A packet of data is modulated into a group of OFDM symbols which are then transmitted across the different sub-bands. Frequency hopping is used to obtain multiple access, as discussed previously.

MB-OFDM has greater flexibility in adapting to the spectral regulation of different countries which makes it attractive especially given that there is still much uncertainty in the worldwide regulation process. Also, MB-OFDM is flexible to provide multiple data rates in the system which makes it capable of meeting the needs of different customers. Due to its multiband scheme, MB-OFDM permits adaptive selection of the sub-bands so as to avoid interference with other systems at certain frequency range. Besides, OFDM technique is well-established and has already gained popularity in WLAN and IEEE 802.11a.

The main disadvantage of MB-OFDM is the inferior multipath resolution compared to DS-UWB due to the narrower bandwidth of each sub-band.

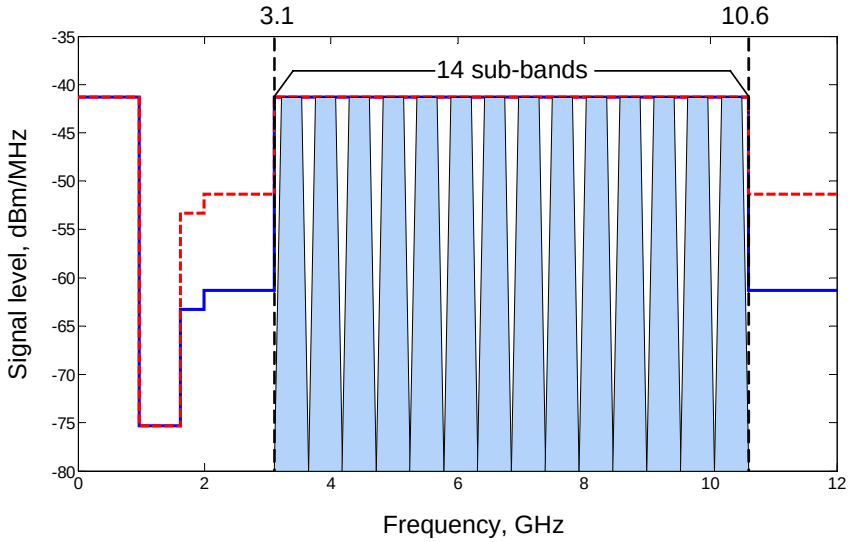


Figure 2.12: Band plan for OFDM UWB system

Currently, both of DS-UWB and MB-OFDM proposals are still under consideration and either of them has its own proponents. It seems that these two proposals will be selected by market forces.

2.5 UWB Applications

As mentioned earlier in this chapter, UWB offers some unique and distinctive properties that make it attractive for various applications.

Firstly, UWB has the potential for very high data rates using very low power at very limited range, which will lead to the applications well suited for WPAN. The peripheral connectivity through cableless connections to applications like storage, I/O devices and wireless USB will improve the ease and value of using Personal Computers (PCs) and laptops. High data rate transmissions between computers and consumer electronics like digital cameras, video cameras, MP3 players, televisions, personal video recorders, automobiles and DVD players will provide new experience in home and personal entertainment.

Secondly, sensors of all types also offer an opportunity for UWB to flourish [2]. Sensor networks is comprised of a large number of nodes within a geographical area. These nodes may be static, when applied for securing home, tracking and monitoring, or mobile, if equipped on soldiers, firemen, automobiles, or robots in military and emergency response situations [19]. The key requirements for sensor networks include low cost, low power and multifunctionality which can be well met by using UWB technology. High data rate UWB systems are capable of gathering and disseminating or exchanging a vast quantity of sensory data in a timely manner. The cost of installation and maintenance can drop significantly by using UWB sensor networks due to being devoid of wires. This merit is especially attractive in medical applications because a UWB sensor network frees the patient from being shackled by wires and cables when extensive medical monitoring is required. In addition, with a wireless solution, the coverage can be expanded more easily and made more reliable.

Thirdly, positioning and tracking is another unique property of UWB. Because of the high data rate characteristic in short range, UWB provides an excellent solution for indoor location with a much higher degree of accuracy than a GPS. Furthermore, with advanced tracking mechanism, the precise determination of the tracking of moving objects within an indoor environment can be achieved with an accuracy of several centimeters [2]. UWB systems can operate in complex situations to yield faster and more effective communication between people. They can also be used to find people or objects in a variety of situations, such as casualties in a collapsed building after an earthquake, children lost in the mall, injured tourists in a remote area, fire fighters in a burning building and so on.

Lastly, UWB can also be applied to radar and imaging applications. It has been used in military applications to locate enemy objects behind walls and around corners in the battlefield. It has also found value in commercial use, such as rescue work where a UWB radar could detect a person's breath beneath rubble, or medical diagnostics where X-ray systems may be less desirable.

UWB short pulses allow for very accurate delay estimates, enabling high definition radar. Based on the high ranging accuracy, intelligent collision-avoidance and cruise-control systems can be envisioned [19]. These systems can also improve airbag deployment and adapt suspension/braking systems depending on road conditions. Besides, UWB vehicular radar is also used to detect the location and movement of objects near a vehicle.

2.6 Summary

The FCC approval of UWB for commercial use has prompted the industry as well as the academia to put significant efforts into this technology.

The future of UWB will heavily depend on the regulatory rulings and standards. Currently, several regulatory bodies around the world are conducting studies to build the UWB regulations. The majority of the debate on UWB centred around the question of whether it will cause harmful interference to other systems and services. Although UWB devices are required to operate with a power level compliant with the emission mask, the concern about the potential interference will continue.

The IEEE has established two task groups working on the UWB standards. In the high data rate case, there are two leading proposals which compete with each other, i.e. DS-UWB and MB-OFDM. Both of these two proposals are still under consideration and probably the market will do the selection.

Owning to its distinctive advantages, UWB technology will be applied in a wide range of areas, including communications, sensors, positioning, radar, imaging and so on.

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Chapter 3

Antenna Theory

The main objective of this thesis is to design antennas that are suitable for the future UWB communication systems. Before the design work, it is necessary to get familiar with the fundamental antenna theory in this chapter. Some important parameters that always have to be considered in antenna design are described. At the same time, the primary requirements for a suitable UWB antenna are discussed. Some general approaches to achieve wide operating bandwidth of antenna are presented. Also, some classic UWB antenna configurations are introduced.

3.1 Introduction

3.1.1 Definition of Antenna

The antennas are an essential part of any wireless system. According to *The IEEE Standard Definitions of terms for Antennas*, an antenna is defined as "a means for radiating or receiving radio waves [1]". In other words, a transmit antenna is a device that takes the signals from a transmission line, converts them into electromagnetic waves and then broadcasts them into free space, as shown in Figure 3.1; while operating in receive

mode, the antenna collects the incident electromagnetic waves and converts them back into signals.

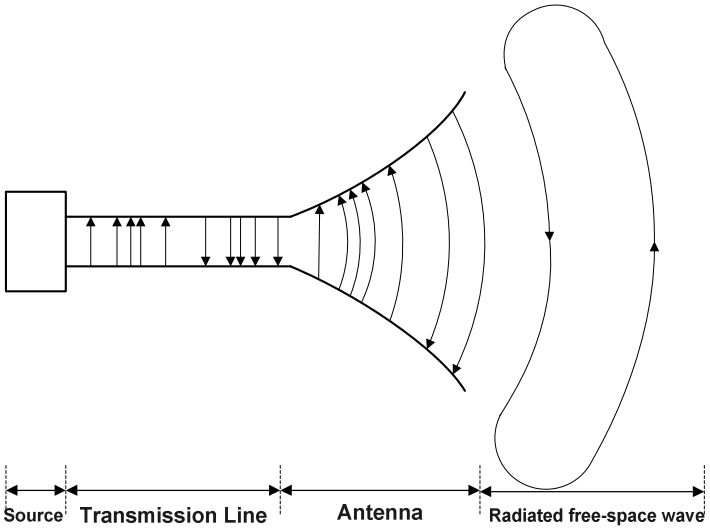


Figure 3.1: Antenna as a transition device

In an advanced wireless system, an antenna is usually required to optimize or accentuate the radiation energy in some directions and suppress it in others at certain frequencies. Thus the antenna must also serve as a directional in addition to a transition device. In order to meet the particular requirement, it must take various forms. As a result, an antenna may be a piece of conducting wire, an aperture, a patch, a reflector, a lens, an assembly of elements (arrays) and so on. A good design of the antenna can relax system requirements and improve overall system performance.

3.1.2 Important Parameters of Antenna

To describe the performance of an antenna, definitions of various parameters are necessary. In practice, there are several commonly used antenna parameters, including frequency bandwidth, radiation pattern, directivity, gain, input impedance, and so on.

3.1.2.1 Frequency Bandwidth

Frequency bandwidth (BW) is the range of frequencies within which the performance of the antenna, with respect to some characteristic, conforms to a specified standard. The bandwidth can be considered to be the range of frequencies, on either side of the center frequency, where the antenna characteristics are within an acceptable value of those at the center frequency. Generally, in wireless communications, the antenna is required to provide a return loss less than -10dB over its frequency bandwidth.

The frequency bandwidth of an antenna can be expressed as either absolute bandwidth (ABW) or fractional bandwidth (FBW). If f_H and f_L denote the upper edge and the lower edge of the antenna bandwidth, respectively. The ABW is defined as the difference of the two edges and the FBW is designated as the percentage of the frequency difference over the center frequency, as given in Equation 3.1 and 3.2, respectively.

$$ABW = f_H - f_L \tag{3.1}$$

$$FBW = 2\frac{f_H - f_L}{f_H + f_L} \tag{3.2}$$

For broadband antennas, the bandwidth can also be expressed as the ratio of the upper to the lower frequencies, where the antenna performance is acceptable, as shown in Equation 3.3.

$$BW = \frac{f_H}{f_L} \tag{3.3}$$

3.1.2.2 Radiation Pattern

The radiation pattern (or antenna pattern) is the representation of the radiation properties of the antenna as a function of space coordinates. In most cases, it is determined in the far-field region where the spatial (angular) distribution of the radiated power does not depend on the distance. Usually, the pattern describes the normalized field (power)

values with respect to the maximum values.

The radiation property of most concern is the two- or three-dimensional (2D or 3D) spatial distribution of radiated energy as a function of the observer's position along a path or surface of constant radius. In practice, the three-dimensional pattern is sometimes required and can be constructed in a series of two-dimensional patterns. For most practical applications, a few plots of the pattern as a function of φ for some particular values of frequency, plus a few plots as a function of frequency for some particular values of θ will provide most of the useful information needed, where φ and θ are the two axes in a spherical coordinate.

For a linearly polarised antenna, its performance is often described in terms of its principle E-plane and H-plane patterns. The E-plane is defined as the plane containing the electric-field vector and the direction of maximum radiation whilst the H-plane is defined as the plane containing the magnetic-field vector and the direction of maximum radiation [1].

There are three common radiation patterns that are used to describe an antenna's radiation property:

(a) *Isotropic* - A hypothetical lossless antenna having equal radiation in all directions. It is only applicable for an ideal antenna and is often taken as a reference for expressing the directive properties of actual antennas.

(b) *Directional* - An antenna having the property of radiating or receiving electromagnetic waves more effectively in some directions than in others. This is usually applicable to an antenna where its maximum directivity is significantly greater than that of a half-wave dipole.

(c) *Omni-directional* - An antenna having an essentially non-directional pattern in a given plane and a directional pattern in any orthogonal plane.

3.1.2.3 Directivity and Gain

To describe the directional properties of antenna radiation pattern, directivity D is introduced and it is defined as the ratio of the radiation intensity U in a given direction from the antenna over that of an isotropic source. For an isotropic source, the radiation intensity U_0 is equal to the total radiated power P_{rad} divided by 4π. So the directivity can be calculated by:

$$D = \frac{U}{U_0} = \frac{4\pi U}{P_{rad}} \tag{3.4}$$

If not specified, antenna directivity implies its maximum value, i.e. D_0.

$$D_0 = \frac{U|_{max}}{U_0} = \frac{U_{max}}{U_0} = \frac{4\pi U_{max}}{P_{rad}} \tag{3.5}$$

Antenna gain G is closely related to the directivity, but it takes into account the radiation efficiency e_{rad} of the antenna as well as its directional properties, as given by:

$$G = e_{rad} D \tag{3.6}$$

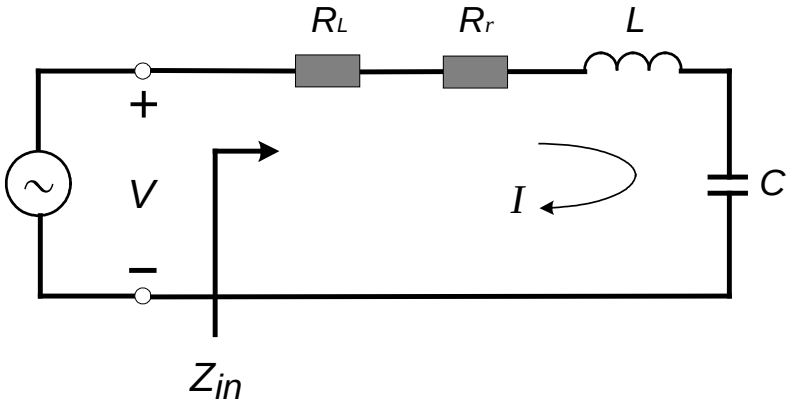


Figure 3.2: Equivalent circuit of antenna

Figure 3.2 shows the equivalent circuit of the antenna, where R_r, R_L, L and C represent the radiation resistance, loss resistance, inductor and capacitor, respectively. The radiation efficiency e_{rad} is defined as the ratio of the power delivered to the radiation

resistance R_r to the power delivered to R_r and R_L. So the radiation efficiency e_{rad} can be written as:

$$e_{rad} = \frac{\frac{1}{2}|I|^2 R_r}{\frac{1}{2}|I|^2 R_r + \frac{1}{2}|I|^2 R_L} = \frac{R_r}{R_r + R_L} \tag{3.7}$$

Similarly, the maximum gain G_0 is related the maximum directivity D_0 by:

$$G_0 = e_{rad} D_0 \tag{3.8}$$

3.1.3 Infinitesimal Dipole (Hertzian Dipole)

The Hertzian Dipole is a dipole whose length dl is much smaller than the wavelength λ of the excited wave, i.e. $dl \ll \lambda$ ($dl < \lambda/50$). Besides, it is very thin, and its radius a is also much smaller than the wavelength λ.

As shown in Figure 3.3, the infinitesimal linear wire is positioned symmetrically at the origin of the coordinate system and oriented along the z-axis.

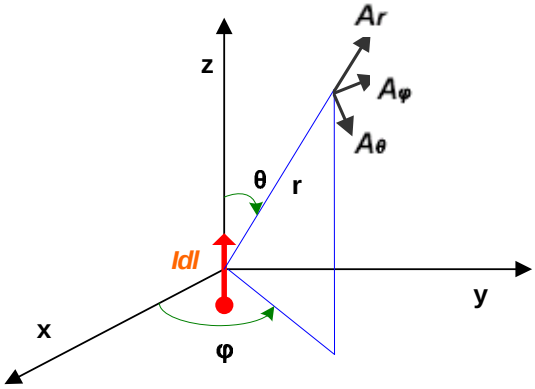


Figure 3.3: Hertzian Dipole

The infinitesimal dipole is equivalent to a current element $I_0 dl$. Since it is very short, the current is assumed to be constant.

Although infinitesimal dipoles are not very practical, they are utilized as building blocks of more complex geometries.

3.1.3.1 Radiated Field

To obtain the fields radiated by the current element, it is required to determine magnetic vector potential $\vec{A}$ first.

For Hertzian Dipole, $\vec{A}$ is expressed as [2]:

$$\vec{A} = \frac{\mu_0 I_0 dl}{4\pi r} e^{-jkr} \vec{z} \tag{3.9}$$

In the spherical coordinate, Equation 3.9 is transformed to:

$$\begin{aligned} A_r &= A_z \cos\theta = \frac{\mu_0 I_0 dl}{4\pi r} e^{-jkr} \cos\theta \\ A_\theta &= -A_z \sin\theta = -\frac{\mu_0 I_0 dl}{4\pi r} e^{-jkr} \sin\theta \\ A_\varphi &= 0 \end{aligned} \tag{3.10}$$

According to Maxwell's equations and the relationship between $\vec{A}$ and $\vec{H}$:

$$\begin{aligned} \nabla \times \vec{E} &= -j\omega\mu\vec{H} \\ \vec{H} &= \frac{1}{\mu} \nabla \times \vec{A} \end{aligned} \tag{3.11}$$

Now E- and H-field can be found:

$$\begin{aligned} H_r &= H_\theta = 0 \\ H_\varphi &= j\frac{kI_0 dl \sin\theta}{4\pi r} \left[1 + \frac{1}{jkr}\right] e^{-jkr} \\ E_r &= \eta\frac{I_0 dl \cos\theta}{2\pi r^2} \left[1 + \frac{1}{jkr}\right] e^{-jkr} \\ E_\theta &= j\eta\frac{kI_0 dl \sin\theta}{4\pi r} \left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2}\right] e^{-jkr} \\ E_\varphi &= 0 \end{aligned} \tag{3.12}$$

In the far-field region where $kr \gg 1$, the E- and H-field can be simplified and approximated by:

$$\begin{aligned} E_\theta &\approx j\eta \frac{kI_0 dl \sin\theta}{4\pi r} e^{-jkr} \\ E_r &\approx E_\varphi = H_r = H_\theta = 0 \\ H_\varphi &\approx j \frac{kI_0 dl \sin\theta}{4\pi r} e^{-jkr} \end{aligned} \tag{3.13}$$

The ratio of E_θ and H_φ is:

$$Z_w = \frac{E_\theta}{H_\varphi} \approx \eta \tag{3.14}$$

where Z_w is the wave impedance; η is the intrinsic impedance of the medium (377≈120π Ohms for free space).

In the far-field region, the E- and H-field components are perpendicular to each other, transverse to the radial direction of propagation, and the r variations are separable from those of θ and φ. The shape of the pattern is not a function of the radial distance r, and the fields form a Transverse Electromagnetic (TEM) wave whose wave impedance equals to the intrinsic impedance of the medium.

3.1.3.2 Radiation Resistance

For a lossless antenna, the real part of the input impedance was designated as radiation resistance R_r. By integrating the Poynting vector over a closed surface, P_{rad}, i.e. the total power radiated by the source, can be found and its real part is related to the radiation resistance.

$$P_{rad} = \frac{1}{2}\mathbf{Re}\int (\vec{E} \times \vec{H}^*) ds = \eta \left(\frac{\pi}{3}\right) \left|\frac{I_0 dl}{\lambda}\right|^2 = \frac{1}{2} |I_0|^2 R_r \tag{3.15}$$

$$R_r = \eta \left(\frac{2\pi}{3}\right) \left|\frac{dl}{\lambda}\right|^2 = 80\pi^2 \left|\frac{dl}{\lambda}\right|^2 \tag{3.16}$$

Thus, the radiation resistance R_r is dependent on dl and the wavelength λ.

3.1.3.3 Directivity

The average power density W_{av} can be obtained by Equation 3.17:

$$W_{av} = \frac{1}{2}\mathbf{Re}[\vec{E} \times \vec{H}^*] = \frac{1}{2\eta}|E_\theta|^2\vec{r} = \frac{\eta}{2}\left|\frac{kI_0dl}{4\pi}\right|^2 \frac{\sin^2\theta}{r^2}\vec{r} \tag{3.17}$$

Associated with the average power density is a radiation intensity U given by:

$$U = r^2 W_{av} = \frac{\eta}{2}\left|\frac{kI_0dl}{4\pi}\right|^2 \sin^2\theta = \frac{r^2}{2\eta}|E_\theta(r,\theta,\varphi)|^2 \tag{3.18}$$

The maximum value occurs at $\theta = \pi/2$ and it equals to:

$$U_{max} = \frac{\eta}{2}\left|\frac{kI_0dl}{4\pi}\right|^2 \tag{3.19}$$

Then the directivity D_0 of Hertzian Dipole reduces to [2]:

$$D_0 = \frac{4\pi U_{max}}{P_{rad}} = 1.5 \tag{3.20}$$

3.2 Requirements for UWB Antennas

As is the case in conventional wireless communication systems, an antenna also plays a crucial role in UWB systems. However, there are more challenges in designing a UWB antenna than a narrow band one [3].

First of all, what distinguishes a UWB antenna from other antennas is its ultra wide frequency bandwidth. According to the FCC's definition, a suitable UWB antenna should be able to yield an absolute bandwidth no less than 500MHz or a fractional

bandwidth of at least 0.2.

Secondly, the performance of a UWB antenna is required to be consistent over the entire operational band. Ideally, antenna radiation patterns, gains and impedance matching should be stable across the entire band. Sometimes, it is also demanded that the UWB antenna provides the band-rejected characteristic to coexist with other narrow-band devices and services occupying the same operational band [4, 5].

Thirdly, directional or omni-directional radiation properties are needed depending on the practical application. Omni-directional patterns are normally desirable in mobile and hand-held systems. For radar systems and other directional systems where high gain is desired, directional radiation characteristics are preferred.

Fourthly, a suitable antenna needs to be small enough to be compatible to the UWB unit especially in mobile and portable devices. It is also highly desirable that the antenna feature low profile and compatibility for integration with printed circuit board (PCB).

Fifthly, a good design of UWB antenna should be optimal for the performance of overall system. For example, the antenna should be designed such that the overall device (antenna and RF front end) complies with the mandatory power emission mask given by the FCC or other regulatory bodies.

Lastly, but not the least important, a UWB antenna is required to achieve good time domain characteristics. For the narrow band case, it is approximated that an antenna has same performance over the entire bandwidth and the basic parameters, such as gain and return loss, have little variation across the operational band. In contrast, UWB systems often employ extremely short pulses for data transmission. In other words, enormous bandwidth has been occupied. Thus the antenna can't be treated as a "spot filter" any more but a "band-pass filter". In this case, the antenna imposes more significant impacts on the input signal. As a result, a good time domain performance, i.e. minimum pulse distortion in the received waveform, is a primary concern of a suitable UWB antenna [6] because the signal is the carrier of useful information. Therefore, it is indispensable

and important to study the antenna's characteristics in time domain.

3.3 Approaches to Achieve Wide Operating Bandwidth

As discussed in previous section, operating bandwidth is one of the most important parameter of an antenna. It is also the main characteristic in which a UWB antenna differs from other antennas. Various methods have already been exploited to achieve bandwidth enhancement for different types of antennas.

3.3.1 Resonant Antennas

Resonant antennas, such as straight wire dipoles and microstrip patch antennas, operate at a single resonance mode at a time and the operating bandwidth is related to the antenna quality factor Q and radiation efficiency e_{rad}.

3.3.1.1 Quality Factor and Bandwidth

The quality factor Q of an antenna is defined as $2\pi f$ times the energy stored over the power radiated and the ohmic losses. It can be calculated through the equivalent circuit of the antenna, as shown in Figure 3.2.

At resonant frequency f_0 ($2\pi f_0 = 1/\sqrt{LC}$),

$$\begin{aligned} Q &= 2\pi f_0 \frac{\frac{1}{4}|I|^2 L + \frac{1}{4}|I|^2 \frac{1}{(2\pi f_0)^2 C}}{\frac{1}{2}|I|^2 (R_r + R_L)} = \frac{2\pi f_0 L}{R_r + R_L} \\ &= \frac{1}{2\pi f_0 (R_r + R_L) C} = \frac{1}{2\pi f_0 R_r C} \cdot \frac{R_r}{R_r + R_L} \\ &= Q_{lossless} \cdot e_{rad} \end{aligned} \tag{3.21}$$

where $Q_{lossless}$ is the quality factor when the antenna is assumed to be lossless, i.e. R_L=0; e_{rad} is the antenna radiation efficiency.

Equation 3.21 indicates that the quantity factor Q is proportion to the antenna radiation efficiency e_{rad}.

Now consider the behavior of the input impedance Z_{in} near resonance f_0 and let $f = f_0 + \Delta f$, where Δf is small. Then Z_{in} is expressed as:

$$Z_{in} = R + j2\pi fL\left(1 - \frac{1}{(2\pi f)^2 LC}\right) = R + j2\pi fL\left(\frac{f^2 - f_0^2}{f^2}\right) \tag{3.22}$$

where $R = R_r + R_L$;

Since $f^2 - f_0^2 = (f - f_0)(f + f_0) = \Delta f(2f - \Delta f) \approx 2f\Delta f$ for small Δf, Equation 3.22 reduces to:

$$Z_{in} \approx R + j4\pi L\Delta f = R + j\frac{2RQ\Delta f}{f} \tag{3.23}$$

At half power frequencies ($f = f_0 + \Delta f_H$), $|Z_{in}|^2 = 2R^2$, i.e.

$$|R + jRQ\frac{2\Delta f_H}{f_0}|^2 = 2R^2 \tag{3.24}$$

Thus,

$$FBW = 2\frac{\Delta f_H}{f_0} = \frac{1}{Q} \tag{3.25}$$

Here, FBW is the half-power fractional bandwidth of the antenna which is inversely proportional to the quality factor Q.

3.3.1.2 Fundamental Limitations for Electrically Small Antennas

Equation 3.25 indicates that the fractional bandwidth of the antenna may be increases by reducing the quality factor Q. However, according to Equation 3.21, reducing the Q factor will also lead to the decrease of the efficiency e_{rad} and hence the decrease of the antenna gain. Furthermore, there are fundamental limitations as to how small the antenna elements can be made, which also affects the lowest achievable Q .

An electrically small antenna is the one whose largest linear dimension $2r$ satisfies: $kr < 1$, where k is the wave number and equal to $2\pi/\lambda$. The limits on electrically small antennas are derived by assuming that the entire antenna structure (with a largest linear dimension of $2r$), and its transmission line and oscillator are all enclosed within a sphere of radius r [1], as shown in Figure 3.4.

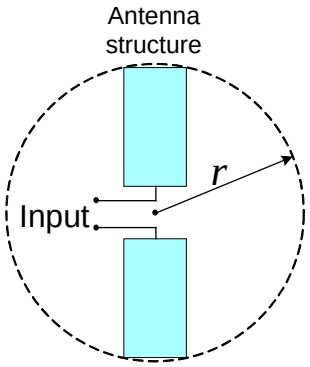


Figure 3.4: Antenna within a sphere of radius r

The fundamental limitations for electrically small antennas were investigated first by Chu [7] and subsequently by Harrington [8], so they are also called Chu-Harrington limitations. Chu's approach uses spherical wave functions to describe the field and calculate the quality factor Q.

When $kr < 1$, the quality factor Q of a small antenna can be expressed as [9]:

$$Q = \frac{1 + 2(kr)^2}{(kr)^3[1 + (kr)^2]} \cdot e_{rad} \tag{3.26}$$

The calculated Q versus kr for different antenna radiation efficiencies are plotted in Figure 3.5.

Equation 3.26 and Figure 3.5 present the relationships between the quality factor Q and the antenna size as well as the radiation efficiency. Since the Q rises rapidly as antenna size decreases, the result relates the lowest achievable Q to the maximum dimension of an electrically small antenna. Because the antenna bandwidth FBW is the reciprocity of Q, the increasing Q with reducing size r indeed implies a fundamental

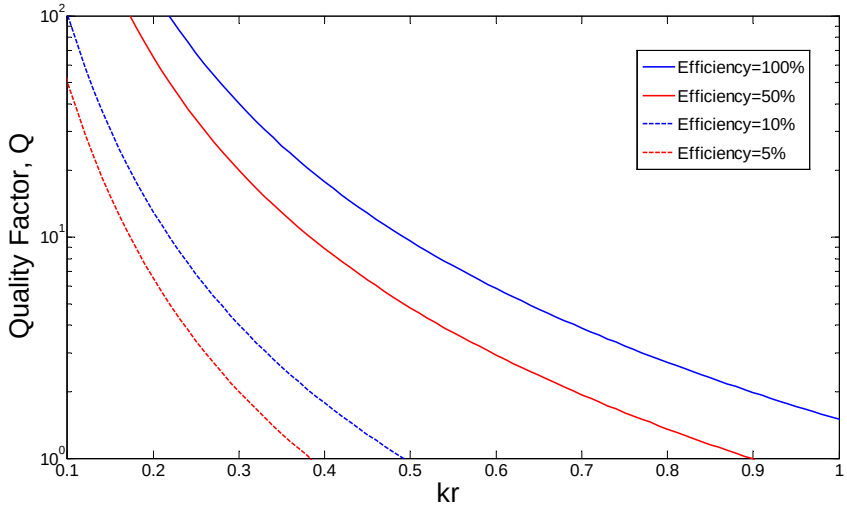


Figure 3.5: Calculated antenna quality factor Q versus kr

limitation on the widest achievable bandwidth FBW. So, the antenna size, quality factor, bandwidth and radiation efficiency are interrelated, and there is no complete freedom to independently optimize each one. Therefore, there is always a tradeoff between them to obtain an optimal antenna performance.

3.3.2 Travelling Wave Antennas

For resonant antennas, such as straight wire dipoles, the wave travelling outward from the feed point to the end of the antenna is reflected, setting up a standing-wave-type current distribution [10]. These antennas are also referred to as standing wave antennas.

If the reflected wave is not strongly present on an antenna such that the antenna acts as a guiding structure for travelling wave, it is referred to as a travelling wave antenna. Travelling wave can be created by using matched load at the ends to prevent reflections. Also, very long antennas may dissipate most of the power, leading to small reflected waves.

The simplest travelling wave antenna is the travelling wave long wire antenna with a length of L greater than one-half wavelength, as illustrated in Figure 3.6. R_L is the

matched load to prevent reflections from the wire end.

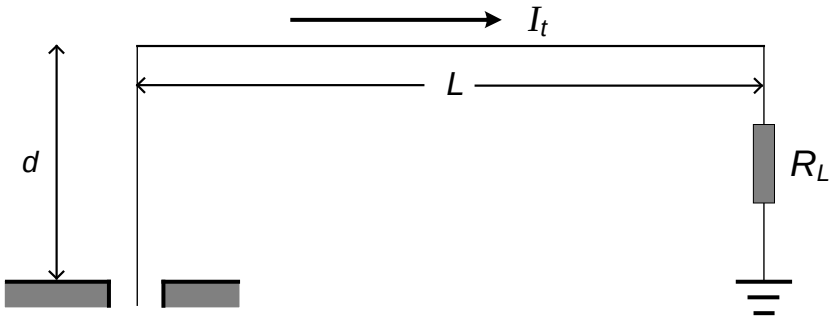


Figure 3.6: Travelling wave long wire antenna

Since a travelling wave antenna carries a pure travelling wave, the current and the voltage have uniform patterns. Thus, the input impedance of a travelling wave antenna is always predominantly real, leading to the wide band characteristic.

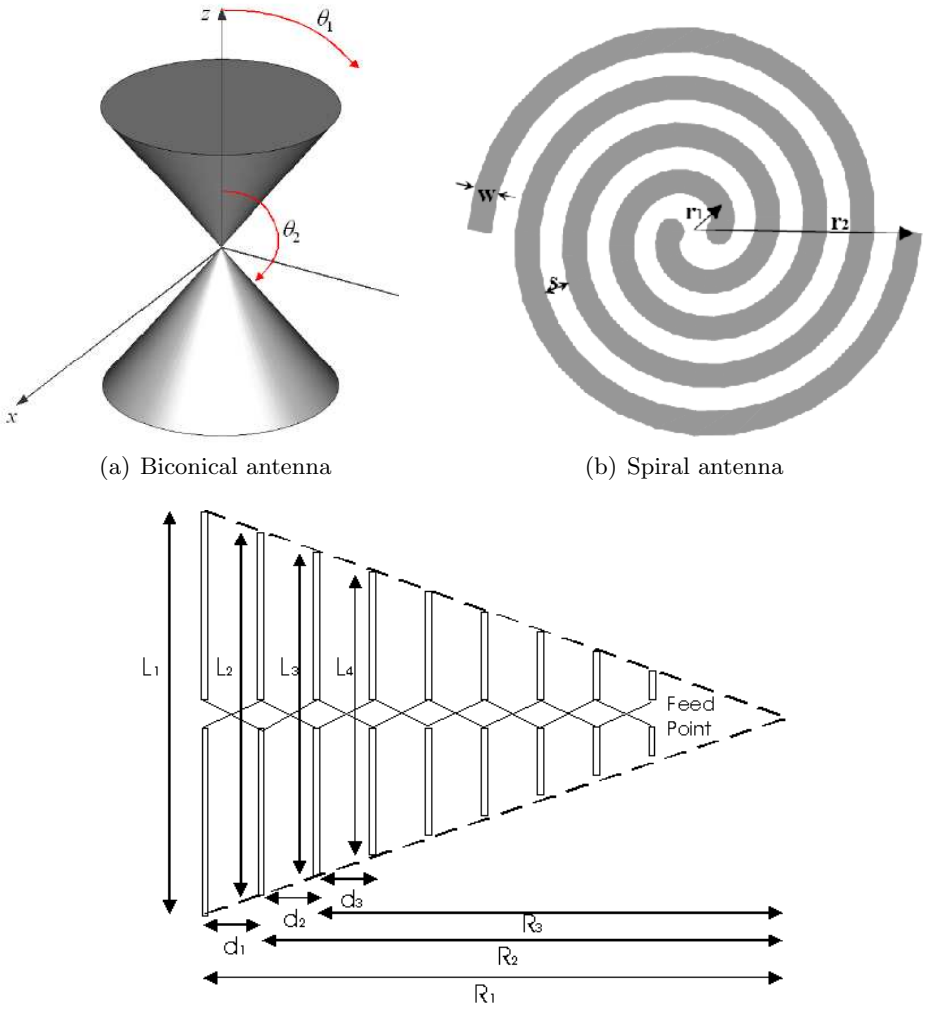
Frequency independent antennas, such as biconical, spiral and log periodic antennas, as illustrated in Figure 3.8, are classic broadband and ultra wideband antennas. They can offer a real constant impedance and constant pattern properties over a frequency bandwidth greater than 10:1. It is suggested by Johnson Wang that these frequency independent antennas are travelling wave antennas in nature [11].

There are two principles for achieving frequency independent characteristics.

The first one was introduced by Victor Rumsey in the 1950s [12]. Rumsey's principle suggests that the impedance and pattern properties of an antenna will be frequency independent if the antenna shape is specified only in terms of angles.

Infinite biconical and spiral antennas are good examples whose shapes are completely described by angles [10]. For log periodic antenna, its entire shape is not solely specified by angles; it is also dependent on the length from the origin to any point on the structure. But log periodic antenna can still exhibit frequency independent characteristics.

The second principle accounting for frequency independent characteristics is self-complementarity, which was introduced by Yasuto Mushiake in the 1940s [13, 14].



(a) Biconical antenna

(b) Spiral antenna

(c) Log periodic antenna

Figure 3.7: Geometries of frequency independent antennas

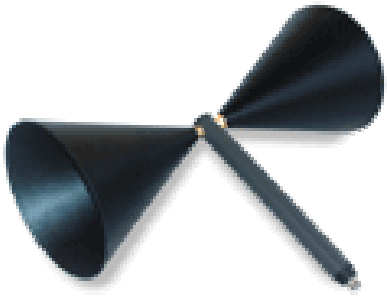
Mushiake discovered that the product of input impedances of a planar electric current antenna (plate) and its corresponding "magnetic current" antenna (slot) was the real constant $\eta^2/4$, where η is the intrinsic impedance ($377 \approx 120\pi$ Ohms for free space). Therefore, if an antenna is its own complement, frequency independent impedance behavior is achieved. The self-complementary antenna has a constant impedance of $\eta/2$ which is equal to 188.5 ohms. In Figure 3.7(b), if $W = S$, i.e. the metal and the air regions of the antenna are equal, the spiral antenna is self-complementary.

Although frequency independent antennas can operate over an extremely wide frequency range, they still have some limitations.

Firstly, to satisfy the Rumsey's requirement, the antenna configuration needs to be infinite in principle, but is usually truncated in size in practice. This requirement makes frequency independent antennas quite large in terms of wavelength.

Secondly, frequency independent antennas tend to be dispersive because they radiate different frequency components from different parts of the antenna, i.e. the smaller-scale part contributes higher frequencies while the large-scale part accounts for lower frequencies. Consequently, the received signal suffers from severe ringing effects and distortions. Due to this drawback, frequency independent antennas can be used only when waveform dispersion may be tolerated.

Some practical frequency independent antennas are shown in Figure 3.8.



(a) Biconical antenna [15]



(b) Spiral antenna [16]



(c) Log periodic antenna [17]

Figure 3.8: Frequency independent antennas

3.3.3 Resonance Overlapping Type of Antennas

Normally, the bandwidth of a resonant antenna is not very broad because it has only one resonance. But if there are two or more resonant parts available with each one operating at its own resonance, the overlapping of these multiple resonances may lead to multi-band or broadband performance. In fact, the technique of using two resonant parts has been widely applied in antenna design especially for mobile handset antennas which are required to operate at various wireless bands.

The two resonant parts can be combined in parallel [18, 19], or one serves as the passive radiator and the other as parasitic element [20, 21].

In [21], a stacked shorted patch antenna is proposed for GSM1800 (1710MHz-1880MHz) handset application. As shown in Figure 3.9, the lower patch (with length of $L_1 = 25mm$ and width $W = 20mm$) is directly fed by a probe feed at the patch edge and the upper patch (with length of $L_2 = 24mm$ and width $W = 20mm$) is parasitically coupled from the lower one. Both of the two patches are printed on a low-loss dielectric substrate and they have a common shorting wall.

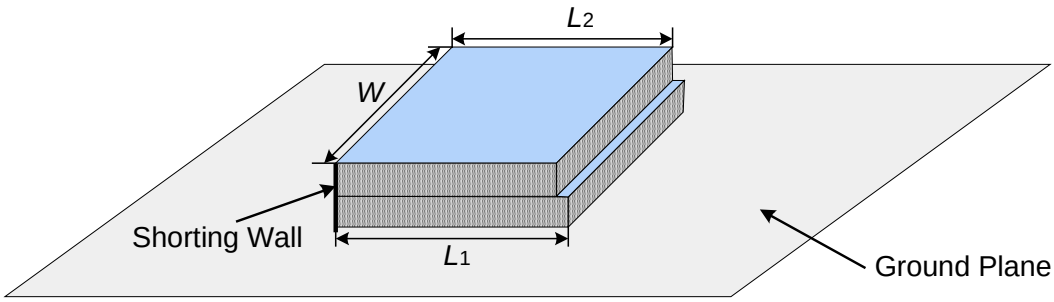


Figure 3.9: Geometry of stacked shorted patch antenna (Reproduced from [21])

Figure 3.10 shows the measured return loss curve of the stacked shorted patch antenna. It is clearly seen that two resonant modes are excited because there are two patch radiators. Furthermore, the two resonant frequencies are closely spaced. Consequently, the overlapping of these two resonances lead to a broad bandwidth, ranging from 1688MHz to 1890MHz.

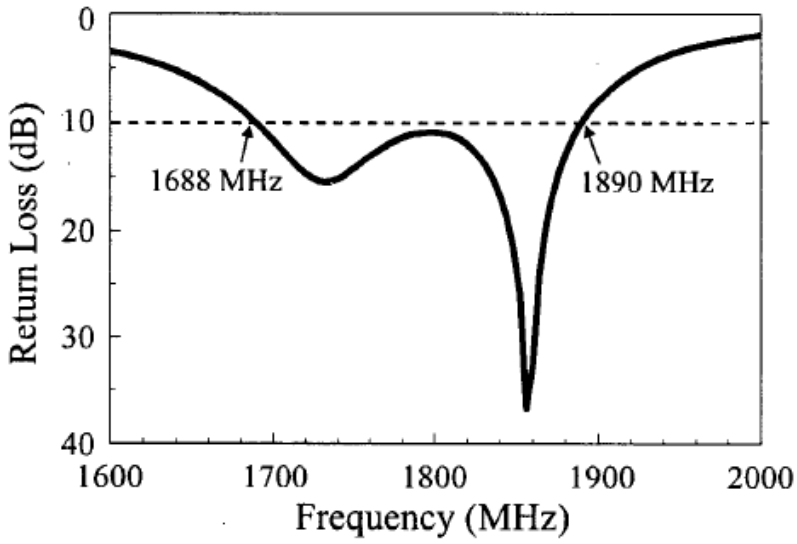


Figure 3.10: Measured return loss curve of stacked shorted patch antenna [21]

The main drawback of the multi-band antennas discussed above is that they can not provide constant radiation patterns over the operational bandwidth, i.e. the patterns differ from each other at different frequencies. This is due to the availability of two different radiating elements.

Theoretically, an ultra wide bandwidth can be obtained if there are a sufficient number of resonant parts and their resonances can overlap each other well. However, in practice, it is more difficult to achieve impedance matching over the entire frequency range when there are more resonant parts. Also, it will make the antenna structure more complicated and more expensive to fabricate. Besides, it is more difficult to achieve constant radiation properties since there are more different radiating elements.

Alternately, if a single radiator itself can support multiple resonances, properly overlapping of these resonances may lead to UWB characteristics with nearly constant radiation patterns across the whole operational band. Such an element can be a circular disc, or an elliptical slot. These two types of UWB antennas will be studied in detail in the later chapters.

3.3.4 "Fat" Monopole Antennas

Conventional monopole has a straight wire configuration against a ground plane, as illustrated in Figure 3.11. It is one of the most widely used antennas for wireless communication systems due to its simple structure, low cost, omni-directional radiation patterns and ease for matching to 50Ω [1]. Besides, it is unbalanced, thus eliminating the need for a balun, which may have a limited bandwidth [22].

The -10dB return loss bandwidth of straight wire monopole is typically around 10%–20%, depending on the radius-to-length ratio of the monopole.

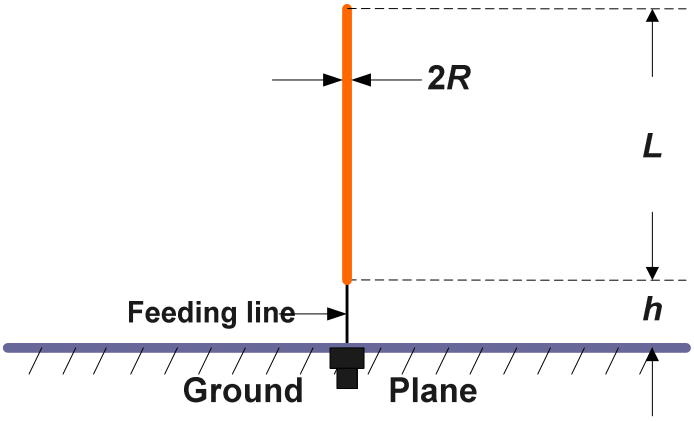


Figure 3.11: Geometry of straight wire monopole

Table 3-A presents the simulated bandwidths of straight wire monopole (given in Figure 3.11) for different radius-to-length ratios when the monopole length L is fixed at 12.5mm and the feed gap h at 2mm, respectively.

It is noticed in Table 3-A that the bandwidth increases with the increase of the radius-to-length ratio. This indicates that a "fatter" structure will lead to a broader bandwidth because the current area and hence the radiation resistance is increased [23].

However, when the monopole radius is too large related to the feeding line, the impedance mismatch between them will become significant and the bandwidth can not be further increased.

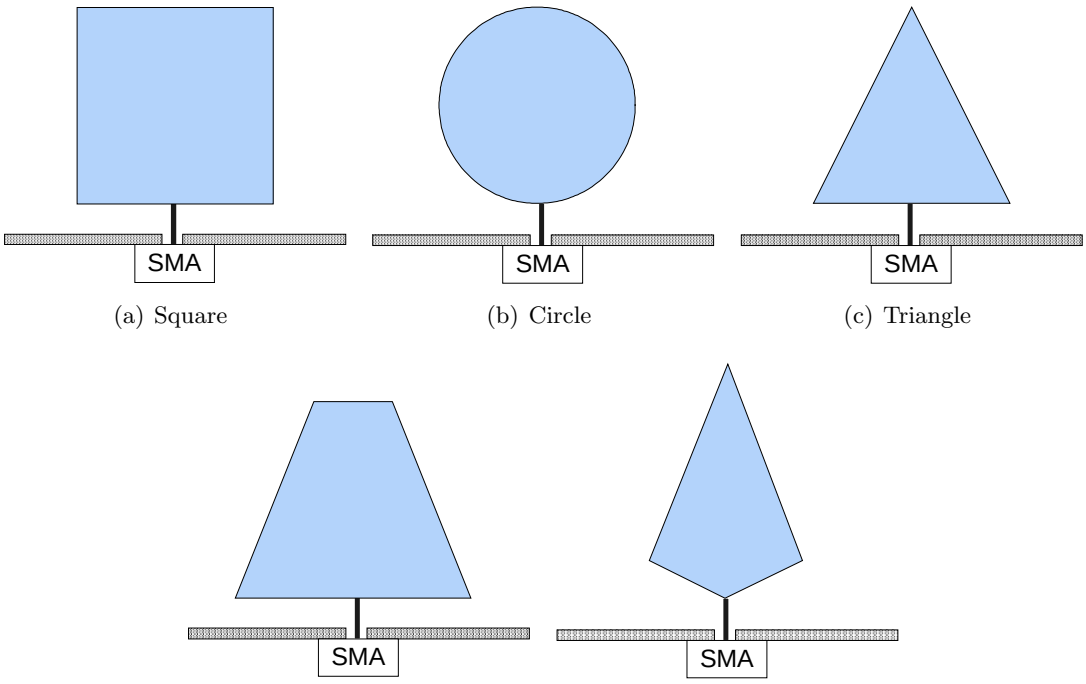
Table 3-A: Simulated -10dB bandwidth of straight wire monopole with L=12.5mm and h=2mm

Radius-to-length ratio R/L (%)	Lower edge of bandwidth (GHz)	Upper edge of bandwidth (GHz)	Absolute bandwidth (GHz)	Fractional bandwidth (%)
0.8	4.95	5.64	0.69	13.03
2.0	4.86	5.85	0.99	18.49
4.0	4.67	5.64	0.97	18.82
6.4	4.65	5.94	1.29	24.36

Another method to obtain bandwidth enhancement is to replace the wire element with a plate which is obviously much "fatter". This plate can take various configurations such as square [22, 24], circle [25], triangle [26], trapezoid [27], "Bishop's Hat" [28] and so on [29, 30], as shown in Figure 3.12.

Several techniques have been proposed to improve the antenna bandwidth, such as the use of a beveling plate [31, 32], a double feed [33] or an asymmetrical feed arrangement [34], a trident-shaped feeding strip [35], and so on. The addition of a second orthogonal element is also reported to achieve satisfactory radiation pattern stability [36, 37].

Compared with other configurations, a circular disc monopole can yield a wider frequency bandwidth. In [38], a bandwidth of 2.25GHz–17.25GHz for voltage standing wave ratio (VSWR)<2 is achieved by using a circular disc with diameter of 25mm and a square ground plane sized 300×300mm^2. This bandwidth has covered the whole FCC defined UWB band. The relationship between the disc dimension and the lower frequency edge of the bandwidth is discussed in [25].

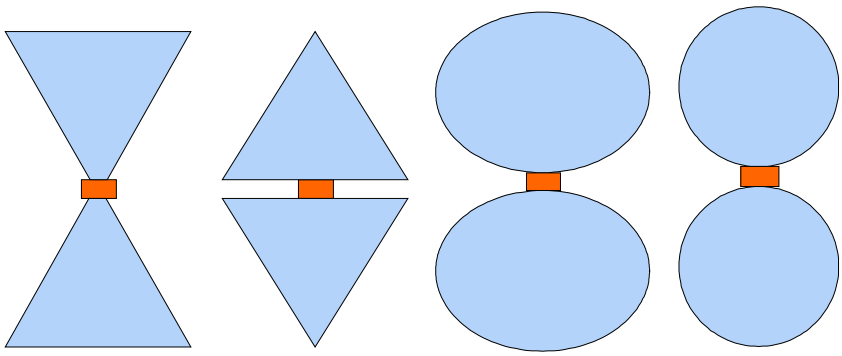


(a) Square (b) Circle (c) Triangle

(d) Trapezoid (e) "Bishop's Hat"

Figure 3.12: Plate monopole antennas with various configurations

Since monopole originates from dipole by removing one element and driving the remaining one against a ground plane. It is understandable that "fat" dipoles, such as bow tie antenna [39], diamond antenna [40], elliptical disc dipole [41] and circular disc dipole [42], can also exhibit UWB characteristics. These UWB dipoles are illustrated in Figure 3.13.



(a) Bow tie (b) Diamond (c) Elliptical (d) Circular

Figure 3.13: UWB dipoles with various configurations

3.4 Summary

As is the case for narrowband systems, antennas are also a key component in UWB systems, but have more stringent requirements.

Several methods have been exploited to widen the antenna operating bandwidth. For resonant antennas, the bandwidth enhancement can be accomplished by lowering the antenna quality factor, but the bandwidth can not be very wide due to the Chu-Harrington limitations. Besides, a low quality factor will lead to low efficiency and low gain.

Frequency independent antennas can yield ultra wide bandwidth owning to either self-complementary or angle-specified structures. But the big size and dispersive performance limit their practical applications.

Broadband or multi-band performances have been achieved through the overlapping of multiple resonances. But it is difficult to obtain UWB performance with constant radiation properties by using multiple radiating elements.

Increasing the surface area so as to make the antenna "fat" will enhance the bandwidth significantly. Several "fat" types of monopoles and dipoles have been demonstrated to exhibit UWB characteristics.

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Chapter 4

UWB Disc Monopole Antennas

As mentioned in Chapter 3, many different types of antennas are currently being considered for UWB applications. Among these antenna configurations, circular disc monopole features simple structure, easy fabrication, wide frequency bandwidth and satisfactory radiation patterns [1, 2].

However, the performances and characteristics of circular disc monopole antennas are not analyzed in detail in either [1] or [2]. How exactly the disc monopole operates across the entire bandwidth, remains a question. It is still not clear why this resonating type of antenna retains a seemingly omni-directional radiation pattern with gain variation less than 10dB over an ultra wide frequency band.

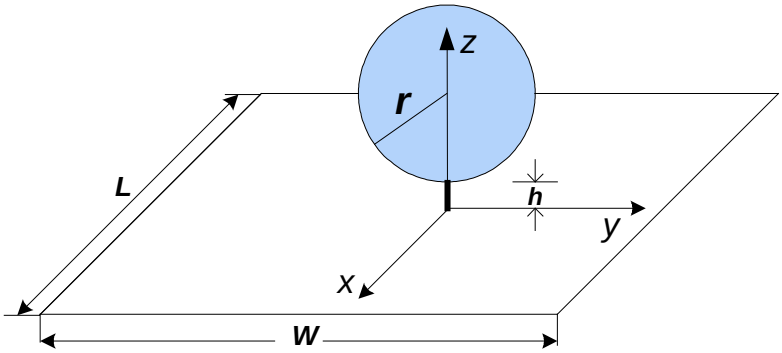
In this Chapter, circular disc monopole antennas will be studied in the frequency domain with an emphasis on the understanding of their operations. The configuration of circular disc monopoles has also evolved from a vertical disc to a printed disc on a printed circuit board (PCB). The important parameters which affect the antenna performances will be investigated both numerically and experimentally to obtain some quantitative guidelines for designing this type of antennas. The time domain behaviors of circular disc monopole antennas will be evaluated in Chapetr 6.

4.1 Vertical Disc Monopole

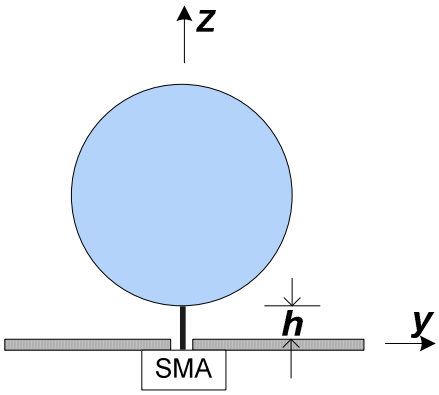
A vertical disc monopole is so called because its disc radiator is vertically placed on the ground plane. It originates from conventional straight wire monopole and is realized by replacing the wire element with a disc copper plate.

4.1.1 Antenna Geometry

The geometry of the vertical disc monopole as well as the coordinate system is illustrated in Figure 4.1.



(a) The coordinate system



(b) Side view

Figure 4.1: Geometry of vertical disc monopole

As shown in Figure 4.1, a circular copper disc with a radius of r and a thickness of 0.4mm is selected as the radiator and mounted vertically above a rectangular copper

ground plane. W and L denote the width and the length of the ground plane, respectively. A 50Ω coaxial probe connects to the bottom of the disc through the ground plane via an SMA connector. h is the height of the feed gap between the feed point and the ground.

Simulations have shown that the performance of the antenna is mainly dependent on the feed gap h and the dimension of the ground plane.

4.1.2 Effect of the Feed Gap

Figure 4.2 illustrates the simulated return loss curves for different feed gaps (h=0.3, 0.7, 1, and 2mm) when both W and L are fixed at 100mm and r at 12.5mm; their corresponding input impedance curves are plotted in Figure 4.3.

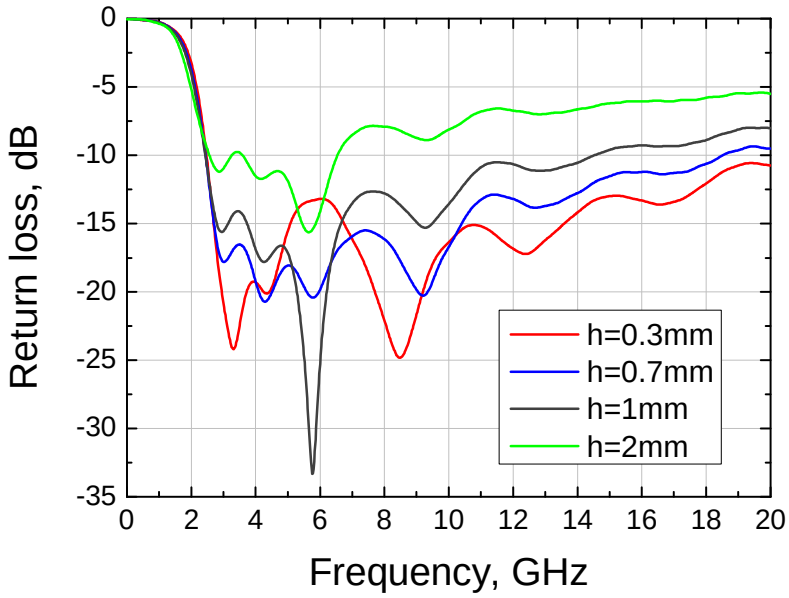
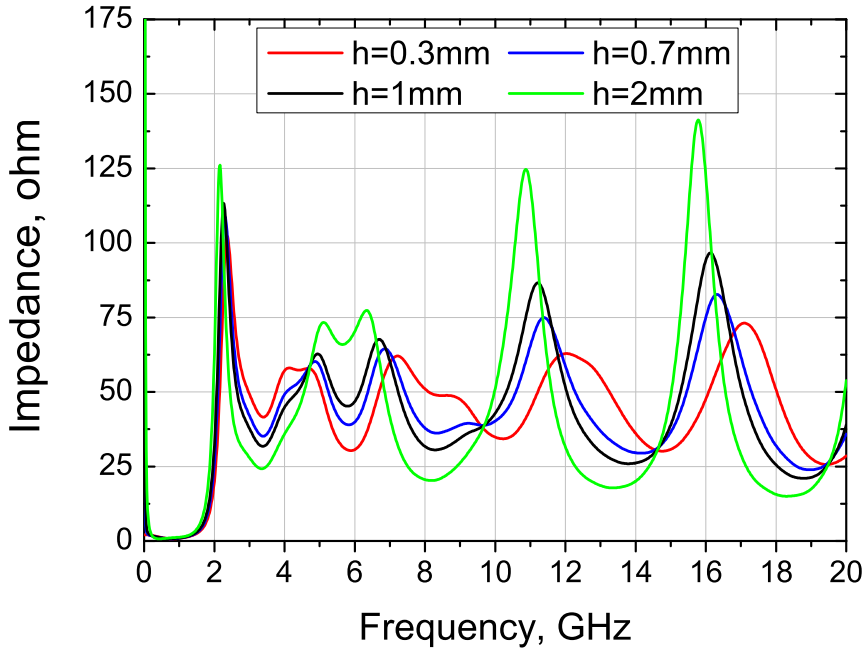
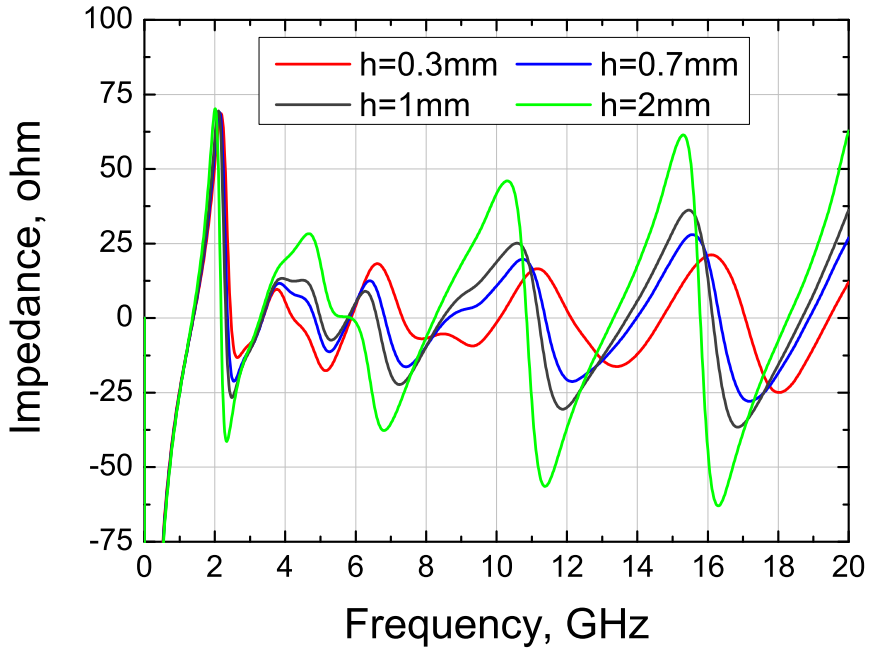


Figure 4.2: Simulated return loss curves of vertical disc monopole for different feed gaps with r=12.5mm and W=L=100mm

It is shown in Figure 4.2 that the -10dB operating bandwidth of the antenna varies remarkably with the variation of the feed gap h. The optimal feed gap is found to be between 0.3-0.7mm with the bandwidth covering an extremely wide frequency range from 2.47GHz to greater than 18GHz.



(a) Resistance R



(b) Reactance X

Figure 4.3: Simulated input impedance curves of vertical disc monopole for different feed gaps with r=12.5mm and W=L=100mm

As shown in Figure 4.3, the low return loss (<-10dB) always occurs over the frequency range when the input impedance is matched to 50Ω, i.e. the input resistance R is close

to 50Ω while the input reactance X is not far from zero for the four different feed gaps. When h=0.3 and 0.7mm, R varies tardily at the level of 50Ω whilst X remains small across an extremely wide frequency range, leading to a UWB characteristic. However, when h rises to 1mm and 2mm, R varies more widely and X also fluctuates significantly across the frequency range, thus resulting in impedance mismatch at the antenna and hence the decrease of the operating bandwidth.

4.1.3 Effect of the Ground Plane

In previous section, it has been demonstrated that the variation of the feed gap h leads to the variations of the input impedance and frequency bandwidth. Since h is the gap between the ground plane and the disc, in a broad sense, the ground plane serves as an impedance matching circuit and also it tunes the resonant frequencies.

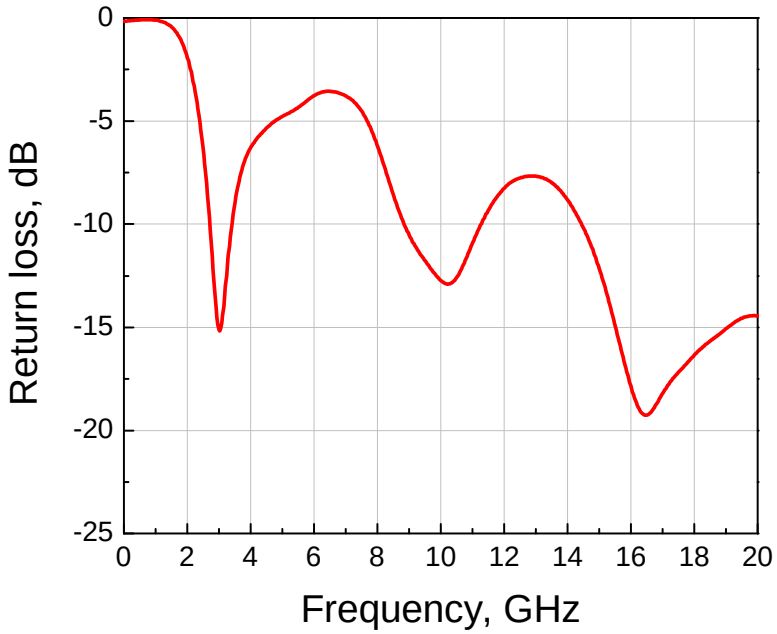


Figure 4.4: Simulated return loss curve of vertical disc monopole without ground plane when r=12.5mm

To confirm this, Figure 4.4 illustrates the simulated return loss curve for the disc monopole without ground plane when r=12.5mm. Its simulated input impedance curve is plotted in Figure 4.5. It is noticed in Figure 4.4 that the first -10dB bandwidth ranges

only from 2.7GHz to 3.4GHz, much narrower than that of a disc with ground plane, as shown in Figure 4.2. This is due to the impedance mismatch over an extremely frequency range resulting from the removal of the ground plane. As shown in Figure 4.5, in most frequency range from 0 to 14GHz, both the resistance and reactance curves fluctuate substantially. The peak value of resistance is as high as 160 ohms, while the maximum reactance is around 86 ohms. Furthermore, at the frequencies where resistance is close to 50 ohms, reactance is far from 0; when reactance reaches 0, resistance is either in its peak or near its minimum value. As a result, the impedance is mismatched to 50 ohms at the antenna, leading to a narrow operating bandwidth.

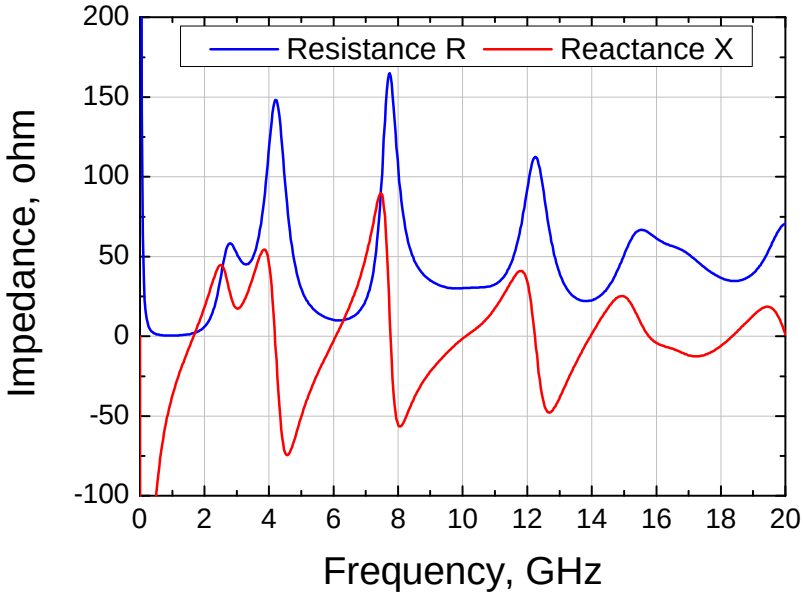


Figure 4.5: Simulated impedance curve of vertical disc monopole without ground plane when r=12.5mm

Besides, it is also shown in the simulation that the -10dB operating bandwidth of vertical disc monopole is critically dependent on the width W of the ground plane and it is not much determined by the length L. The simulated return loss curves for different lengths of the ground plane with r=12.5mm, L=10mm and the optimal feed gap h=0.7mm are given in Figure 4.6.

It is noticed in Figure 4.6 that the -10dB operating bandwidth decreases greatly with W being reduced from 100mm to 25mm, and the first -10dB bandwidth for W=25mm

only ranges from 2.7GHz to 3.9GHz.

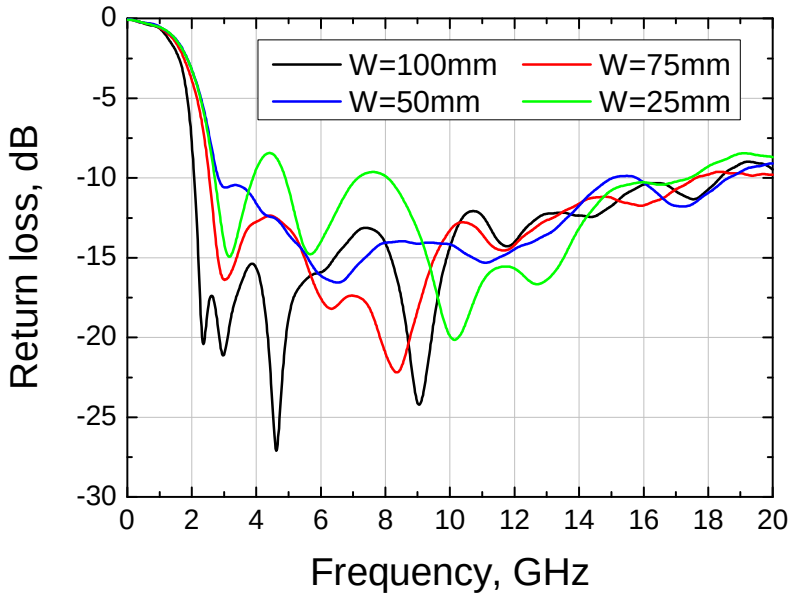


Figure 4.6: Simulated return loss curves of vertical disc monopole for different widths of the ground plane with r=12.5mm, h=0.7mm and L=10mm

The simulated return loss curves when L=5, 10, 25 and 100mm with r=12.5mm, W=100mm and h=0.7mm are illustrated in Figure 4.7. Table 4-A presents the -10dB bandwidths for different lengths of the ground plane.

It can be seen in Figure 4.7 and Table 4-A that the -10 dB bandwidth of vertical disc monopole does not change much with the variation of the length of the ground plane. When L rises from 5mm to 100mm with W fixed at 100mm, the lower edge of the bandwidth increases tardily from 1.98GHz to 2.44GHz, while the upper edge is almost constant at around 18.5GHz. This result demonstrates that the antenna bandwidth is not heavily dependent on the length L of the ground plane.

Since the ground plane serves as an impedance matching circuit, Figure 4.6 and Figure 4.7 indicate that the impedance of the ground plane is mainly determined by its width W. This is because the current is mainly distributed along the y-direction, as will be discussed in later sections. So the length L of the ground plane can be reduced significantly without any sacrifice of the operating bandwidth. This result is

very important and encouraging for the miniaturization of the antenna.

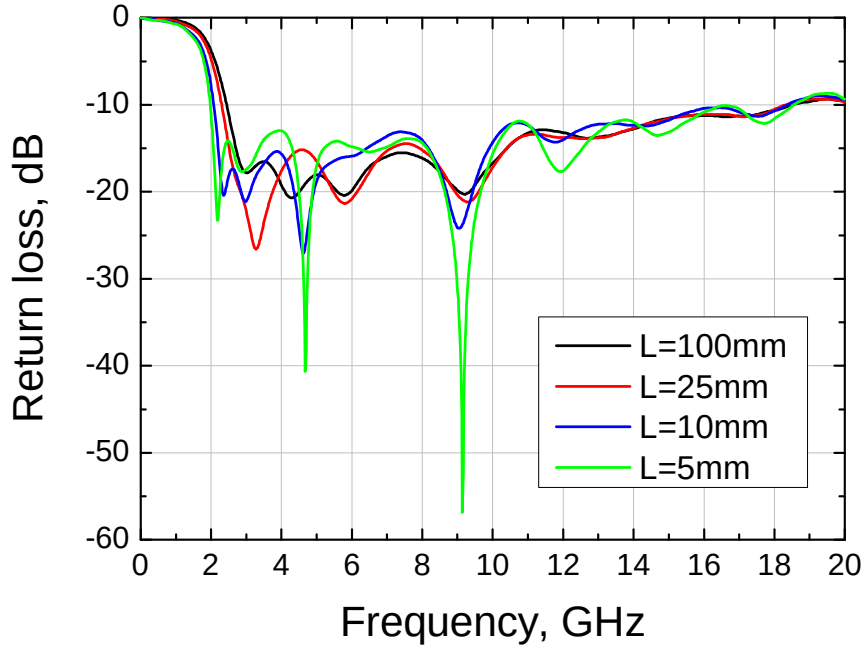


Figure 4.7: Simulated return loss curves of vertical disc monopole for different lengths of the ground plane with r=12.5mm, h=0.7mm and W=100mm

Table 4-A: Simulated -10dB bandwidths of vertical disc monopole for different lengths of the ground plane with r=12.5mm, h=0.7mm and W=100mm

L (mm)	Lower edge of bandwidth (GHz)	Upper edge of bandwidth (GHz)	Absolute bandwidth (GHz)	Fractional bandwidth (%)
5	1.98	18.55	16.57	161
10	2.08	18.37	16.29	159
25	2.30	18.33	16.03	155
50	2.53	18.64	16.11	152
75	2.56	18.58	16.02	152
100	2.44	18.51	16.07	153

4.1.4 Effect of Tilted Angle

The disc monopole vertically mounted above the ground plane, as shown in Figure 4.1, has been discussed in previous sections. In this section, the disc is tilted from the z-direction, as shown in Figure 4.8. Simulations have shown that the tilted angle θ between the disc and the z-direction also has a significant effect on the performance of the antenna.

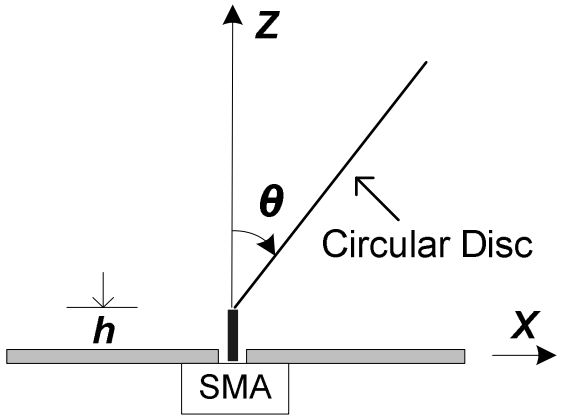


Figure 4.8: Geometry of the tilted disc monopole

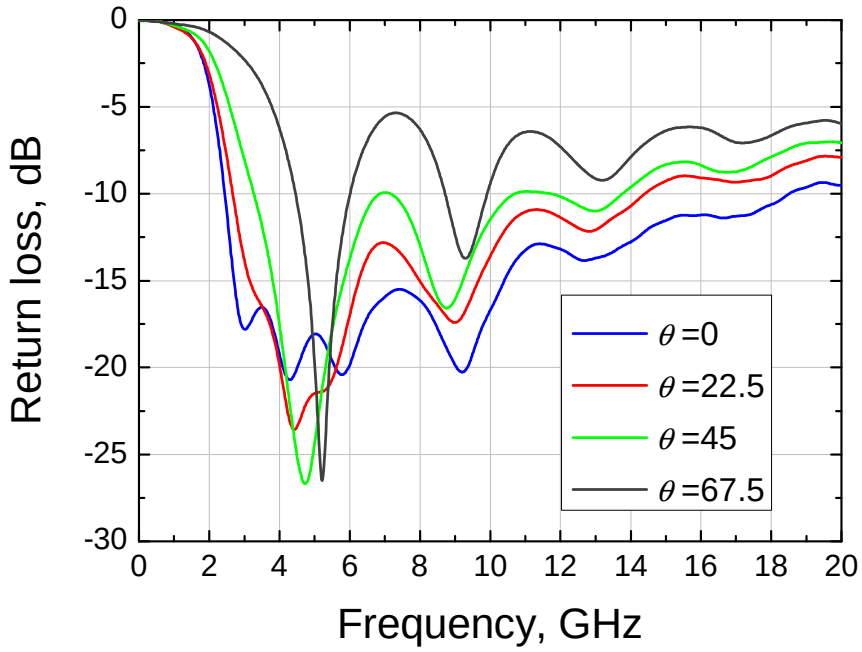
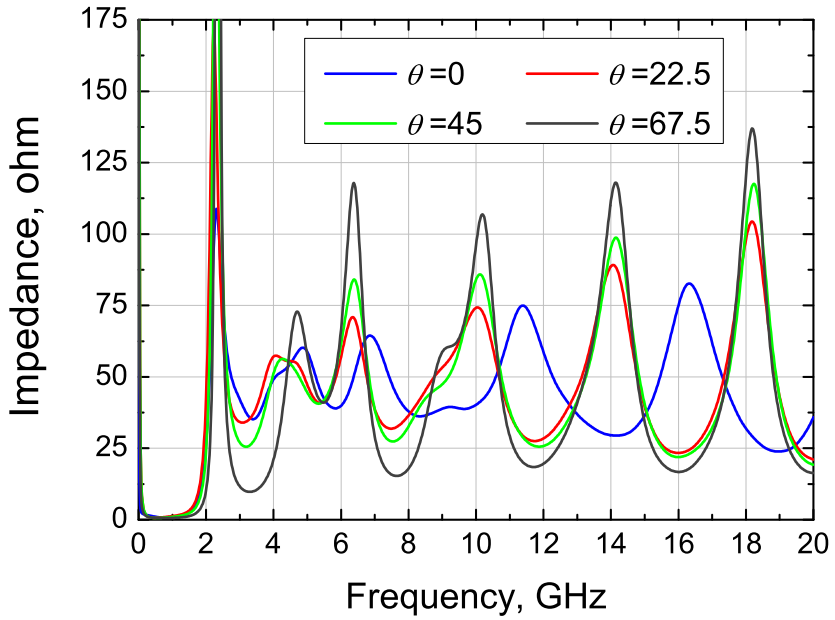
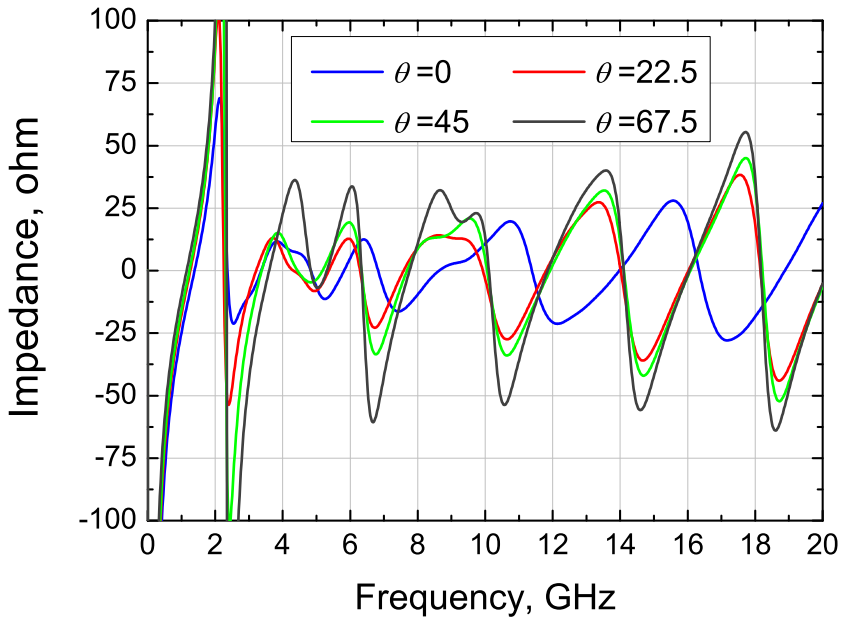


Figure 4.9: Simulated return loss curves of the disc monopole for different tilted angles with r=12.5mm, h=0.7mm and W=L=100mm

Figure 4.9 illustrates the simulated return loss curves of the disc monopole for different tilted angles when r=12.5mm, h=0.7mm and W=L=100mm. Their respective input impedance curves are given in Figure 4.10.



(a) Resistance R



(b) Reactance X

Figure 4.10: Simulated input impedance curves of the disc monopole for different tilted angles with r=12.5mm, h=0.7mm and W=L=100mm

It is observed in Figure 4.9 that the operating bandwidth is getting narrower with the increase of the tilted angle θ. When θ=0, the vertically mounted disc monopole can provide a -10dB bandwidth of 16.07GHz (from 2.44GHz to 18.51GHz, as shown in Table 4-A); when θ rises to 22.5 degrees, the antenna can still exhibit UWB characteristic with an absolute bandwidth of 11.74GHz (from 2.67GHz to 14.41GHz); when θ is changed to 45 degrees, the -10dB bandwidth is reduced to 3.6GHz (from 3.27GHz to 6.87GHz); with the further increase of θ to 67.5GHz, the first -10dB bandwidth only ranges from 4.45GHz to 6.01GHz, leading to a bandwidth of 1.56GHz.

This is due to the sensitivity of the input impedance of the disc to the tilted angle, as shown in Figure 4.10. With the increase of tilted angle θ, both of the resistance and the reactance of the antenna vary more widely across the frequency range, which means the impedance mismatch is getting worse, thus leading to the narrowing of the operating bandwidth, as illustrated in Figure 4.9 and 4.10.

4.1.5 Mechanism of the UWB characteristic

There is an important phenomenon in Figure 4.4 that the first resonance occurs at 3GHz when the ground plane is removed. Here, the resonance frequencies are defined where the dips on the return loss curve are located. When the vertical disc monopole has a square ground plane with W=L=100mm, the first resonance is also fixed at around 3GHz for all of the four different feed gaps, as shown in Figure 4.2. When the ground plane is reduced in either direction (length or width), this resonance is shifted slightly, but still not far from 3GHz, as shown in Figure 4.6 and Figure 4.7.

In fact, the quarter wavelength (25mm) at this first resonant frequency (3GHz) just equals to the diameter of the disc. This implies that this resonant frequency is mostly determined by the circular disc and slightly detuned by the size of the ground plane.

Figure 4.2 and Figure 4.3 show that the circular disc is capable of supporting multiple resonance modes, the higher order modes (f_2, f_3...f_n) being the harmonics of

the fundamental mode of the disc. So the wavelengths of the higher order modes satisfy: $2r = n\lambda_n /4 = \lambda_1/4$, where n is the mode number. It is also indicated that these higher order modes are closely spaced. Hence, the overlapping of these resonance modes leads to the UWB characteristic, as illustrated in Figure 4.11.

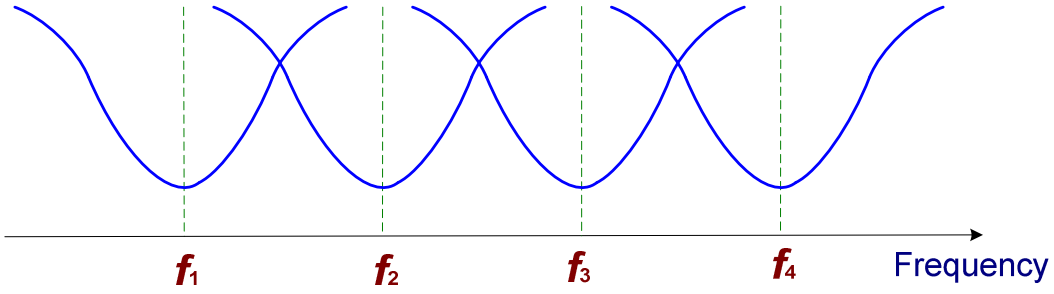
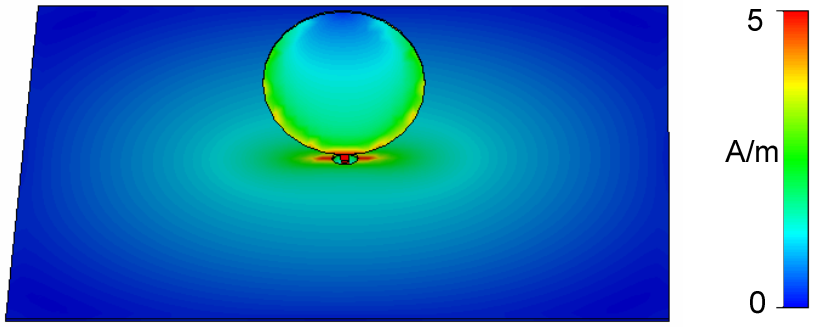


Figure 4.11: Overlapping of the multiple resonance modes

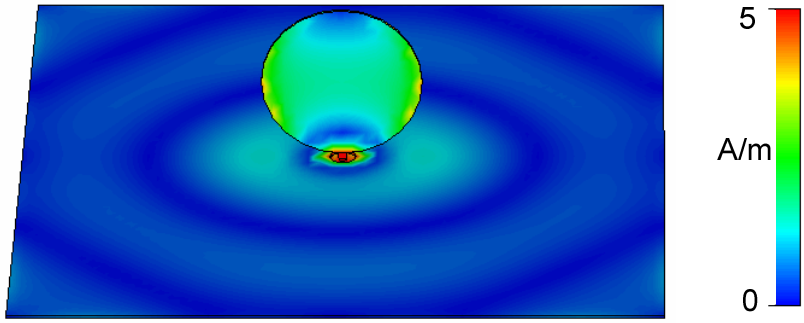
4.1.6 Current Distributions

The simulated current distributions of vertical disc monopole at different frequencies with r=12.5mm, h=0.7mm and W=L=100mm are presented in Figure 4.12. Figure 4.12(a) shows the current pattern near the first resonance at 3GHz. The current pattern near the second resonance at around 6GHz is given in Figure 4.12(b), indicating approximately a second order harmonic. Figure 4.12(c) illustrates a more complicated current pattern at 9GHz, corresponding to the third order harmonic.

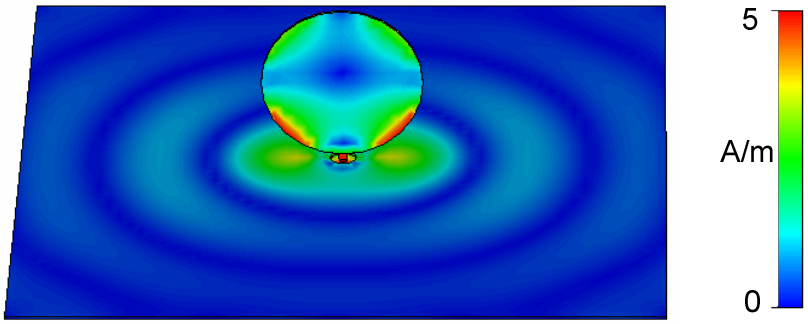
As shown in Figure 4.12, the current is mainly distributed along the edge of the circular disc for all of the three different frequencies. This is the reason why the first resonant frequency is associated with the dimension of the disc. On the ground plane, the current is mainly distributed along the y-direction within a narrow area, which confirms our previous observation that the performance of the antenna is critically dependent on the width of the ground plane W and it is not very sensitive to the length of the ground plane L, as discussed in previous subsection.



(a) At 3GHz



(b) At 6GHz

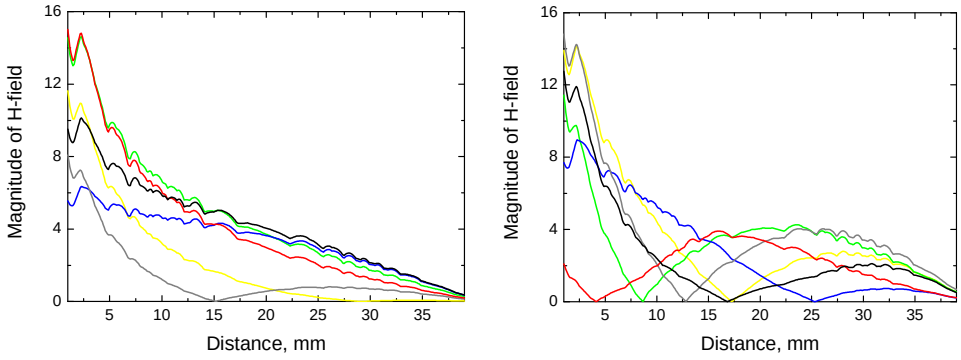


(c) At 9GHz

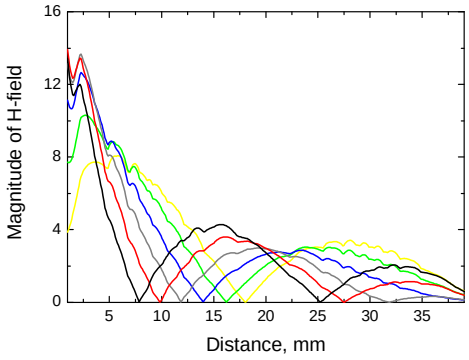
Figure 4.12: Simulated current distributions of vertical disc monopole with r=12.5mm, h=0.7mm and W=L=100mm

The snapshots of the magnetic field distributions corresponding to the currents along the half disc edge D (D = 0 - 39mm: bottom to top) with different phases at each resonance are plotted in Figure 4.13 (a) - (c), respectively.

Figure 4.13(a) shows that at the first resonance the current is oscillating and has a pure standing wave pattern along most part of the disc edge. So the disc behaves like an oscillating monopole. But the variation of the currents becomes more complicated



(a) at 3GHz (b) at 6GHz



(c) at 9GHz

Figure 4.13: Simulated magnetic field distributions along the edge of the half disc D (D = 0 - 39mm: bottom to top) with different phases at each resonance

at higher resonance harmonics. Figure 4.13(b) indicates a complex current variation patterns at the second harmonic. The current is travelling along the lower disc edge, but oscillating at the top edge. There is a broad current envelope peak formed around D = 25mm. In Figure 4.13(c), at the third harmonic, the feature of travelling wave seems more prominent at the lower edge of the disc, while standing wave retains on the top edge with an envelope peak.

4.1.7 Experimental Verification

Two prototypes of vertical disc monopole antenna with r=12.5mm and the optimal feed gap of h=0.7mm were built and tested in the Antenna Measurement Laboratory

at Queen Mary, University of London. One antenna is bigger and has a square ground plane with W=L=100mm; the other one is smaller, exhibiting a narrower ground plane with L=10mm and W=100mm, as shown in Figure 4.14 and Figure 4.15, respectively.

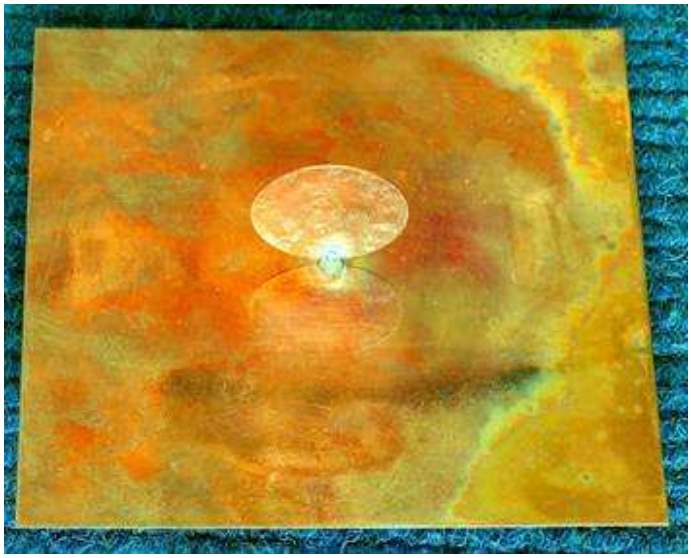


Figure 4.14: Vertical disc monopole with r=12.5mm, h=0.7mm and W=L=100mm

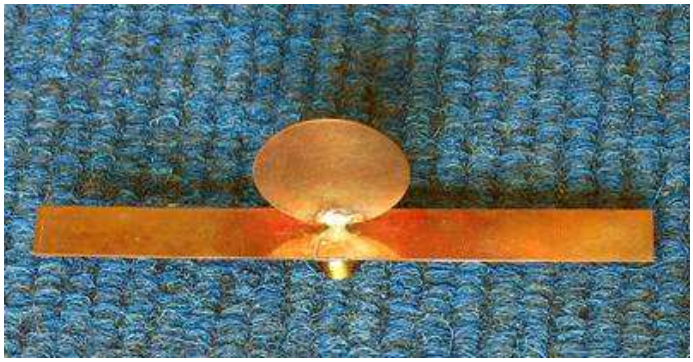


Figure 4.15: Vertical disc monopole with r=12.5mm, h=0.7mm, W=100mm and L=10mm

The return losses were measured by using a HP 8720ES network analyzer inside an anechoic chamber.The measured and simulated return loss curves of the two antennas are plotted in Figure 4.16 and Figure 4.17, respectively.

For the antenna with a square ground plane, as shown in Figure 4.16, the measured return loss agrees very well with the simulated one in the whole frequency range from 0

to 20GHz; the measured resonant frequencies are very close to the simulated ones with differences less than 5%.

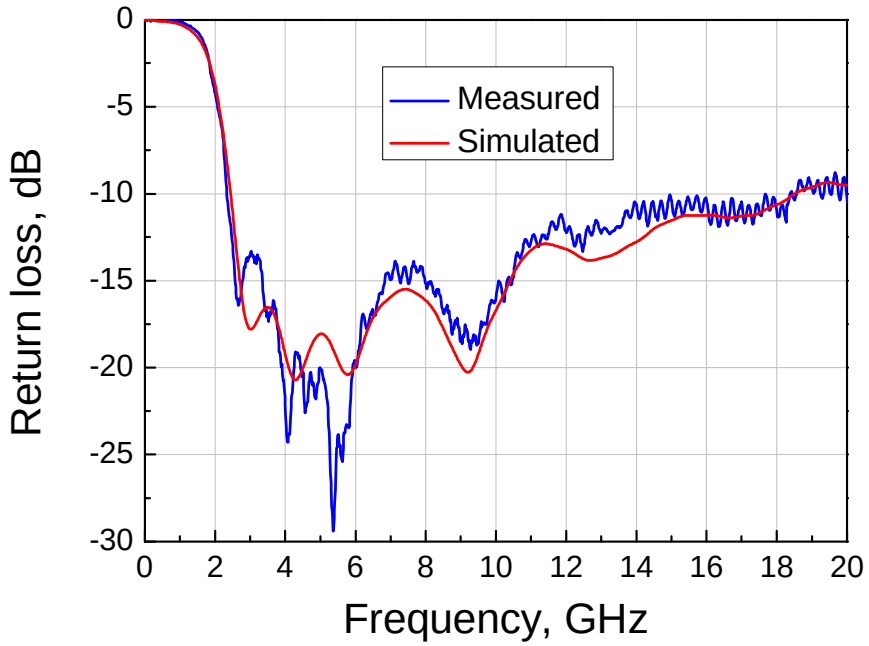


Figure 4.16: Measured and simulated return loss curves of vertical disc monopole with r=12.5mm, h=0.7mm and W=L=100mm

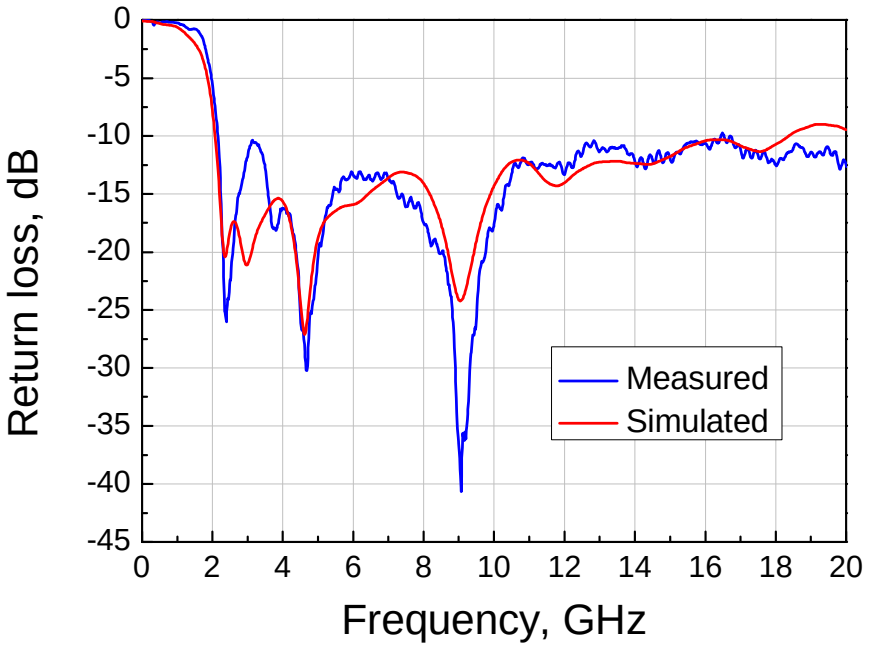
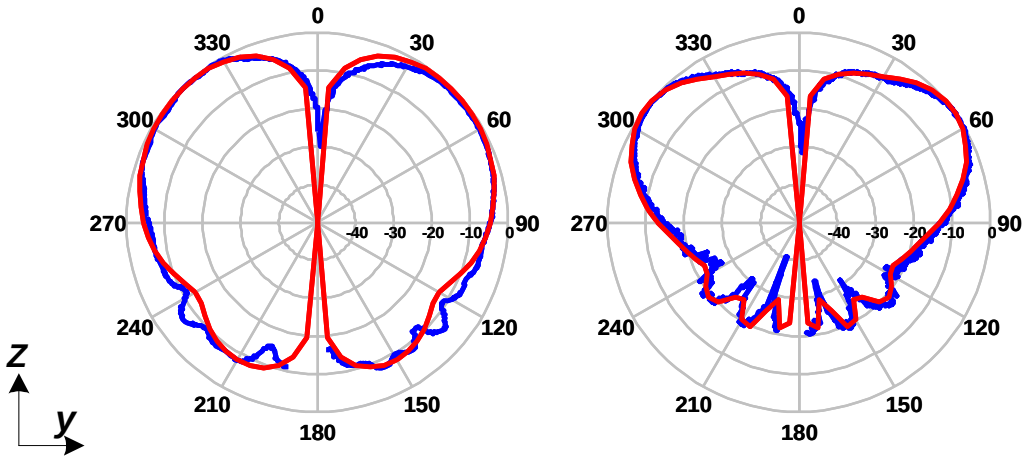
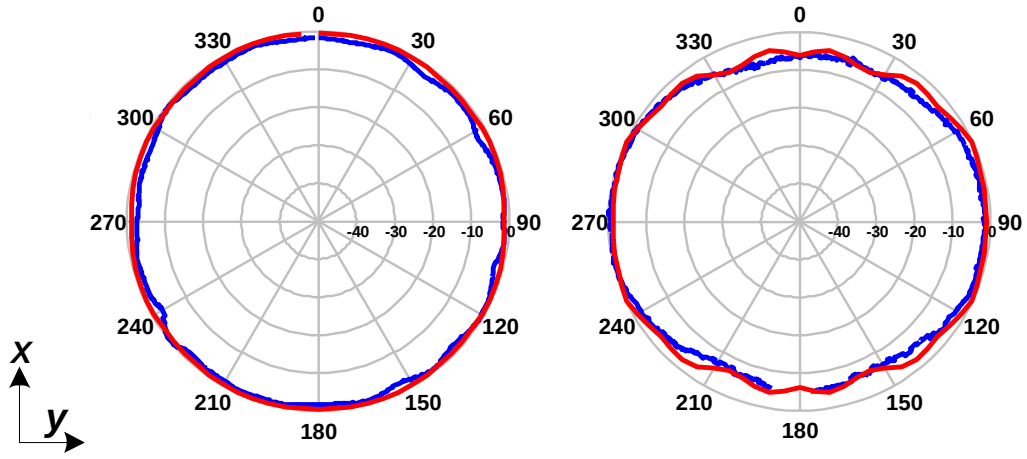


Figure 4.17: Measured and simulated return loss curves of vertical disc monopole with r=12.5mm, h=0.7mm, W=100mm and L=10mm

For the antenna with a narrower ground plane, as shown in Figure 4.17, the measured return loss is also very close to the simulated one in most range of the frequency band except at around 3GHz where the simulated return loss is -21.08dB, but the measured result is -10.55dB. In spite of the mismatch at this frequency, the measured resonance frequencies are nearly identical to the simulated ones with differences less than 2%. The measurements confirm the UWB characteristic of the circular disc monopoles, as predicted in the simulations. It is also confirmed that reducing the length of the ground plane will not lead to any sacrifice of the -10dB operating bandwidth.



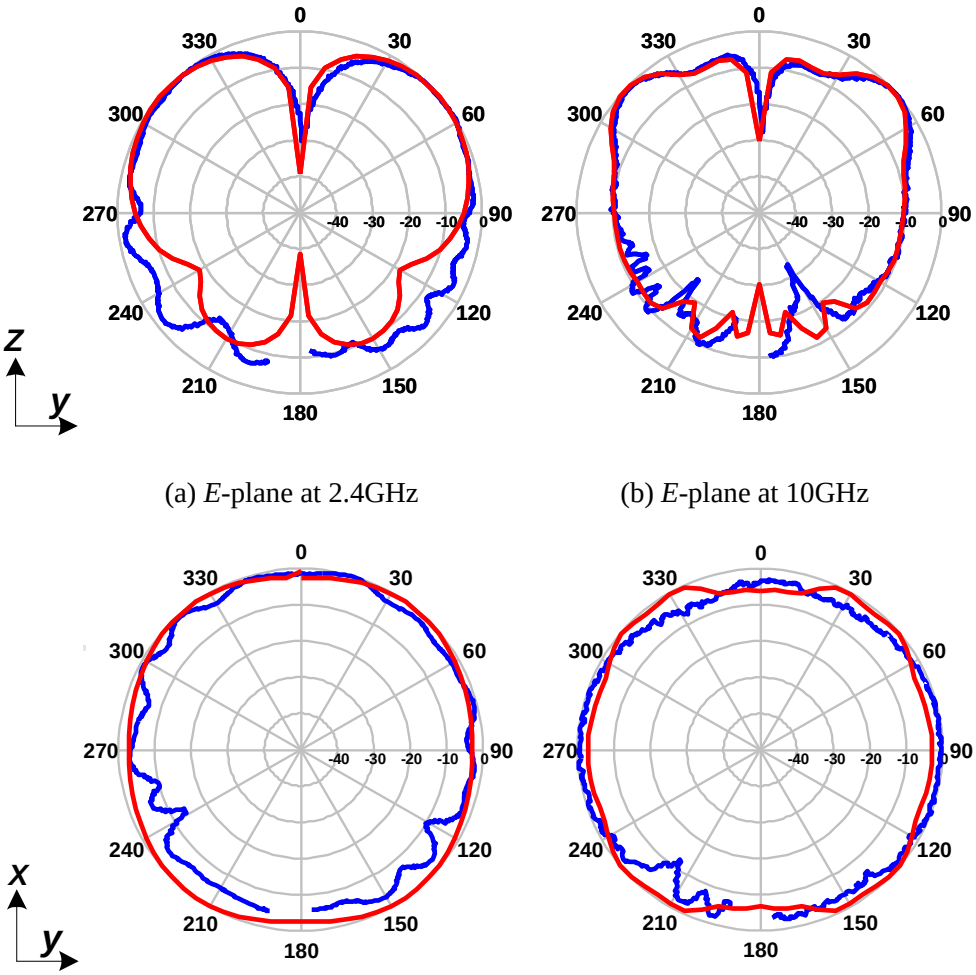
(a) *E*-plane at 2.8GHz (b) *E*-plane at 10GHz



(c) *H*-plane at 2.8GHz (d) *H*-plane at 10GHz

Figure 4.18: Measured (blue line) and simulated (red line) radiation patterns of vertical disc monopole with r=12.5mm, h=0.7mm and W=L=100mm

The radiation pattern measurements of the two antennas were also carried out inside an anechoic chamber. Figure 4.18 illustrates the radiation patterns for the antenna with a 100mm square ground plane. The measured and simulated radiation patterns also agree very well at 2.8GHz and 10GHz. The E-plane patterns have large back lobes and with increasing frequency they become smaller, splitting into many minor ones. The maximum direction is shifted from ± 60 degrees to ± 50 degrees when frequency varies from 2.8GHz to 10GHz. The H-plane pattern is omni-directional at 2.8GHz and is only distorted slightly at 10GHz (the gain being reduced less than 10dB in the x direction).



(a) E-plane at 2.4GHz (b) E-plane at 10GHz

(c) H-plane at 2.4GHz (d) H-plane at 10GHz

Figure 4.19: Measured (blue line) and simulated (red line) radiation patterns of vertical disc monopole with r=12.5mm, h=0.7mm, W=100mm and L=10mm

The measured and simulated radiation patterns for the antenna with a narrower ground plane (L=10mm) are given in Figure 4.19. The patterns obtained in the measurement are close to those in the simulation. Compared to the disc monopole with a square ground plane, this one has slightly larger back lobes in the E-plane patterns due to the reducing of the length of the ground plane. Also, the back lobe is bigger at low frequency than at high frequency because the ground plane is electrically small compared to the wavelength at low frequency. The H-plane patterns are still generally omni-directional over the entire bandwidth, like a conventional monopole. This is because the disc dimension is within the wavelengths of the first four resonance harmonics.

4.2 Coplanar Waveguide Fed Disc Monopole

As discussed in previous section, vertical disc monopole is capable of yielding ultra wide bandwidth with nearly omni-directional radiation patterns. However, this type of antenna is not a planar structure because it requires a ground plane which is perpendicular to the disc. Although the ground plane can be miniaturized significantly, it is still not suitable for integration with a PCB. This drawback limits its practical application. There is great demand for UWB antennas that offer a fully planar structure.

To derive a planar version of the UWB disc monopole, a printed feeding structure is required. There are two choices; one is microstrip line [3–6] and the other coplanar waveguide (CPW) feeding structure [7–11]. This section will focus on CPW fed disc monopole, while microstrip line fed disc monopole will be studied in next section.

4.2.1 Antenna Design and Performance

The CPW fed disc monopole antenna studied in this section has a single layer-metallic structure, as shown in Figure 4.20. A circular disc monopole with a radius of r and a 50Ω CPW are printed on the same side of a dielectric substrate. W_f is the width of

the metal strip and g is the gap of distance between the strip and the coplanar ground plane. W and L denote the width and the length of the ground plane, respectively, h is the feed gap between the disc and the ground plane. In this study, a dielectric substrate with a thickness of H=1.6mm and a relative permittivity of ε_r=3 is chosen, so W_f and g are fixed at 4mm and 0.33mm, respectively, in order to achieve 50Ω impedance.

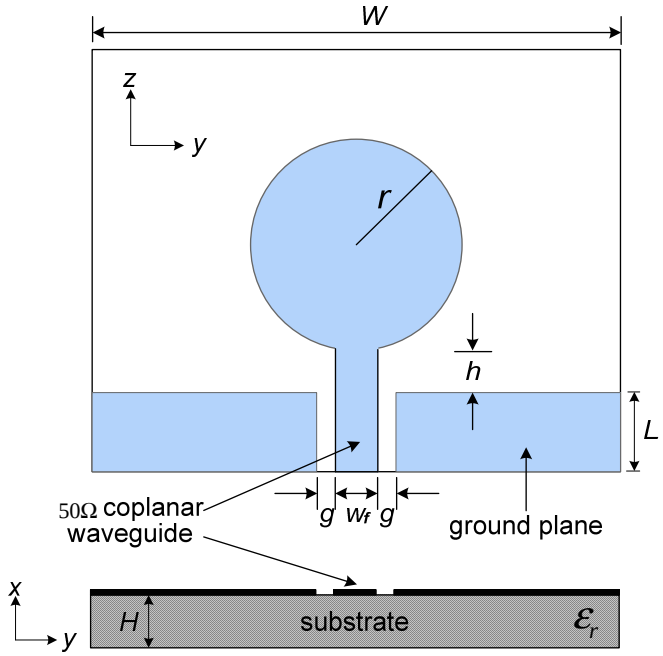


Figure 4.20: The geometry of the CPW fed circular disc monopole

The simulations were performed using the CST Microwave Studio^{TM} package, which utilizes Finite Integration Technique for electromagnetic computation, as mentioned in Appendix B. The complete configuration of the antenna was simulated using this package. The effect of the 50Ω SMA (SubMiniature version A) feeding port should not be neglected because it is no longer shadowed by the ground plane for this planar type of antenna. Besides, it is positioned near to the disc and the slot of the coplanar waveguide. So the SMA port has been included in the simulation, but this does lead to a substantial computing overhead.

A prototype of the CPW fed circular disc monopole antenna with optimal design, i.e. r=12.5mm, h=0.3mm, L=10mm and W=47mm, was built and tested in the Antenna

Measurement Laboratory at Queen Mary, University of London, as shown in Figure 4.21. The return losses were measured in an anechoic chamber by using a HP 8720ES network analyzer.

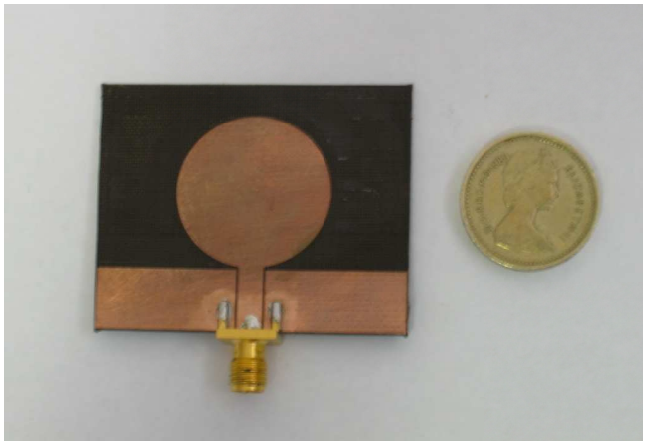


Figure 4.21: Photo of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

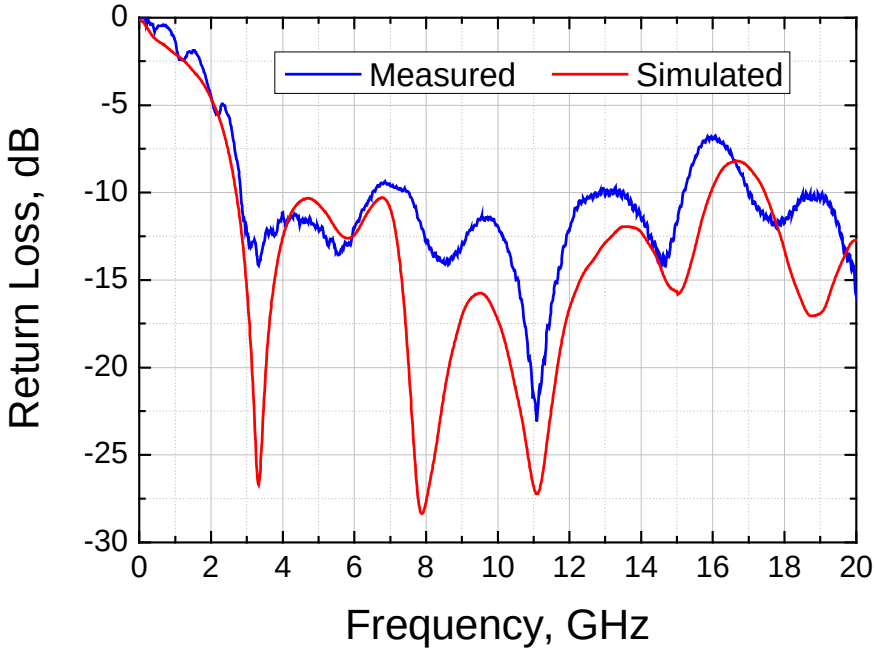
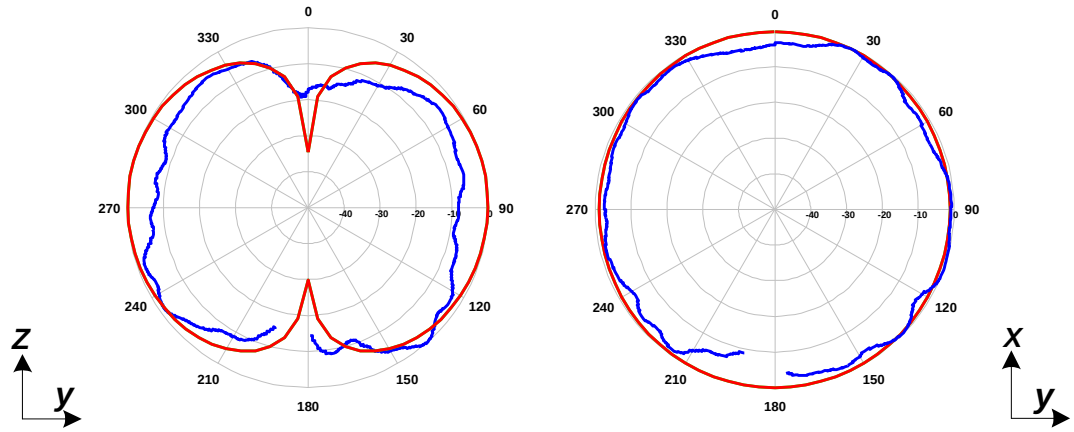


Figure 4.22: Measured and simulated return loss curves of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

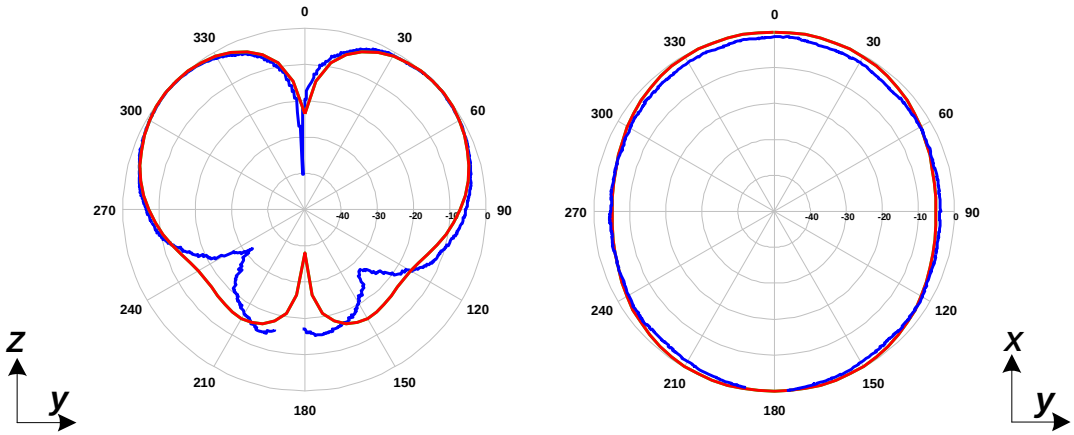
As illustrated in Figure 4.22, the measured return loss curve agrees very well with the simulated one in most range of the frequency band. The measured five resonances (at

around 3.3GHz, 5.6GHz, 8.4GHz 11.1GHz and 14.6GHz) are very close to those obtained in the simulation with differences less than 8%. Generally speaking, the -10dB bandwidth spans an extremely wide frequency range in both simulation and measurement. The simulated bandwidth ranges from 2.69GHz to 15.95GHz. This UWB characteristic of the CPW fed circular disc monopole antenna is confirmed in the measurement except at around 7GHz where the measured return loss is -9.6dB. Besides, the lower and higher frequency edges are slightly shifted to 2.84GHz and 15.17GHz in the measurement.



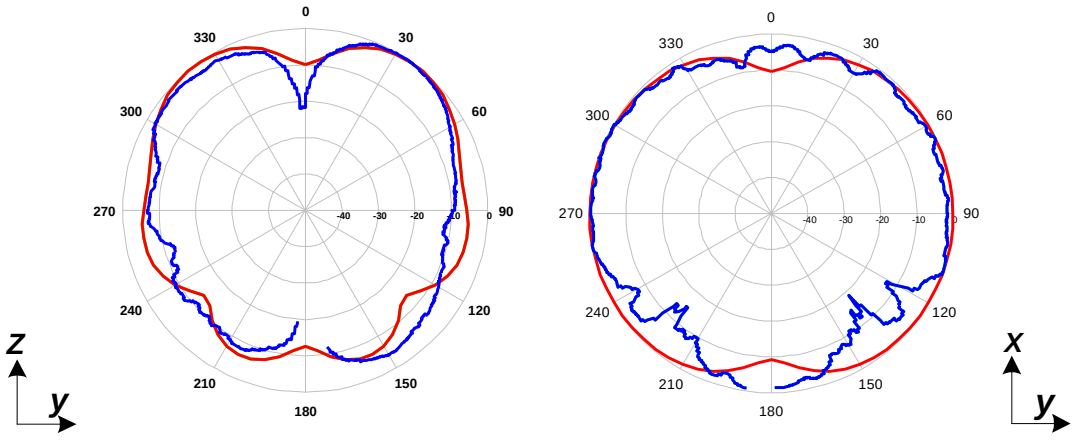
(a) *E*-plane (b) *H*-plane

Figure 4.23: Measured (blue line) and simulated (red line) radiation patterns of the CPW fed circular disc monopole at 3GHz



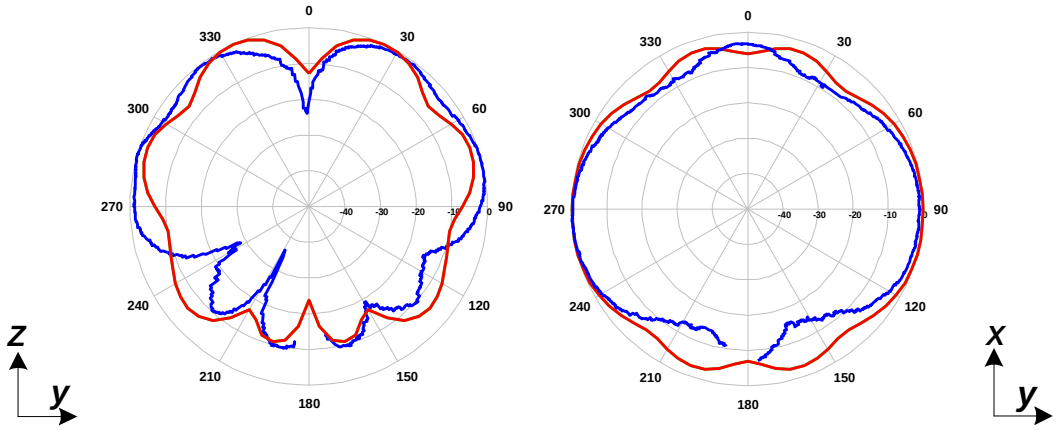
(a) *E*-plane (b) *H*-plane

Figure 4.24: Measured (blue line) and simulated (red line) radiation patterns of the CPW fed circular disc monopole at 5.6GHz



(a) *E*-plane (b) *H*-plane

Figure 4.25: Measured (blue line) and simulated (red line) radiation patterns of the CPW fed circular disc monopole at 7.8GHz



(a) *E*-plane (b) *H*-plane

Figure 4.26: Measured (blue line) and simulated (red line) radiation patterns of the CPW fed circular disc monopole at 11GHz

The radiation patterns of the antenna at the frequencies close to the resonances have been measured inside an anechoic chamber. The measured and simulated radiation patterns at 3GHz, 5.6GHz, 7.8GHz and 11.0GHz are plotted in Figures 4.23–Figure 4.26, respectively.

It is noticed that the measured radiation patterns are generally very close to those obtained in the simulation. The *E*-plane patterns have large back lobes and look like a donut or a slightly pinched donut at lower frequencies. With the increase of the frequency, the back lobes become smaller, splitting into many minor ones, while the front lobes start

to form humps and notches.

The H-plane pattern is omni-directional at low frequency (3GHz) and only distorted slightly at 5.6GHz. With the increase of frequency to the third and fourth resonances (7.8GHz and 11GHz), there are more distortions in the measured patterns compared with the simulated ones due to an enhanced perturbing effect on the antenna performance caused by the feeding structure and cable at these frequencies. Though the overall radiation pattern of the antenna has gone through a notable transformation, the H-plane pattern retains a satisfactory omni-directionality (less than 10dB gain variation in most directions) over the entire bandwidth in both simulation and experiment.

4.2.2 Antenna Characteristics

CPW fed disc monopole has been demonstrated to yield UWB characteristic with nearly omni-directional radiation patterns. It is necessary to gain some insights into its operation.

From this subsection, the 50Ω SMA feeding port is not taken into account in all of the simulations so as to ease the computational requirements. It is noticed that this SMA port mainly affects the higher order resonances by shifting their resonant frequencies.

Figure 4.27 illustrates the simulated return loss curve of the optimal design of the antenna. The corresponding input impedance and Smith Chart curves are plotted in Figure 4.28 and Figure 4.29, respectively.

As shown in Figure 4.27, the first resonance occurs at around 3.0GHz, the second resonance at 5.6GHz, the third one at 8.6GHz, the fourth one at 12.8GHz and the fifth one at 17.7GHz. Compared with Figure 4.22, the third, fourth and fifth resonances are up-shifted to higher frequencies because of the removal of the SMA feeding port.

It is evident that the overlapping of these resonance modes which are closely distributed across the spectrum results in an ultra wide -10dB bandwidth. It is noticed on

the Smith Chart that the input impedance loops around the impedance matching point within the circle of voltage standing wave ratio (VSWR)=2, but doesn't settle down to a real impedance point with the increase of frequency.

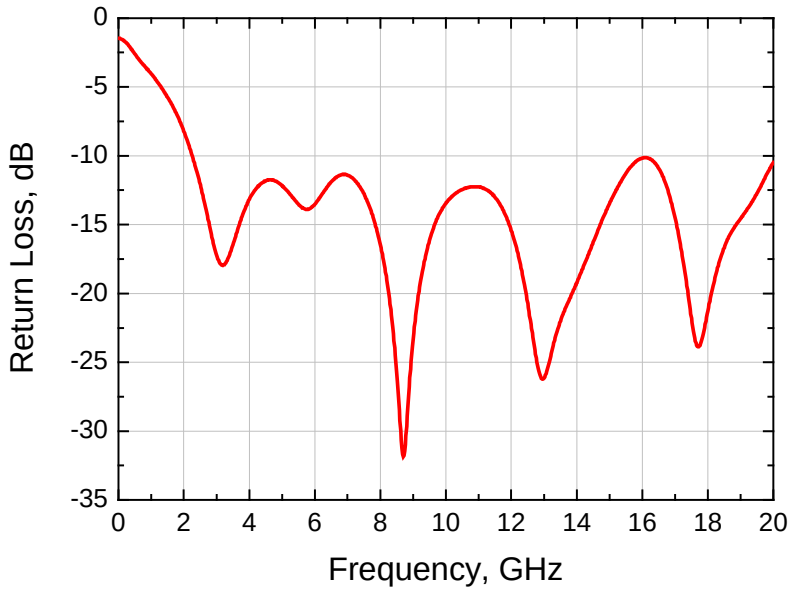


Figure 4.27: Simulated return loss curve of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

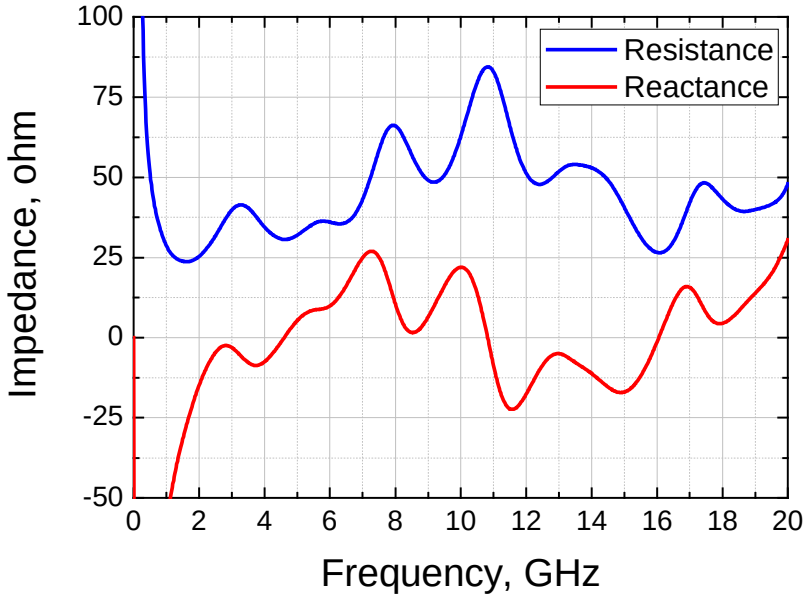


Figure 4.28: Simulated input impedance curve of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

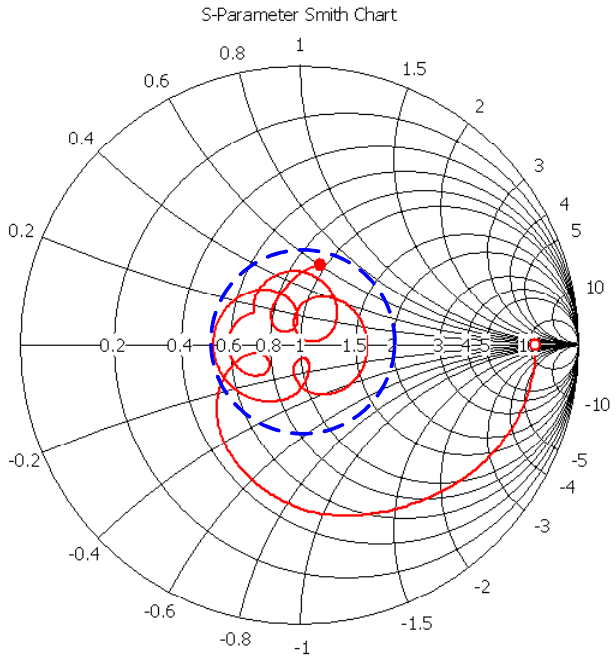


Figure 4.29: Simulated Smith Chart of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

The return loss or input impedance can only describe the behavior of an antenna as a lumped load at the end of feeding line. The detailed EM behavior of the antenna can only be revealed by examining the field/current distributions or radiation patterns. The typical current distributions on the antenna close to the resonance frequencies are plotted in Figure 4.30.

Figure 4.30(a) shows the current pattern near the first resonance at 3.0GHz. The current pattern near the second resonance at 5.6Hz is given in Figure 4.30(b), indicating approximately a second order harmonic. Figures 4.30(c), (d) and (e) illustrate three more complicated current patterns at 8.6GHz, 12.8GHz and 17.7GHz, corresponding to the third, fourth and fifth order harmonics, respectively. These current distributions support that the UWB characteristic of the antenna is attributed to the overlapping of a sequence of closely spaced resonance modes.

The simulated 3D radiation patterns close to these resonances are plotted in Figure 4.31. The radiation pattern looks like a donut, similar to a dipole pattern, at the first

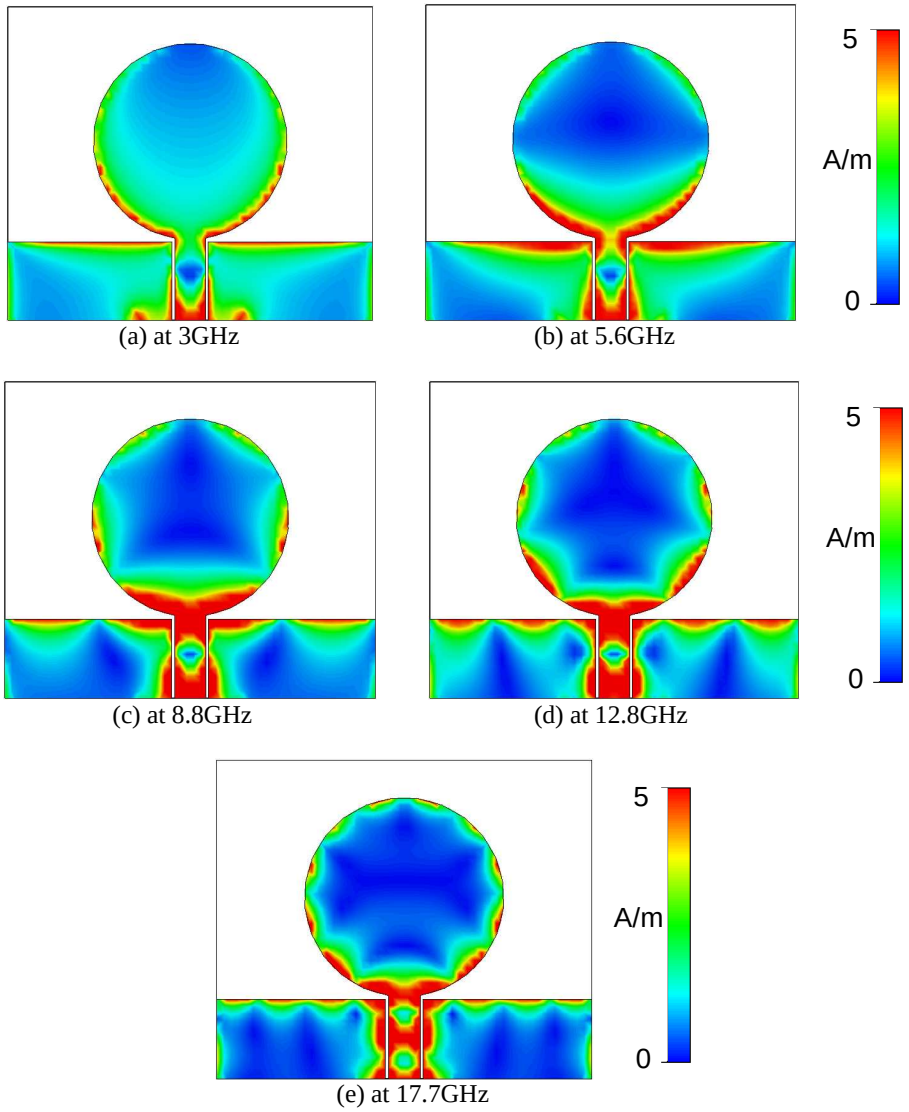
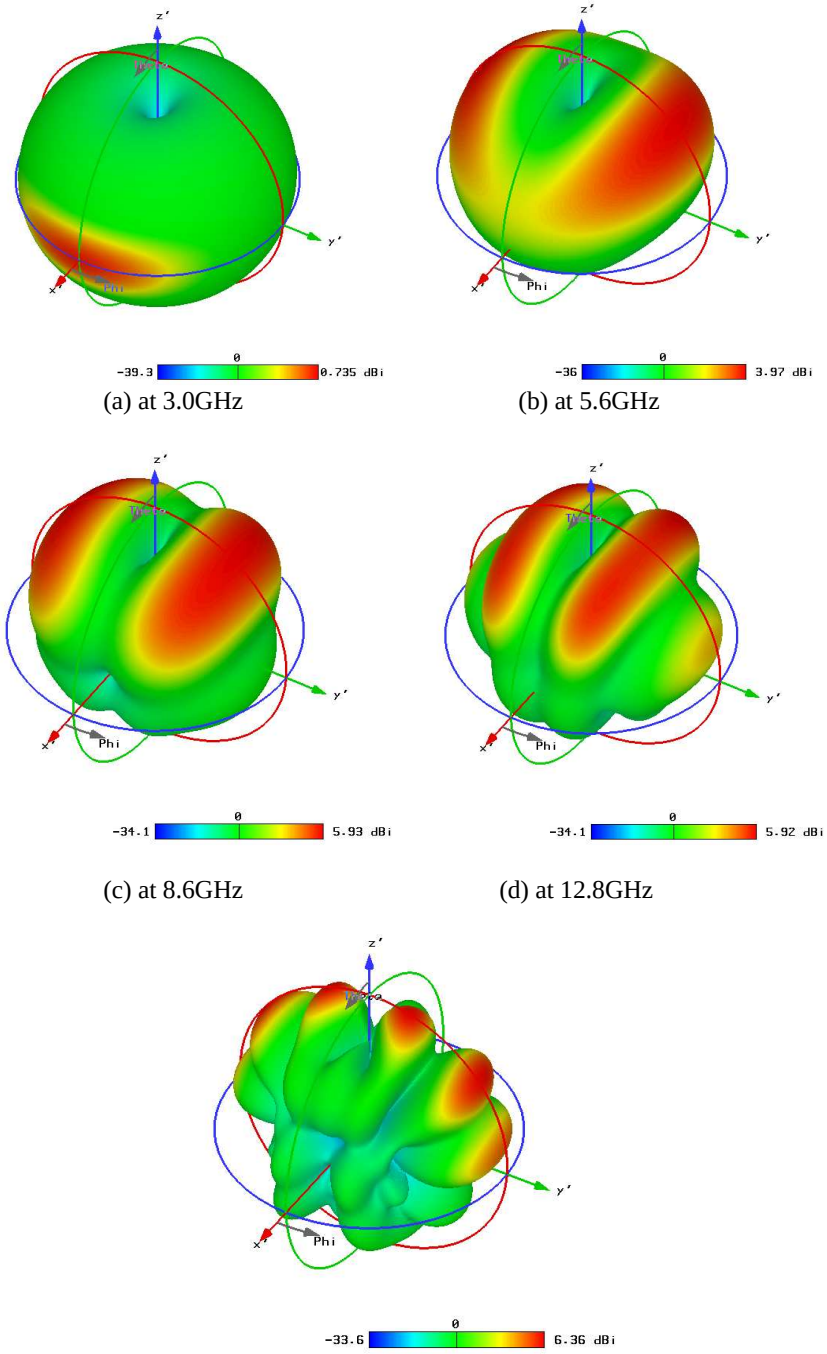


Figure 4.30: Simulated current distributions of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

resonant frequency, as shown in Figure 4.31(a). At the second harmonic, the pattern changes its shape to a slightly pinched donut with the gain increase around θ=45^0 in Figure 4.31(b). When at the third, fourth and fifth harmonics, the patterns are squashed in x-direction and humps form in the up-right directions (gain increasing), as shown in Figures 4.31(c) - 4.31(e), respectively. It is also noticed that the patterns on the H-pane are almost omni-directional at lower resonances (1st and 2nd harmonics) and become distorted at the higher harmonics.



(a) at 3.0GHz (b) at 5.6GHz

(c) at 8.6GHz (d) at 12.8GHz

(e) at 17.7GHz

Figure 4.31: Simulated 3D radiation patterns of the CPW fed circular disc monopole with r=12.5mm, h=0.3mm, L=10mm and W=47mm

The transition of the radiation patterns from a simple donut pattern at the first resonance to the complicated patterns at higher harmonics indicates that this antenna must have gone through some major changes in its behavior.

In order to gain further insight into the antenna operation, the animation of current variation at difference resonances were generated and observed in the CST Microwave Studio. Also, the magnetic field distributions corresponding to the currents along the half disc edge with different phases at each resonance are analyzed and plotted in Figure 4.32 (a) - (e), respectively.

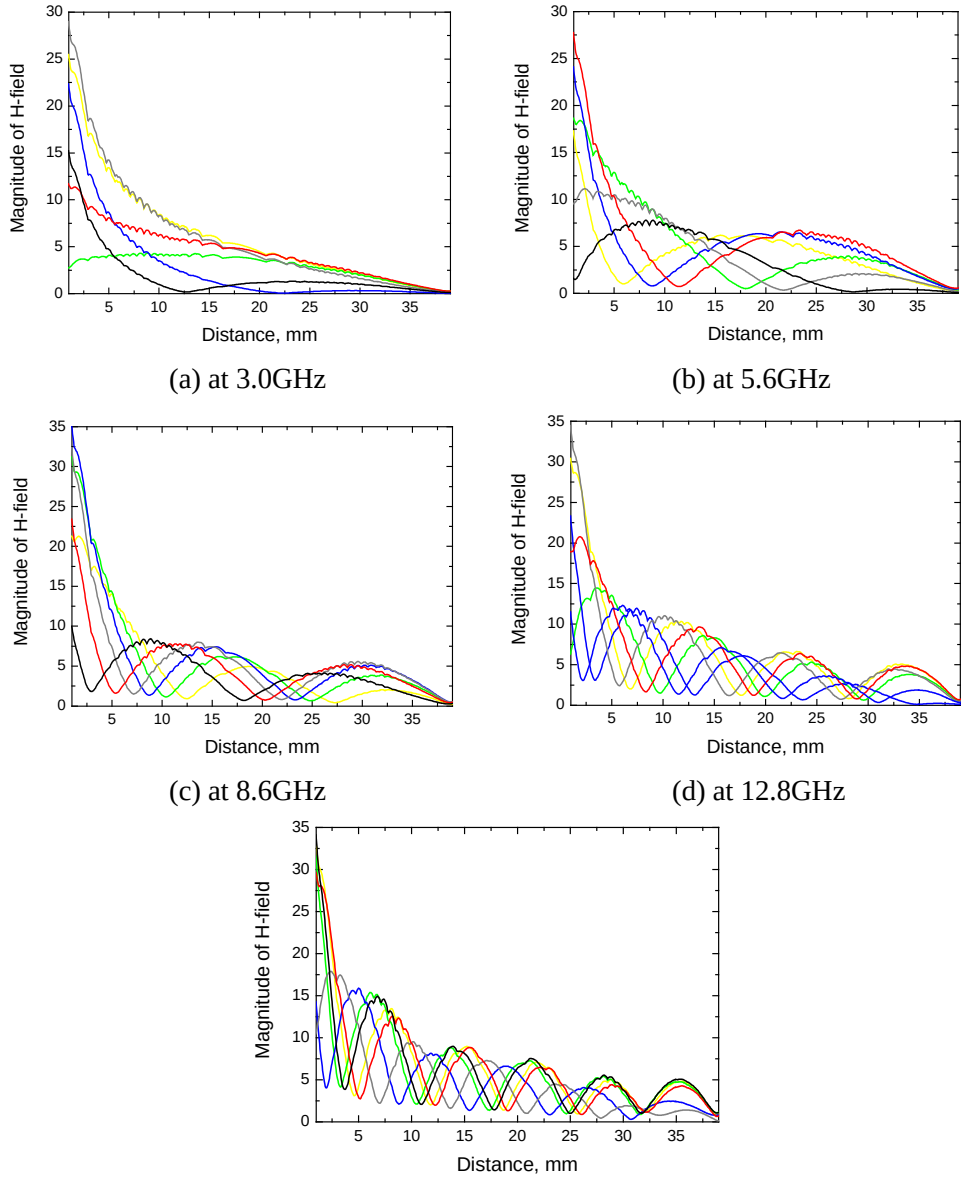


Figure 4.32: Simulated magnetic field distributions along the edge of the half disc D (D = 0 - 39mm: bottom to top) with different phases at each resonance

It is observed in Figure 4.32 that the magnetic field distributions of CPW fed disc monopole are quite similar to those of vertical disc monopole, as given in Figure 4.13.

At the first resonance, as shown in Figure 4.32(a), the current is oscillating and has a pure standing wave pattern along most part of the disc edge which indicates the disc behaves like an oscillating monopole. With the increase of frequency to the second resonance, the current is travelling along the lower disc edge, but oscillating at the top edge, as illustrated in Figure 4.32(b). The broad current envelope peak at around D = 25mm corresponds well to the gain increase in the radiation pattern, Figure 4.31(b). In Figure 4.32(c), (d) and (e), travelling wave seems more prominent at the lower edge of the disc, while standing wave retains on the top edge at the third, fourth and fifth resonances. Again, the envelope peaks correspond well to the humps (gain peaks) on the radiation patterns in Figure 4.31(c), (d) and (e), respectively.

4.2.3 Design Parameters

In this subsection, the important parameters which affect the antenna performance will be analysed to derive some design rules.

The first parameter is the feed gap h. As shown in Figure 4.33, when r is fixed at 12.5mm, L at 10mm and W at 47mm, the performance of the CPW fed disc monopole is quite sensitive to h. It can be seen that the return loss curves have similar shape for the four different feed gaps, but the -10dB bandwidth of the antenna varies significantly with the change of h. When h becomes bigger, the -10dB bandwidth is getting narrower due to the fact that the impedance matching of the antenna is getting worse. Looking across the whole spectrum, it seems that a bigger gap doesn't affect the 1st resonance very much, but has a much larger impact on the high harmonics. This suggests that the feed gap affects more the travelling wave operation of the antenna. The optimal feed gap is found to be at h=0.3mm, which is close to the CPW line gap. It makes perfect sense that the optimal feed gap should have a smooth transition to the CPW feed line.

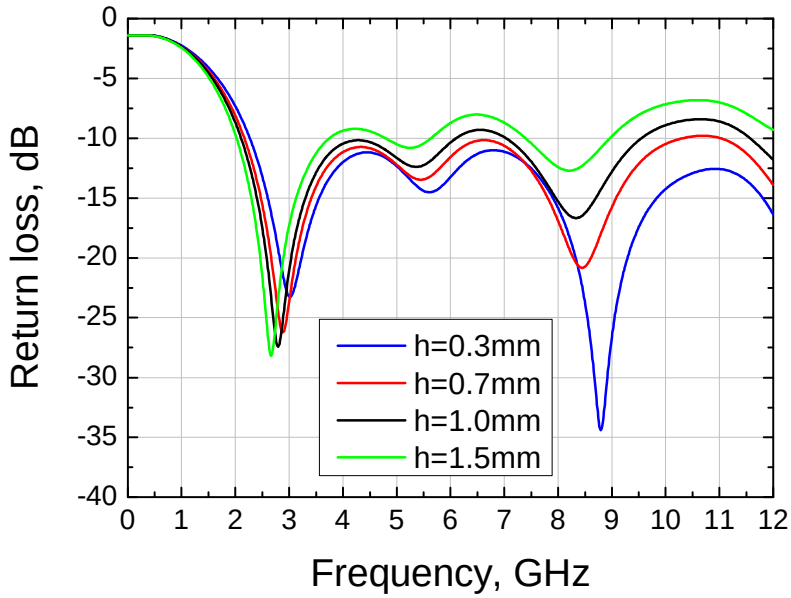


Figure 4.33: Simulated return loss curves of the CPW fed circular disc monopole for different feed gaps with r=12.5mm, L=10mm and W=47mm

Another design parameter influencing the antenna operation is the width of the ground plane W. The simulated return loss curves with r=12.5mm, L=10mm and optimal feed gap h of 0.3mm for different widths W are presented in Figure 4.34. It can be seen that the variation of the ground plane width shifts all the resonance modes across the spectrum. It is interesting to notice that the -10 dB bandwidth is reduced when the width of the ground is either too wide or too narrow. The optimal width of the ground plane is found to be at W=47mm. Again, this phenomenon can be explained when the ground plane is treated as a part of the antenna. When the ground plane width is either reduced or increased from its optimal size, so does the current flow on the top edge of the ground plane. This corresponds to a decrease or increase of the inductance of the antenna if it is treated as a resonating circuit, which causes the first resonance mode either up-shifted or down-shifted in the spectrum. Also, this change of inductance causes the frequencies of the higher harmonics to be unevenly shifted. Therefore, the change of the ground plane width makes some resonances become not so closely spaced across the spectrum and reduces the overlapping between them. Thus, the impedance matching becomes worse (return loss>-10dB) in these frequency ranges.

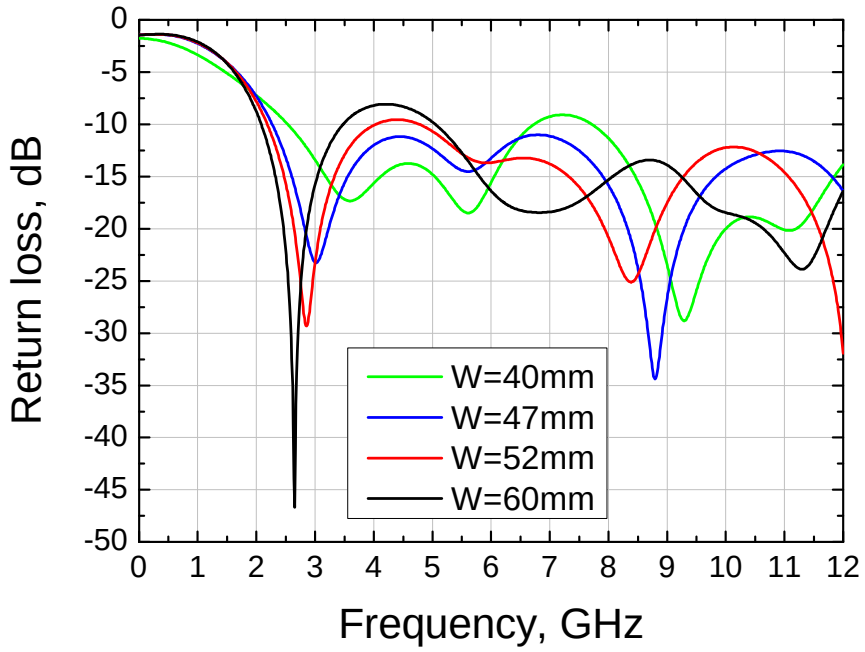


Figure 4.34: Simulated return loss curves of the CPW fed circular disc monopole for different widths of the ground plane with r=12.5mm, h=0.3mm and L=10mm

It is also noticed that the performance of the antenna is almost independent of the length L of the ground plane. This is understandable by inspecting current distributions in Figure 4.30 that the current is mostly distributed on the top edge of ground plane.

It has been established that the ground plane (both its relative position and its width) plays an important role on the antenna bandwidth. Besides, there is an interesting phenomenon in Figures 4.33 and 4.34 that the first resonance always occurs at around 3GHz for different feed gaps and some widths of the ground plane when the disc radius is fixed at 12.5mm. In fact, the quarter wavelength at the first resonant frequency (25mm) just equals to the diameter of the disc. This suggests that the first resonance occurs when the disc behaves like a quarter wave monopole.

Figure 4.35 shows the simulated return loss curves for different dimensions of the disc with their respective optimal designs, which are given in Table 4-B. It can be seen that the ultra wide impedance bandwidth can obtained in all of these designs.

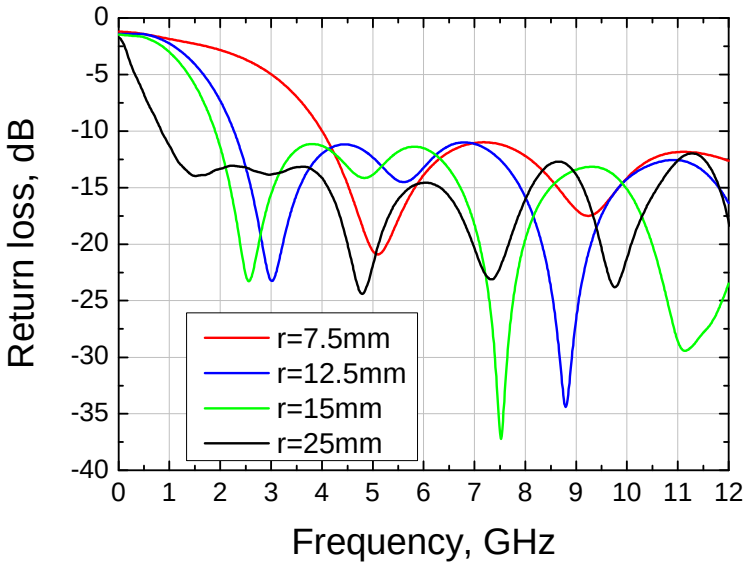


Figure 4.35: Simulated return loss curves for different disc dimensions of the circular disc in the optimal designs

Table 4-B: Optimal design parameters of the CPW fed disc monopole and relationship between the diameter and the first resonance

Diameter $2r$ (mm)	15	25	30	50
First resonance f_1 (GHz)	5.09	3.01	2.57	1.52
Wavelength λ at f_1 (mm)	58.9	99.7	116.7	197.4
$2r/\lambda$	0.25	0.25	0.26	0.25
Optimal W (mm)	28	47	56	90
$W/2r$	1.87	1.88	1.87	1.80
Optimal h (mm)	0.1	0.3	0.3	0.5

The relationship between the disc diameters and the first resonances is also listed in Table 4-B. It is demonstrated that the first resonant frequency is determined by the diameter of the disc, which approximately corresponds to the quarter wavelength at this frequency. In addition, it is shown that the optimal width of the ground plane is just less than twice the diameter of the disc, ranging from 1.80 to 1.88. The optimal feed gap h is around 0.3mm, which is close to the CPW line gap (0.33mm), with slight variations for the big and small discs. Table 4-B is thus a good summary of the design rules for achieving the ultra wide impedance bandwidth in a CPW fed disc monopole.

4.2.4 Operating Principle

It has been demonstrated that the overlapping of closely distributed resonance modes in both vertical and CPW fed disc monopoles is responsible for an ultra wide -10dB bandwidth.

At the low frequency end (the first resonance) when the wavelength is bigger than the antenna dimension, the EM wave can easily 'couple' into the antenna structure so it operates in an oscillating mode, i.e. a standing wave. With the increase of the frequency, the antenna starts to operate in a hybrid mode of standing and travelling waves.

At the high frequency end, travelling wave becomes more critical to the antenna operation since the EM wave needs to travel down to the antenna structure which is big in terms of the wavelength.

For the CPW fed antenna, the slots formed by the lower edge of the disc and the ground plane with a proper dimension can support travelling wave very well. So an optimally designed CPW fed circular disc monopole can exhibit an extremely wide -10dB bandwidth. The operation principle of UWB disc monopole in Figure 4.11 now can be revised as the following schematic, Figure 4.36.

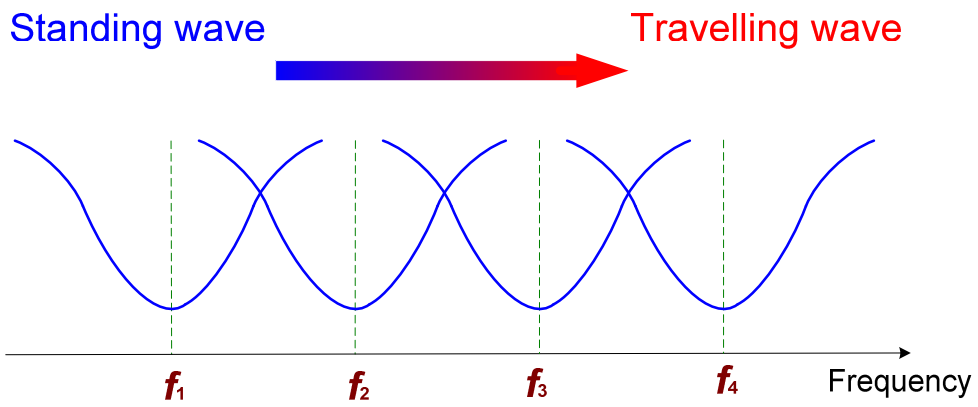


Figure 4.36: Operation principle of CPW fed disc monopole

4.3 Microstrip Line Fed Disc Monopole

Other than CPW feeding structure, a planar circular disc monopole can also be realized by using microstrip feed line, as illustrated in Figure 4.37.

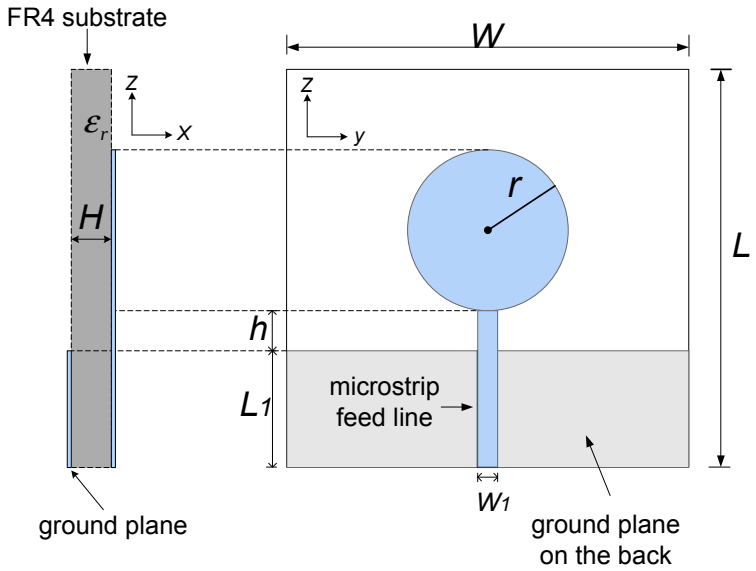


Figure 4.37: Geometry of microstrip line fed disc monopole

The circular disc monopole with a radius of r and a 50Ω microstrip feed line are printed on the same side of the FR4 (Flame Resistant 4) substrate (the substrate has a thickness of H=1.5mm and a relative permittivity of ε_r=4.7). L and W denote the length and the width of the substrate, respectively. The width of the microstrip feed line

is fixed at W_1=2.6mm to achieve 50Ω impedance. On the other side of the substrate, the conducting ground plane with a length of L_1=20mm only covers the section of the microstrip feed line. h is the feed gap between the feed point and the ground plane.

Figure 4.38 presents the photo of microstrip line fed disc monopole in the optimal design, i.e. r=10mm, h=0.3mm, W=42mm and L=50mm. Its measured return loss curve against simulated one is plotted in Figure 4.39.

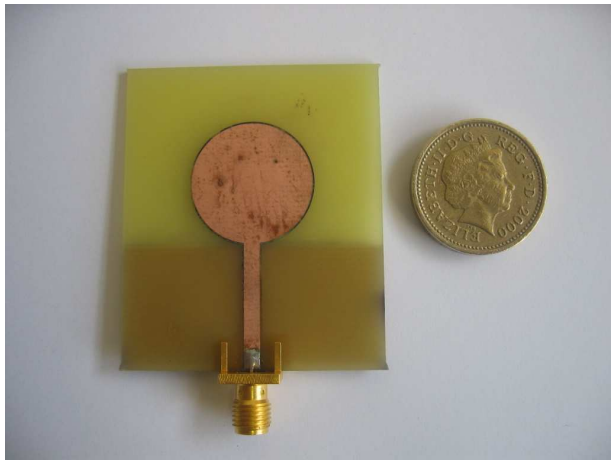


Figure 4.38: Photo of microstrip line fed disc monopole with r=10mm, h=0.3mm, W=42mm and L=50mm

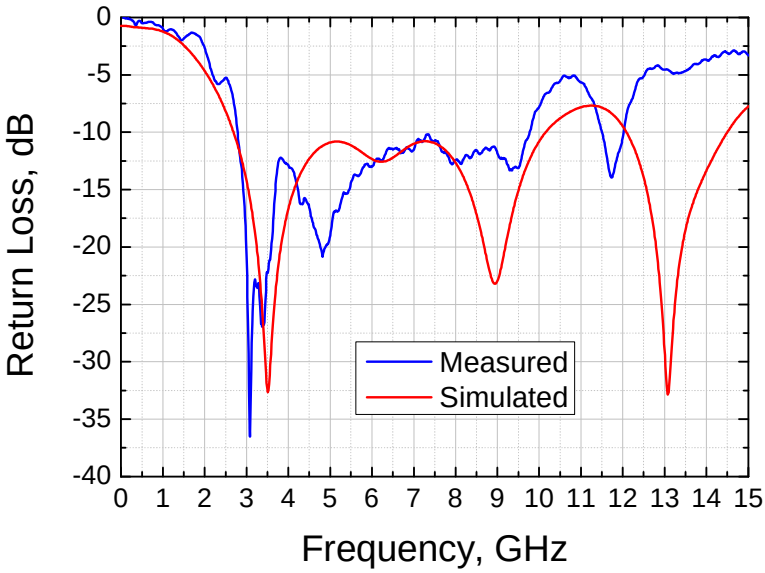


Figure 4.39: Measured and simulated return loss curves of microstrip line fed disc monopole with r=10mm, h=0.3mm, W=42mm and L=50mm

It is observed in Figure 4.39 that the measured -10dB bandwidth ranges from 2.78GHz to 9.78GHz, while in simulation from 2.69GHz to 10.16GHz. The measurement confirms the UWB characteristic of the proposed printed circular disc monopole, as predicted in the simulation.

The UWB characteristic of the antenna can be again attributed to the overlapping of the first three resonances which are closely distributed across the spectrum. However, despite considerable efforts having been spent in tuning the design parameters, a good overlapping between the third and fourth harmonics can not be achieved. Thus, the -10dB bandwidth of this antenna is always limited at the high end around 10GHz. By viewing the simulated input impedance plotted on the Smith Chart, given in Figure 4.40, it is noticed that at the high frequency limit the input impedance loops out of the VSWR=2 circle, i.e. the impedance matching is getting worse. This can be understood by examining the variation in current distribution at resonances.

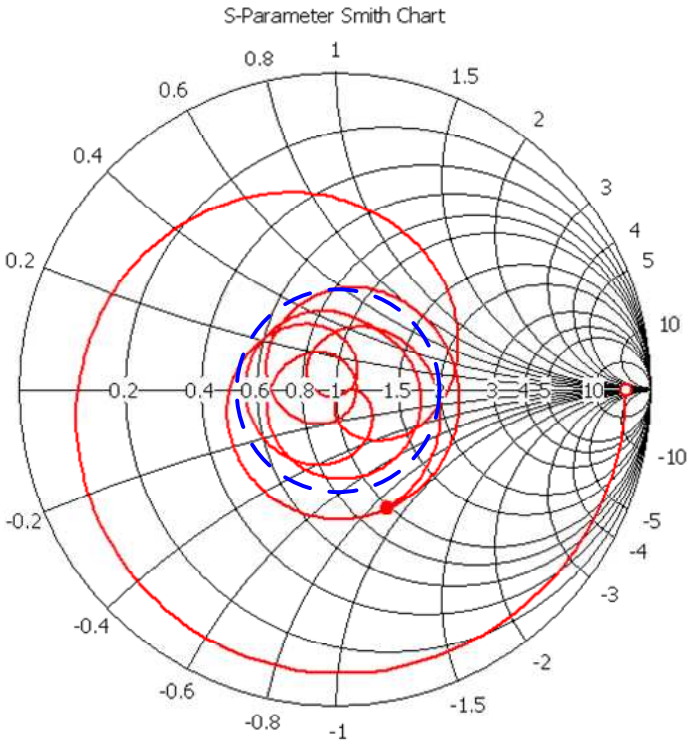
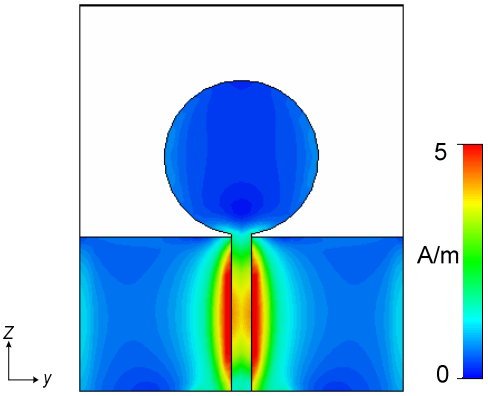
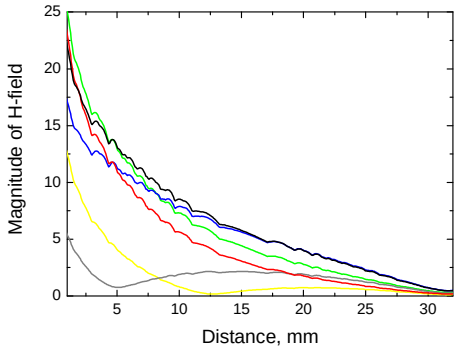


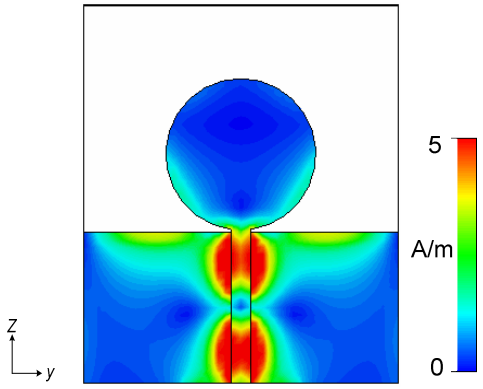
Figure 4.40: Simulated Smith Chart of microstrip line fed disc monopole with r=10mm, h=0.3mm, W=42mm and L=50mm



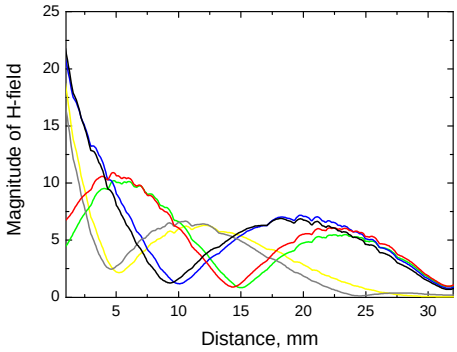
(a) Current distribution at 3GHz



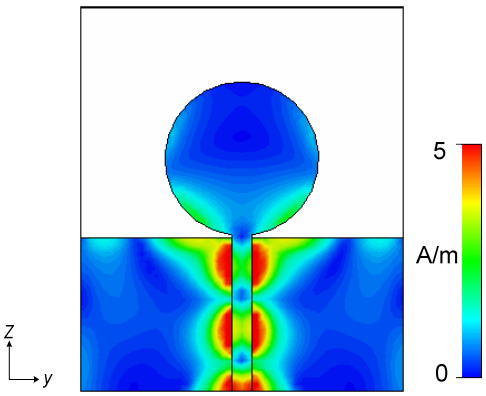
(d) Magnetic field variation at 3GHz



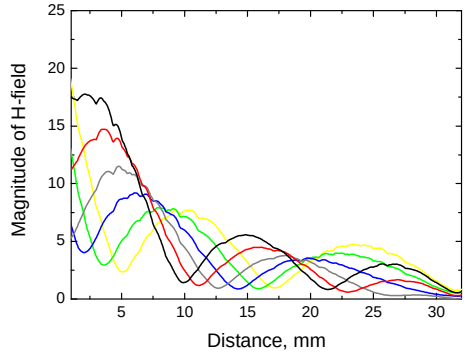
(b) Current distribution at 6.5GHz



(e) Magnetic field variation at 6.5GHz



(c) Current distribution at 9GHz



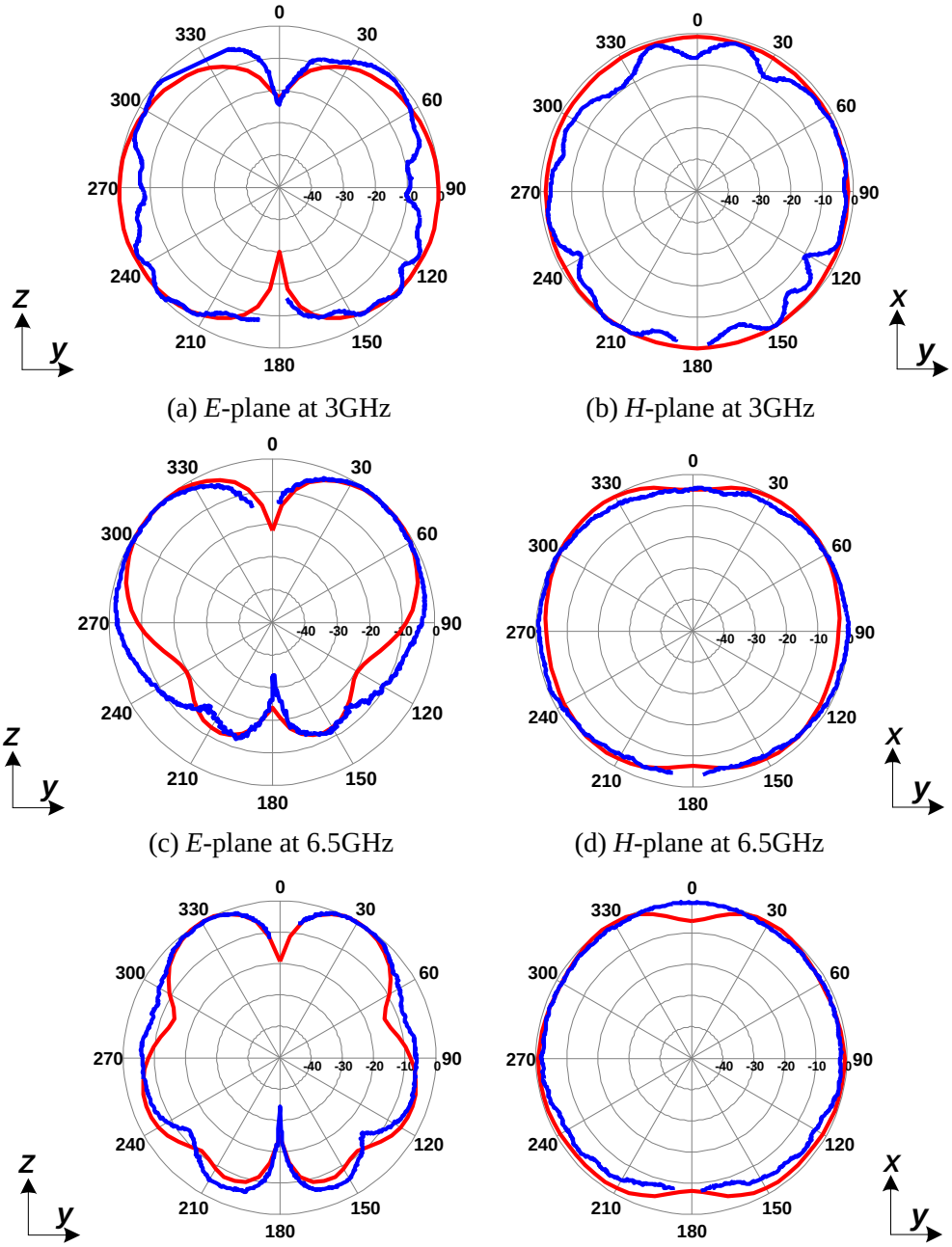
(f) Magnetic field variation at 9GHz

Figure 4.41: Simulated current distributions (a-c) and magnetic field distributions along the edge of the half disc D ($D = 0$ - 33mm: bottom to top) at different phases (d-f) of microstrip line fed disc monopole with r=10mm, h=0.3mm, W=42mm and L=50mm

Simulated current distributions and magnetic field variations of the microstrip line fed disc monopole at different frequencies are presented in Figure 4.41. Figure 4.41(a) shows the current pattern near the first resonance at 3GHz. The current pattern near the second resonance at around 6.5GHz is given in Figure 4.41(b), indicating approximately a second order harmonic. Figure 4.41(c) illustrates a more complicated current pattern at 9GHz, corresponding to the third order harmonic. The magnetic field variation patterns, as shown in Figures 4.41 (d) - (f), indicate that the antenna also operates in a hybrid mode of travelling and standing waves at higher frequencies, like vertical and CPW fed disc monopoles. However, the ground plane on the other side of the substrate can not form a good slot with the disc to support travelling waves as well as CPW fed disc monopole. Therefore, the impedance matching becomes worse for the travelling wave dependent modes at high frequencies, as indicated in Figures 4.39 and 4.40.

The measured and the simulated radiation patterns at 3GHz, 6.5GHz and 9GHz are plotted in Figures 4.42. The measured H-plane patterns are very close to those obtained in the simulation. It is noticed that the H-plane pattern is omni-directional at lower frequency (3GHz) and still retains a good omni-directionality at higher frequencies (6.5GHz and 9GHz).

The measured E-plane patterns also follow the shapes of the simulated ones well. The E-plane pattern is like a donut at 3GHz. With the increase of frequency (6.5GHz and 9GHz), it starts to form humps and get more directional at around 45 degrees from the z-direction. In general, the shapes of the E-plane patterns correspond well to the current patterns on the disc, as shown in Figure 4.41.



(a) *E*-plane at 3GHz (b) *H*-plane at 3GHz

(c) *E*-plane at 6.5GHz (d) *H*-plane at 6.5GHz

(e) *E*-plane at 9GHz (f) *H*-plane at 9GHz

Figure 4.42: Measured (blue line) and simulated (red line) radiation patterns of microstrip line fed disc monopole with r=10mm, h=0.3mm, W=42mm and L=50mm

The design rules for the microstrip fed disc monopole, tabulated in Table 4-C, are also similar to those of the CPW fed disc monopole.

Table 4-C: Optimal design parameters of microstrip line fed disc monopole and relationship between the diameter and the first resonance

Diameter $2r$ (mm)	20	25	30	40
First resonance f_1 (GHz)	3.51	2.96	2.56	1.95
Wavelength λ at f_1 (mm)	85.5	101.4	117.2	153.8
$2r/\lambda$	0.23	0.25	0.26	0.26
Optimal W (mm)	42	50	57	75
$W/2r$	2.1	2	1.9	1.9
Optimal h (mm)	0.3	0.3	0.3	0.4

For the benefit of design, four discs with different diameters have been investigated in simulation. It is observed in the simulation that, in a manner similar to the CPW fed disc monopole, the -10dB bandwidth of the microstrip fed monopole is critically dependent on the feed gap h and the width of the ground plane W. Again, the first resonance frequency can be estimated by treating the disc as a quarter wave monopole. The optimal width of the ground plane is found to be around two times of the diameter and the optimal feed gap is still around 0.3mm, as shown in Table 4-C.

4.4 Other Shape Disc Monopoles

4.4.1 Circular Ring Monopole

It is seen in Figure 4.30 and Figure 4.41 that the current is mainly distributed along the edge of the disc for both CPW fed and microstrip line fed disc monopoles. This implies that the performance of the antenna is independent of the central part of the disc, and hence cutting this part to form a ring monopole can still achieve an ultra wide bandwidth.

For microstrip line fed disc monopole given in Figure 4.37, we retain the optimal parameters, i.e. r=10mm, h=0.3mm, W=42mm and L=50mm, but cut a hole with radius of r_1 in the disc center. Figure 4.43 illustrates the simulated return loss curves for different inner radii r_1.

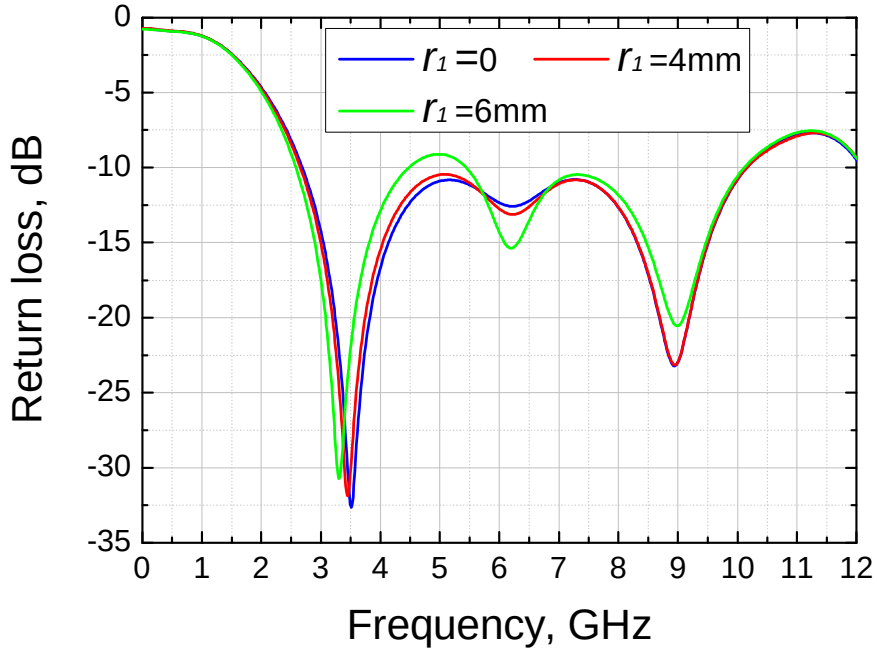


Figure 4.43: Simulated return loss curves of microstrip line fed ring monopole for different inner radii r_1 with r=10mm, h=0.3mm, W=42mm and L=50mm

It is noticed in Figure 4.43 that when r_1 is no more than 4mm, the return loss curve does not change much with the variation of r_1. When r_1 rises to 6mm, the return loss

shifts up to higher than -10dB at around 5GHz. As a result, the -10dB bandwidth is narrowed remarkably. This indicates that a microstrip line fed circular ring monopole with inner radius r_1 up to 4mm can provide similar operating bandwidth as its counterpart circular disc monopole (r_1=0).

In a similar way, it is found that for CPW fed case, a circular ring monopole with inner radius r_1 up to 5mm can provide nearly identical return loss to that of the circular disc monopole with radius of 12.5mm.

Two prototypes of circular ring monopole antennas in their respective optimal designs were built and tested. One is fed by microstrip line, and the other by CPW, as shown in Figure 4.44 and Figure 4.45, respectively.

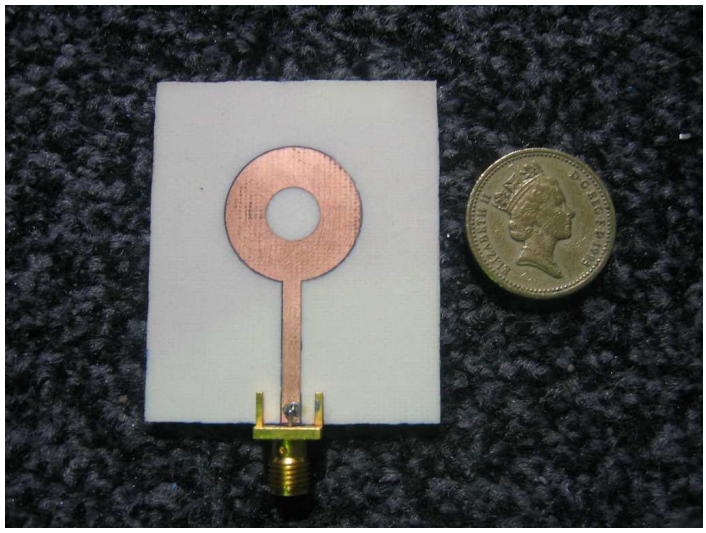


Figure 4.44: Photo of microstrip line fed ring monopole with r=10mm, r_1=4mm, h=0.3mm, W=42mm and L=10mm

The measured and simulated return loss curves of the two circular ring monopoles are illustrated in Figure 4.46 and Figure 4.47, respectively. The measured return loss curves of their respective circular disc counterparts are also plotted in the figures for the ease of comparison.

It is evident in Figure 4.46 and Figure 4.47 that the measured return loss curve is close to the simulated one quite well for both of the two circular ring monopoles.

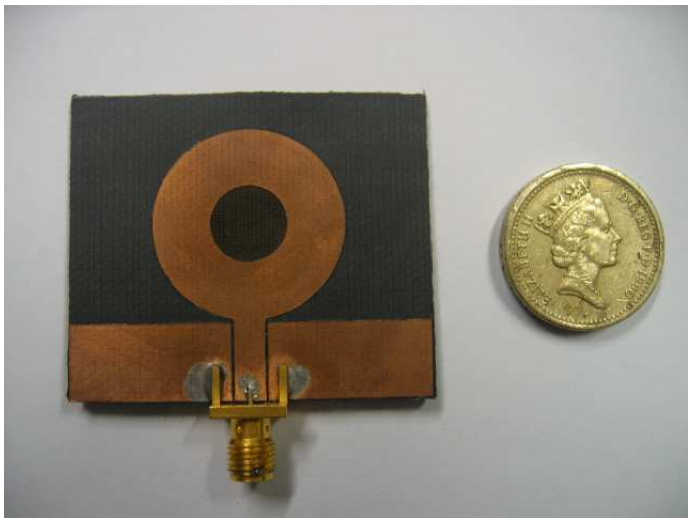


Figure 4.45: Photo of CPW fed ring monopole with r=12.5mm, r_1=5mm, h=0.3mm, W=47mm and L=50mm

Furthermore, the measured curve of ring monopole is almost identical to that of its disc counterpart for both microstrip line fed and CPW fed cases. This confirms that circular ring monopole can provide a similar UWB performance as its counterpart disc monopole.

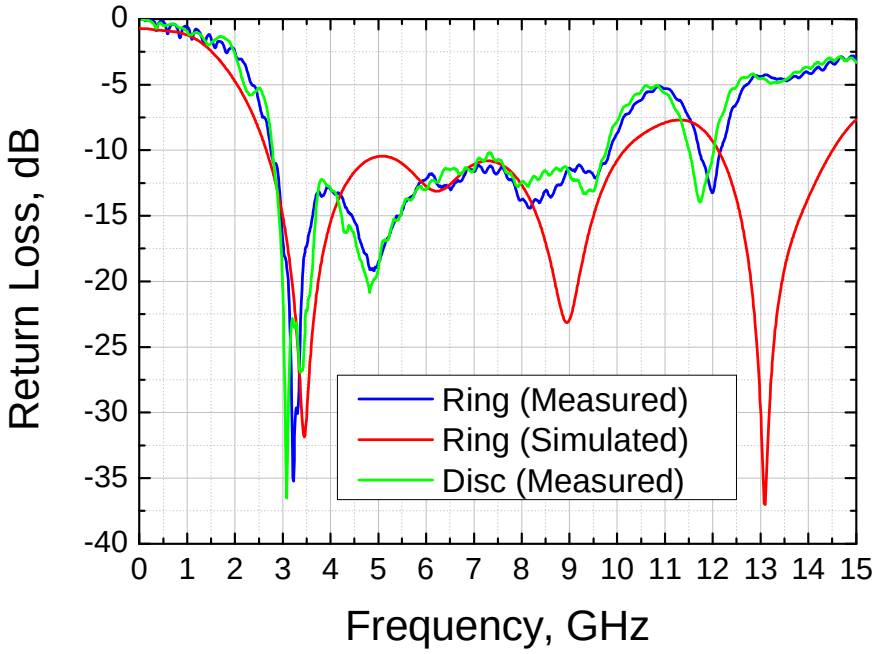


Figure 4.46: Measured and simulated return loss curves of microstrip line fed ring monopole with r=10mm, r_1=4mm, h=0.3mm, W=42mm and L=50mm

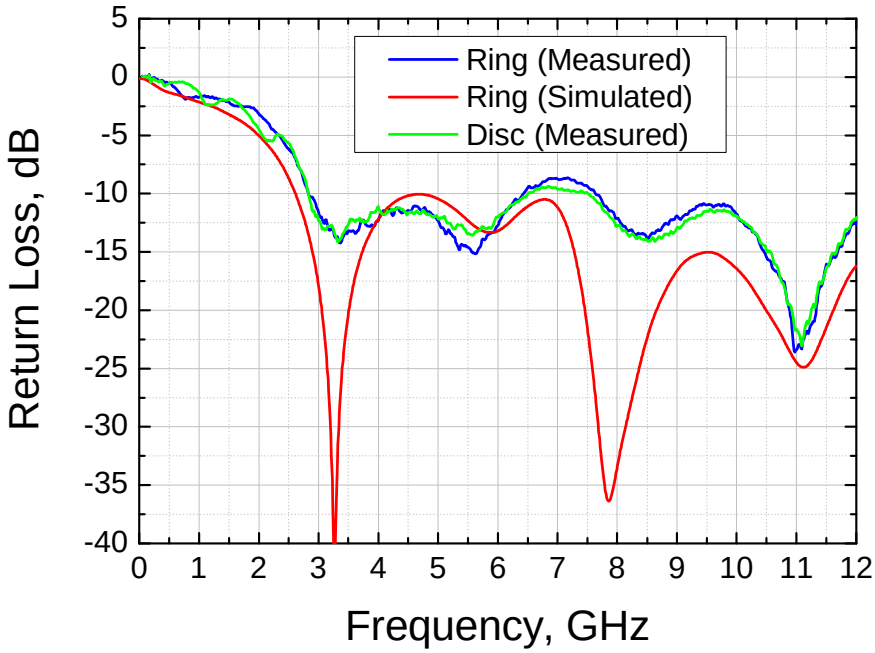
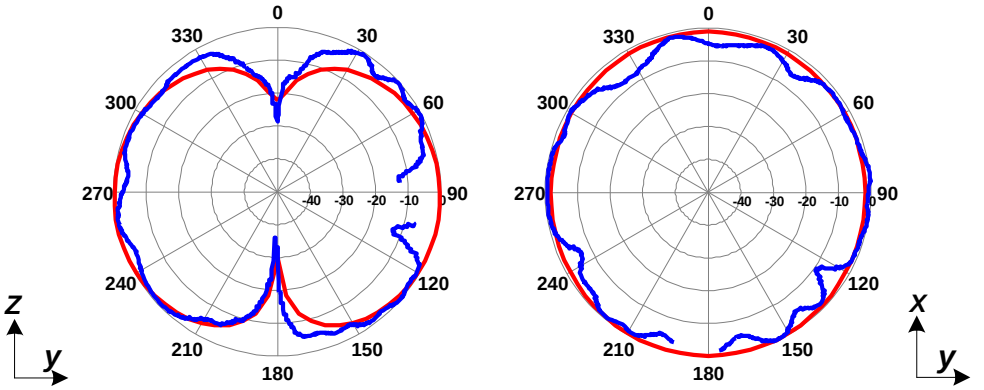


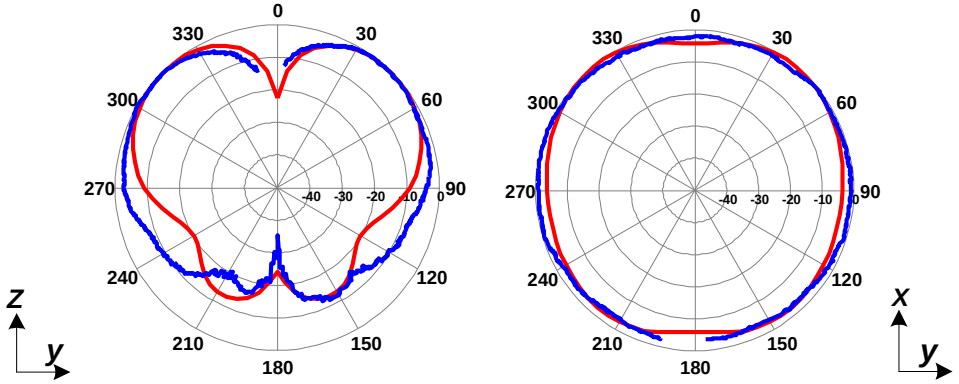
Figure 4.47: Measured and simulated return loss curves of CPW fed ring monopole with r=12.5mm, r_1=5mm, h=0.3mm, W=47mm and L=10mm

The radiation patterns of microstrip line fed circular ring monopole are presented in Figure 4.48. They were simulated and measured at 3GHz, 6.5GHz and 9GHz to compare with the disc counterpart. It is observed in Figure 4.48 that the measured patterns are close to those obtained in the simulation. Compared with the printed disc monopole, as shown in Figure 4.42, the proposed ring monopole exhibits almost the same radiation properties, and it is also nearly omni-directional over the entire bandwidth.

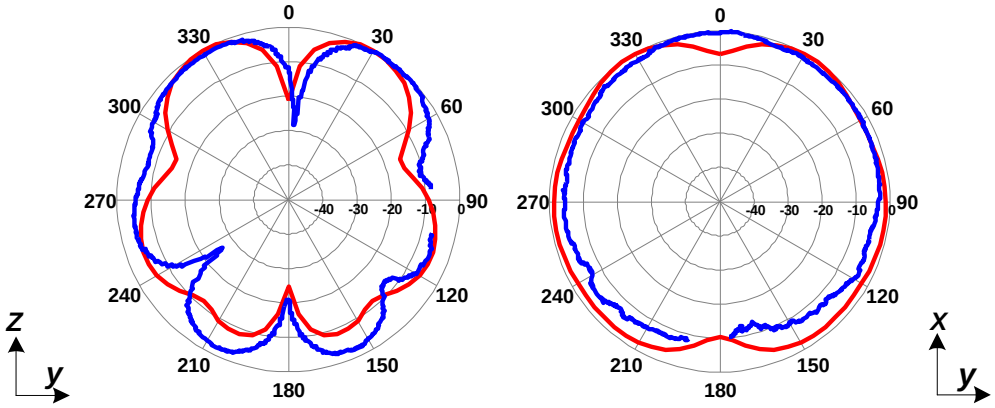
The measured results confirmed that circular ring monopole can exhibit nearly same characteristics as its disc counterpart if the inner radius is below a certain value, as predicted in the simulation.



(a) *E*-plane at 3GHz (b) *H*-plane at 3GHz



(c) *E*-plane at 6.5GHz (d) *H*-plane at 6.5GHz



(e) *E*-plane at 9GHz (f) *H*-plane at 9GHz

Figure 4.48: Measured (blue line) and simulated (red line) radiation patterns of microstrip line fed circular ring monopole with r=10mm, r_1=4mm, h=0.3mm, W=42mm and L=50mm

4.4.2 Elliptical Disc Monopole

A planar elliptical disc monopole is also designed on the FR4 substrate by using a microstrip line fed to provide an ultra wide operating bandwidth, as shown in Figure 4.49. The substrate has a thickness of H=1.5mm and a relative permittivity of ε_r=4.7. Thus, the width of the microstrip feed line is fixed at W_1=2.6mm to achieve 50Ω impedance. The ground plane with a width W and a length of L_1=20mm only covers the section of the microstrip feed line. A and B represent the long-axis and the short-axis of the elliptical disc, respectively.

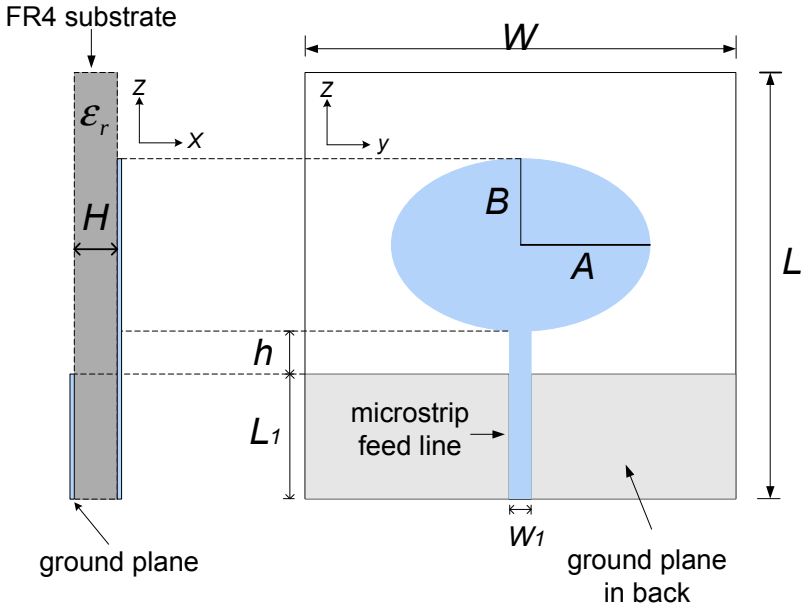


Figure 4.49: Geometry of microstrip line fed elliptical disc monopole

As is the case for circular disc monopole, the feed gap h and the ground plane width W are among the most important design parameters. Their optimal values are found to be at h=0.7mm and W=44mm, respectively. Besides, the elliptical ratio A/B also influences the antenna performance.

As shown in Figure 4.50, when h=0.7mm, W=44mm, L=44mm and B=7.8mm, the return loss curve varies substantially for different elliptical ratios A/B. When A/B=1.7, the first -10dB bandwidth is only 3.51GHz, ranging from 3.15GHz to 6.76GHz; when

A/B is changed to 1.1, the bandwidth is even narrower, from 3GHz to 4.77GHz. While an optimal ratio of A/B=1.4 can yield an ultra wide frequency band of 6.51GHz.

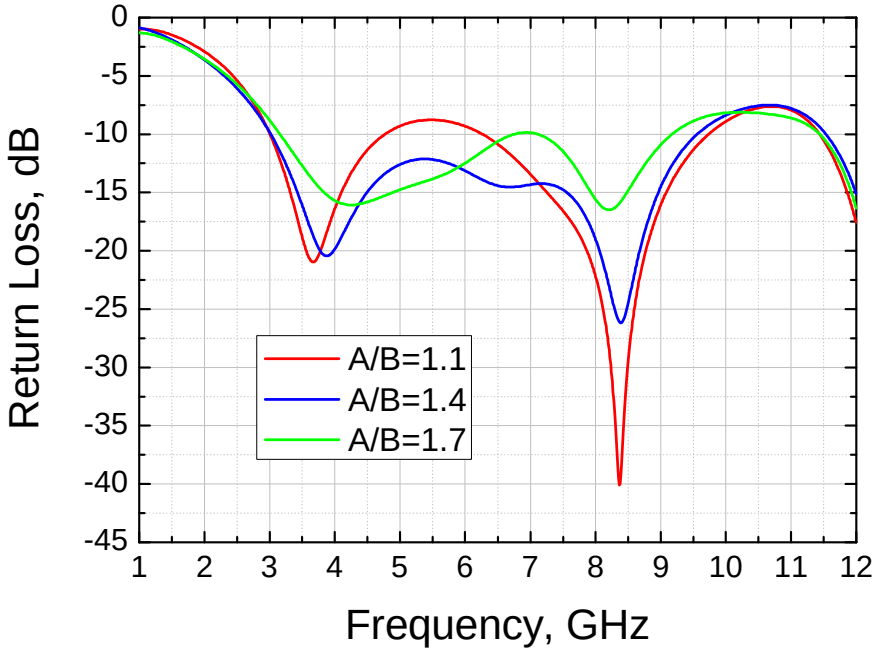


Figure 4.50: Simulated return loss curves of microstrip line fed elliptical disc monopole for different elliptical ratio A/B with h=0.7mm, W=44mm, L=44mm and B=7.8mm

A prototype of the proposed elliptical disc monopole in the optimal design, i.e. h=0.7mm, W=44mm, L=44mm, B=7.8mm and A/B=1.4, was fabricated and tested.

The measured and simulated return loss curves, as given in Figure 4.51, show a good agreement. The simulated -10dB bandwidth ranges from 3.07GHz to 9.58GHz. This UWB characteristic is confirmed in the measurement, with only a slight shift of the upper edge frequency to 9.89GHz.

It can be seen that the UWB characteristic of the antenna is mainly attributed to the overlapping of the first three resonances, which are closely spaced across the spectrum. For the similar reason as that in a microstrip fed circular disc monopole, the -10dB bandwidth is limited at the high end of the frequency due to the increase of impedance mismatching.

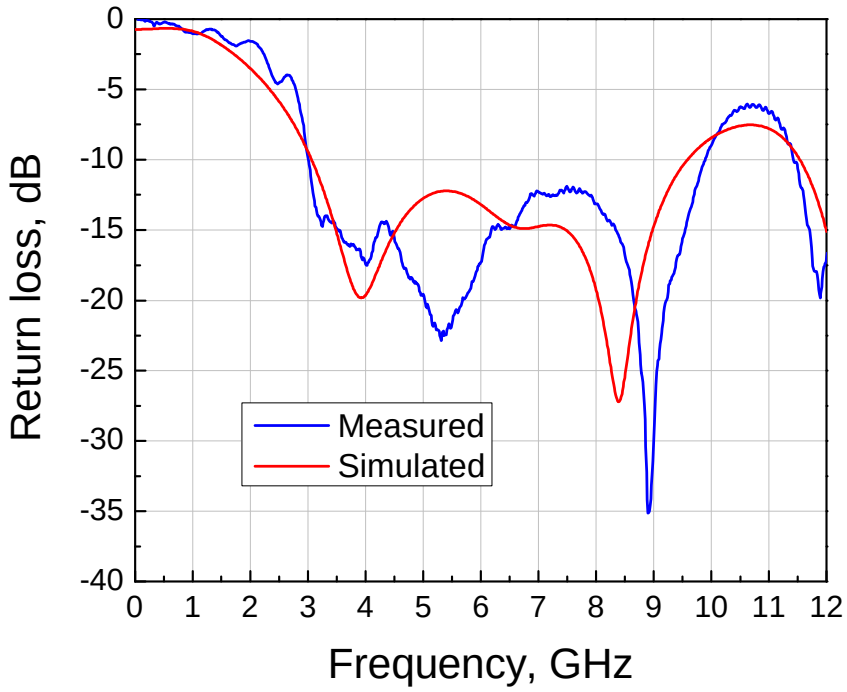


Figure 4.51: Measured and simulated return loss curves of microstrip line fed elliptical disc monopole with h=0.7mm, W=44mm, L=44mm, B=7.8mm and A/B=1.4

It has also been demonstrated both numerically and experimentally that the proposed elliptical disc monopole can provide similar radiation patterns as circular disc monopole, i.e. nearly omni-directional radiation patterns over the entire operational band.

4.5 Summary

This chapter investigates the frequency domain performances of UWB circular disc monopole antennas. This type of antenna is initially realized by replacing the wire element of a conventional monopole with a circular disc element. The antenna configuration has also evolved from a vertical disc to a planar version by using microstrip line and CPW feeding structure to achieve low profile and compatibility with printed circuit board.

It has been demonstrated that the overlapping of closely spaced multiple resonances

accounts for the UWB characteristics of disc monopole antenna. Normally, the first three or four resonances are required to overlap each other properly in order to provide a bandwidth covering the whole FCC defined UWB band from 3.1GHz to 10.6GHz. At the first resonance, the antenna behaves like a quarter wavelength monopole. With the increase of frequency, it starts to operate in a hybrid mode of standing and travelling waves. Travelling wave becomes dominant in the antenna operation at higher order (third and above) resonances. So it is essential to design a smooth transition between the feeding line and the antenna for good impedance matching over the entire operational bandwidth.

The important parameters of disc monopole antennas are also studied to derive the design rules. Both simulation and experiment have shown that disc monopole antennas exhibit nearly omni-directional radiation patterns in the H-plane over the entire operational frequency band.

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Chapter 5

UWB Slot Antennas

In Chapter 4, disc monopole antennas have been demonstrated to exhibit UWB characteristics. These antennas have relatively large electric near-fields that are prone to undesired coupling with near-by objects. In contrast, slot antennas have relatively large magnetic near-fields that tend not to couple strongly with near-by objects [1]. Thus, slot antennas are well suited for applications wherein near-field coupling is required to be minimized. In addition, printed slot antennas feature low profile, lightweight, ease of fabrication and wide frequency bandwidth, so they have attracted significant interest.

This Chapter studies UWB elliptical/circular slot antennas with an emphasis on their frequency domain performances. Their time domain characteristics will be discussed in Chapter 6.

5.1 Introduction

Slot antennas were developed in the late 1940s to early 1950s [2]. In recent years, printed slot antennas are under consideration for use in UWB systems and are getting more and more popular due to their attractive merits of simple structure and low profile. This type of antenna can take various configurations such as rectangle [3–5], circle [6, 7],

arc-shape [8], triangle [8, 9], annular-ring [10] and others [11, 12]. These printed slot antennas have been realized by using either microstrip line [13, 14], or CPW feeding structure [15, 16].

Various techniques have been proposed to broaden the bandwidth of printed slot antennas and improve their performances. In [6], a microstrip line fed circular slot can operate over the entire UWB band, i.e. from 3.1GHz to 10.6GHz. However, the antenna size is big and the slot diameter is 65.2mm. In [13], a round corner rectangular slot antenna which is etched on a substrate with dimension of 68mm×50mm can achieve a -10 dB bandwidth of 6.17GHz. In [14], a fork-like tuning stub is used to enhance the bandwidth of microstrip line fed wide-slot antenna. A bandwidth of 1.1GHz (1.821GHz-2.912GHz) has been achieved with gain variation less than 1.5dBi (3.5dBi-5dBi) over the entire operational band. In [15], a CPW fed square slot antenna with a widened tuning stub can yield a bandwidth of 60%. The antenna has a dimension of 72mm×72mm and its gain ranges from 3.75dBi to 4.88dBi within the operational band. In [16], a CPW fed rectangular slot antenna with a substrate of 100mm×100mm can provide a bandwidth of 110% with gain varying from 1.9dBi to 5.1dBi.

It is shown that the achieved bandwidths of these antennas can not cover the whole FCC defined UWB frequency band from 3.1GHz to 10.6GHz. Besides, the sizes of the antennas are not very small.

In this Chapter, two novel designs of printed elliptical/circular slot antennas are proposed for UWB applications. One is fed by microstrip line, and the other by CPW. In both designs, a U-shaped tuning stub is introduced to enhance the coupling between the slot and the feed line so as to broaden the operating bandwidth of the antenna. Furthermore, an additional bandwidth enhancement can be achieved by tapering the feeding line.

5.2 Antenna Geometry

The proposed printed elliptical/circular slot antennas with two different feeding structures are illustrated in Figure 5.1 and 5.2, respectively. For the microstrip line fed elliptical/circular slot antenna, the slot and the feeding line are printed on different sides of the dielectric substrate; for the CPW fed one, they are printed on the same side of the substrate.

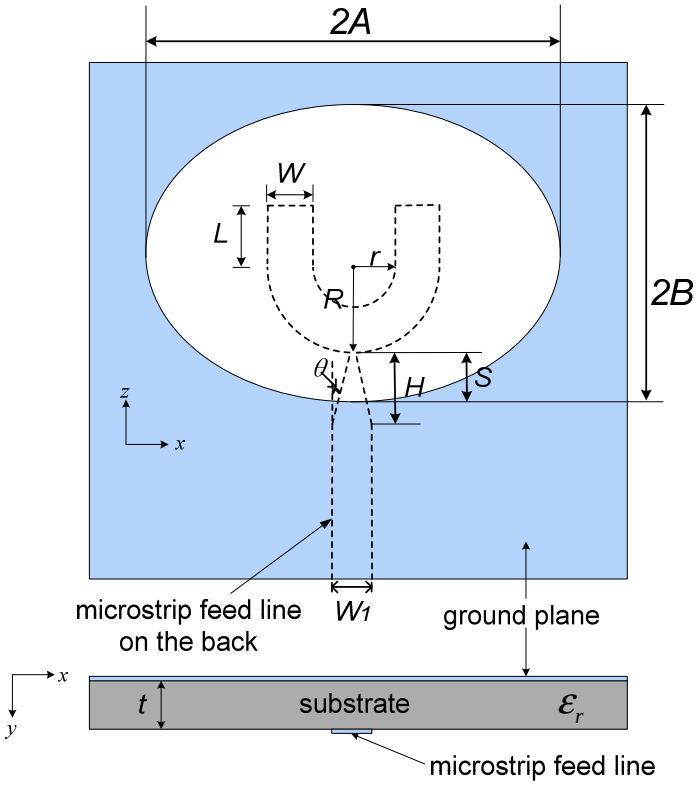


Figure 5.1: Geometry of microstrip line fed elliptical/circular slot antennas

In both designs, the elliptical/circular radiating slot has a long axis radius A and a short axis radius B (for circular slot, A=B) and is etched on a rectangular FR4 substrate with a thickness t=1.5mm and a relative dielectric constant ε_r=4.7. The feed line is tapered with a slant angle θ=15 degrees for a length H to connect with the U-shaped tuning stub which is all positioned within the elliptical/circular slot and symmetrical with respect to the short axis of the elliptical/circular slot. The U-shaped tuning stub consists of three sections: the semi-circle ring section with an outer radius R and an

inner radius r, and two identical branch sections with equal heights L and equal widths W. S represents the distance between the bottom of the tuning stub and the lower edge of the elliptical/circular slot.

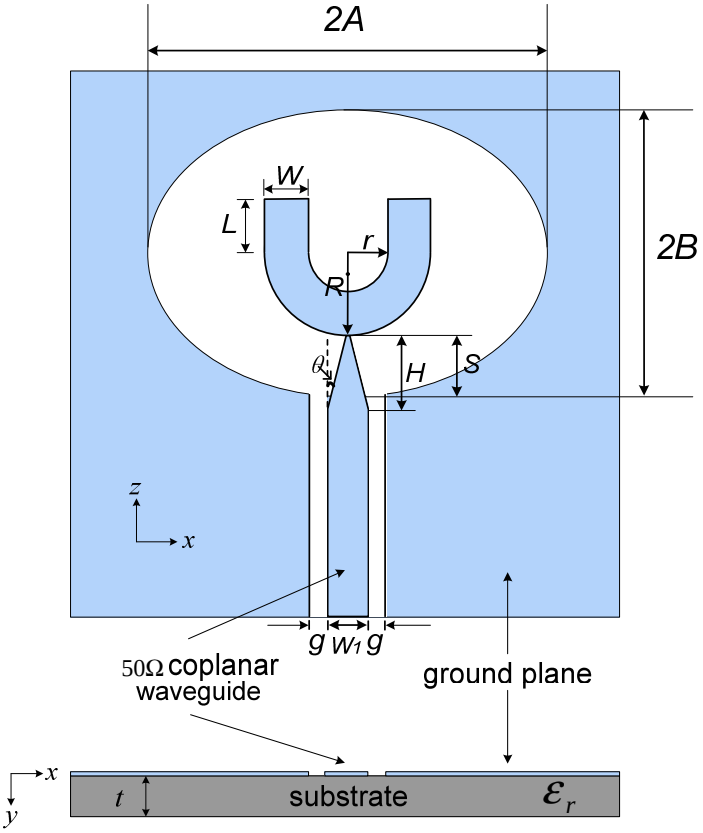
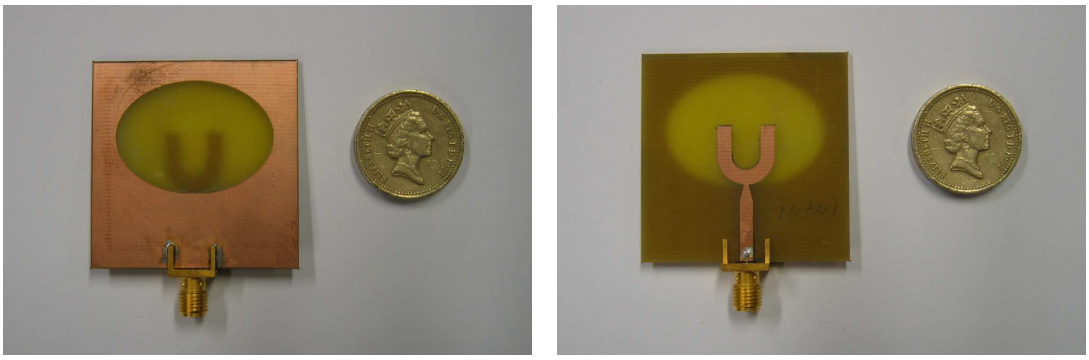


Figure 5.2: Geometry of CPW fed elliptical/circular slot antennas

5.3 Performances and Characteristics

Four printed elliptical/circular slot antennas with the optimal designs were fabricated and tested in the Antennas Laboratory at Queen Mary, University of London, as illustrated in Figure 5.3–Figure 5.6. Their respective dimensions are given in Table 5-A. It is noticed that all of these four antennas feature a small size and even the largest one, i.e. microstrip line fed circular slot, is still 37% less than the antenna presented in [13].



(a) Top view (b) Back view

Figure 5.3: Photo of microstrip line fed elliptical slot antenna



(a) Top view (b) Back view

Figure 5.4: Photo of microstrip line fed circular slot antenna

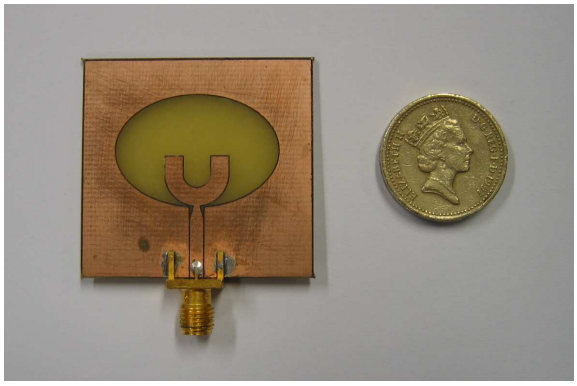


Figure 5.5: Photo of CPW fed elliptical slot antenna

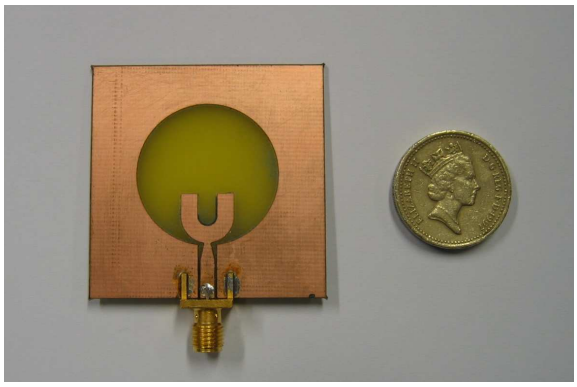


Figure 5.6: Photo of CPW fed circular slot antenna

Table 5-A: Optimal dimensions of the printed elliptical/circular slot antennas

	Microstrip line fed elliptical slot	Microstrip line fed circular slot	CPW fed elliptical slot	CPW fed circular slot
A(mm)	16	13.3	14.5	13.3
B(mm)	11.5	13.3	10	13.3
S(mm)	0.6	0.5	0.4	0.4
R(mm)	5.9	5	5.5	5
r(mm)	2.9	1.8	2.5	1.8
H(mm)	3.3	3.1	2.5	3.1
W(mm)	3	3.2	3	3.2
L(mm)	6	6.7	3	4.3
Substrate Size (mm^2)	42×42	43×50	40×38	44×44

5.3.1 Return Loss and Bandwidth

The return losses of the four antennas were measured by using an HP8720ES vector network analyzer. The measured return loss curves against the simulated ones are plotted

in Figure 5.7—-Figure 5.10. Their respective -10dB bandwidths are tabulated in Table 5-B.

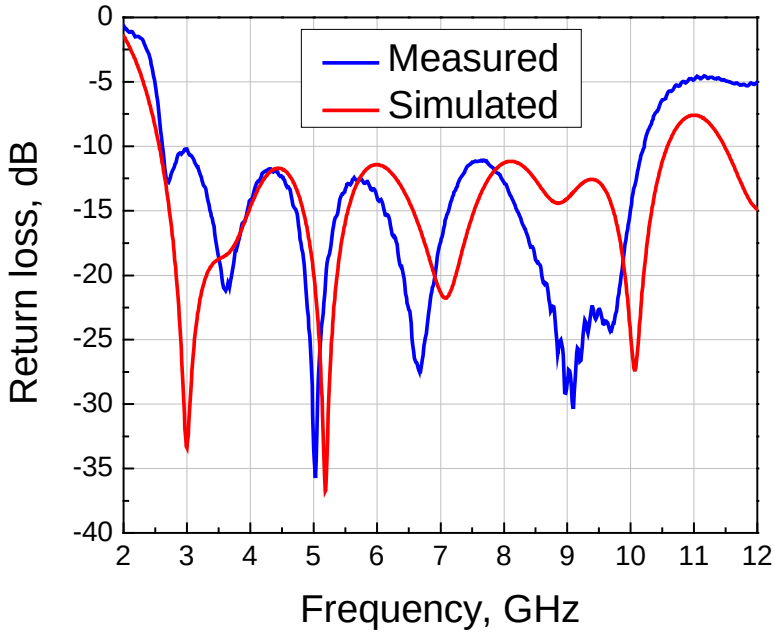


Figure 5.7: Measured and simulated return loss curves of microstrip line fed elliptical slot antenna

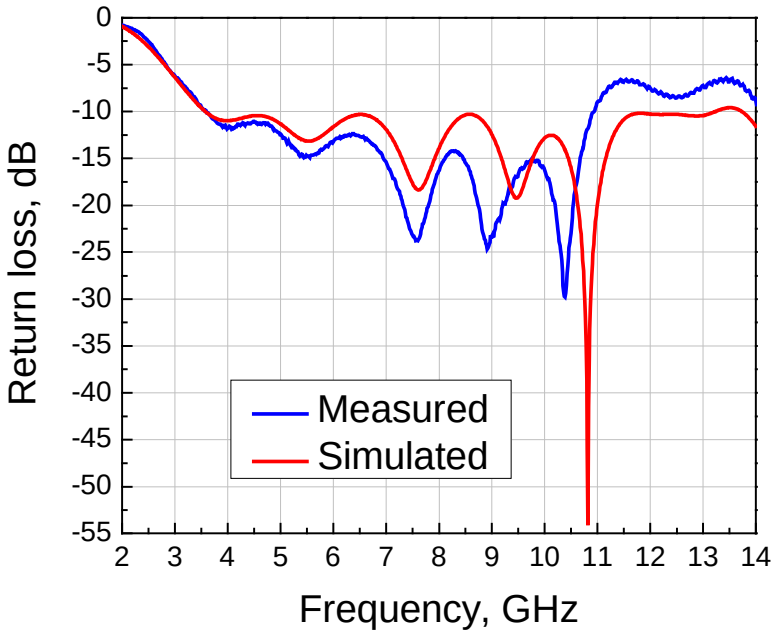


Figure 5.8: Measured and simulated return loss curves of microstrip line fed circular slot antenna

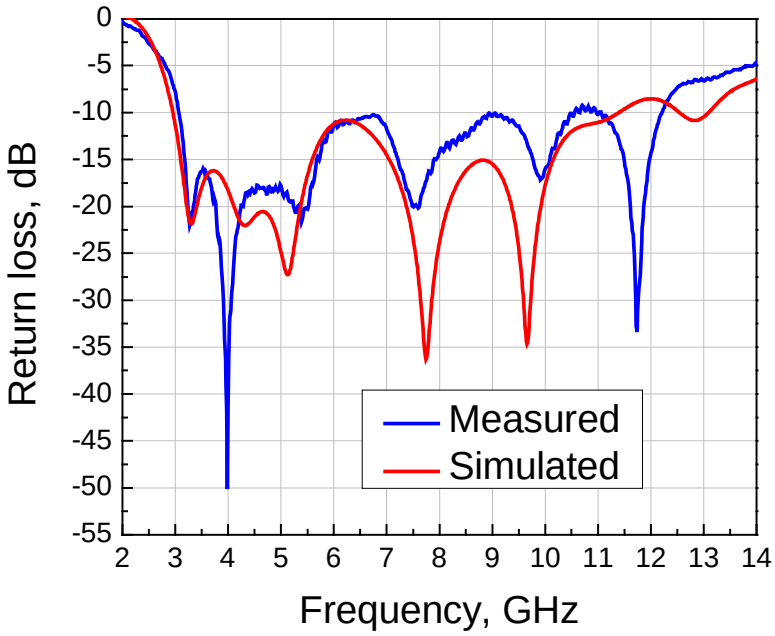


Figure 5.9: Measured and simulated return loss curves of CPW fed elliptical slot antenna

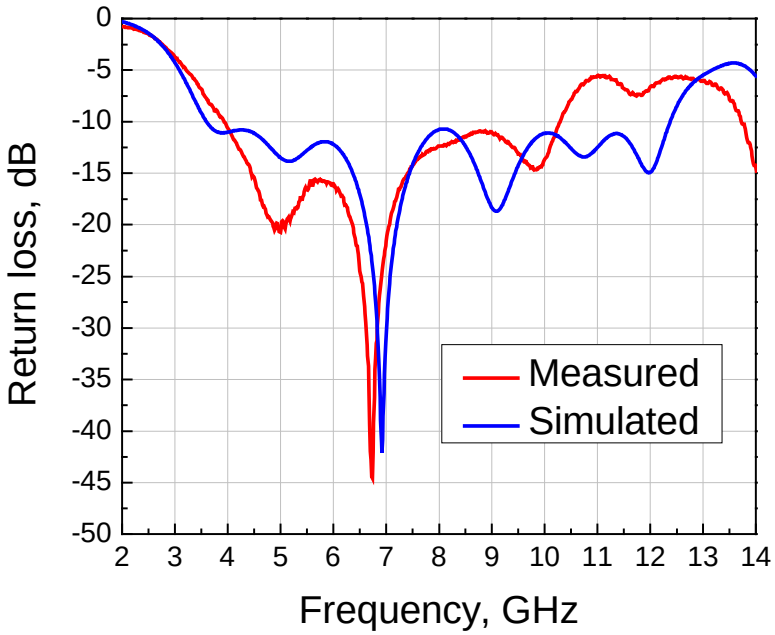


Figure 5.10: Measured and simulated return loss curves of CPW fed circular slot antenna

As shown in Figure 5.7—-Figure 5.10 and Table 5-B, good agreement has been achieved between the measurement and experiment for each of the antennas.

Table 5-B: Measured and simulated -10dB bandwidths of printed elliptical/circular slot antennas

	Microstrip line fed elliptical slot	Microstrip line fed circular slot	CPW fed elliptical slot	CPW fed circular slot
Simulated (GHz)	8 (2.6–10.6)	9.77 (3.45–13.22)	8.4 (3.0–11.4)	8.8 (3.5–12.3)
Measured (GHz)	7.62 (2.6–10.22)	7.44 (3.46–10.9)	7.5 (3.1–10.6)	6.55 (3.75–10.3)

It is also noticed in Figure 5.7—Figure 5.10 that the UWB operation of these slot antennas is all due to the overlapping of the closely spaced resonances over the frequency band. It is interesting to note that the -10 dB bandwidth is always limited at the high frequency end, no matter for a CPW fed or a microstrip line fed slot in either elliptical or circular shape.

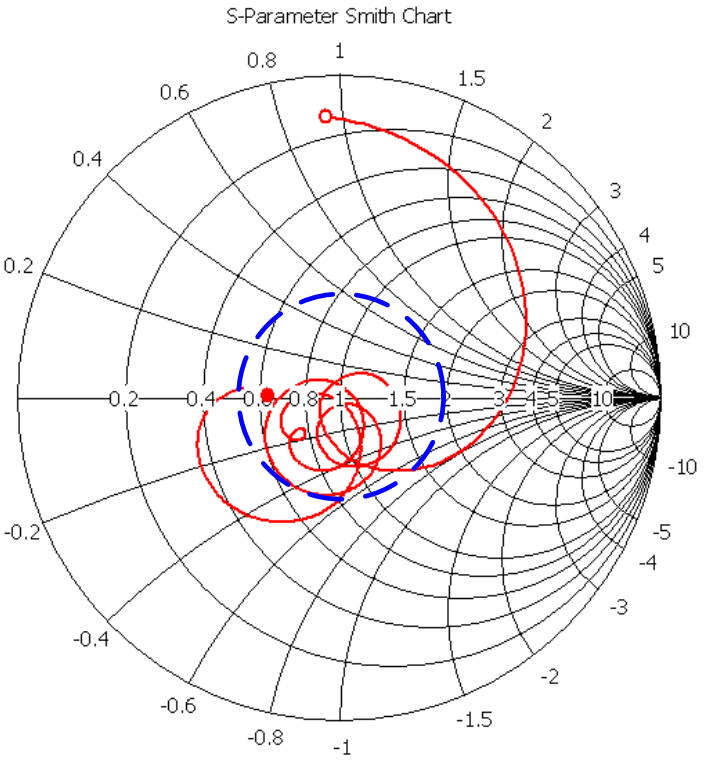


Figure 5.11: Simulated Smith Charts of microstrip line fed elliptical slot antenna (2-12GHz)

The input impedances for the microstrip line fed and CPW fed elliptical slots are plotted on Smith Chart, as shown in Figure 5.11 and Figure 5.12, respectively. It is clearly shown that at the high frequency limit the input impedance spirals out of VSWR=2 circle (the blue line in Figure 5.11 and Figure 5.12), i.e. the impedance matching is getting worse.

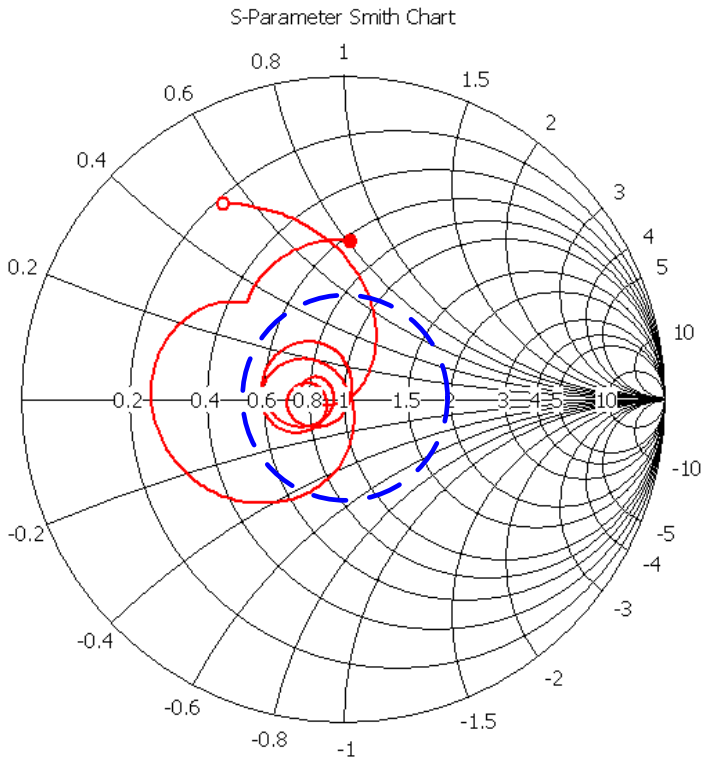
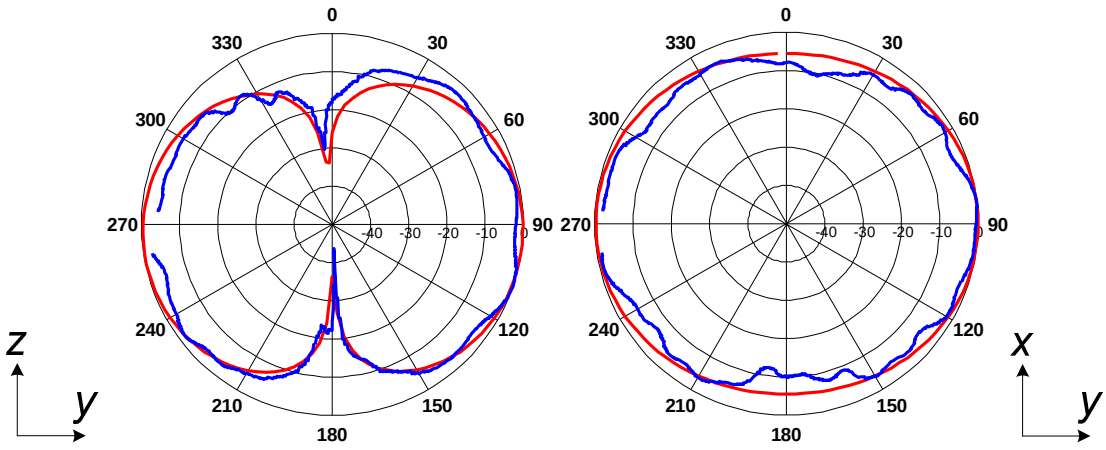


Figure 5.12: Simulated Smith Charts of CPW fed elliptical slot antenna (2-14GHz)

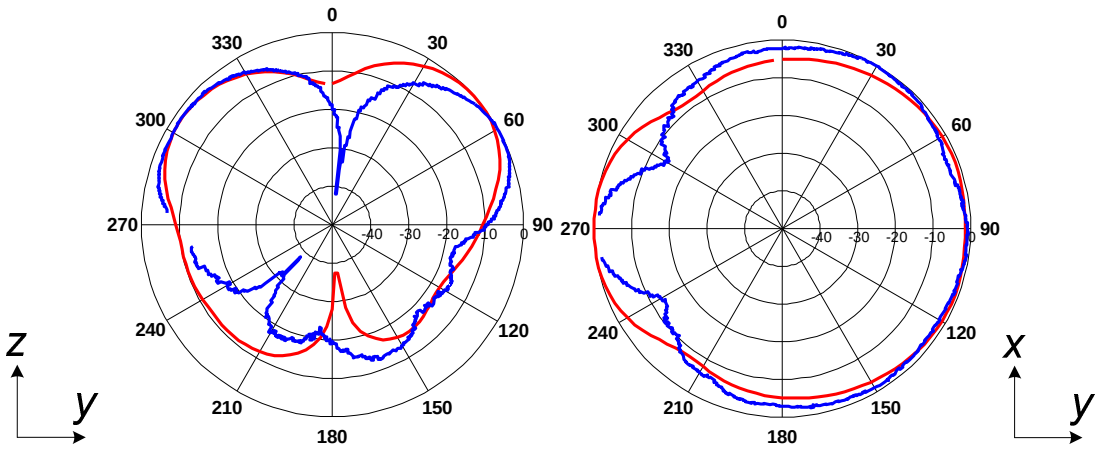
5.3.2 Radiation Patterns

The radiation patterns of the antennas were also measured inside an anechoic chamber. As shown in Figure 5.13 and Figure 5.14, the measured patterns agree well with the simulated ones for both elliptical slot antennas. It is noticed that the printed elliptical/circular slot antennas with different feeding structures can provide similar radiation patterns. The *E*-plane pattern is monopole-like, and the number of lobes rises with the increase of frequency which means the antenna gets more directional. The slight asym-

metry on the E-plane pattern is due to the imperfection of fabrication of the antennas. The H-plane pattern is nearly omni-directional at lower frequency, but becomes more asymmetrical to x-axis at higher frequency. This is due to the tuning stub acting as a radiator itself and its effect becoming more prominent at high frequency.

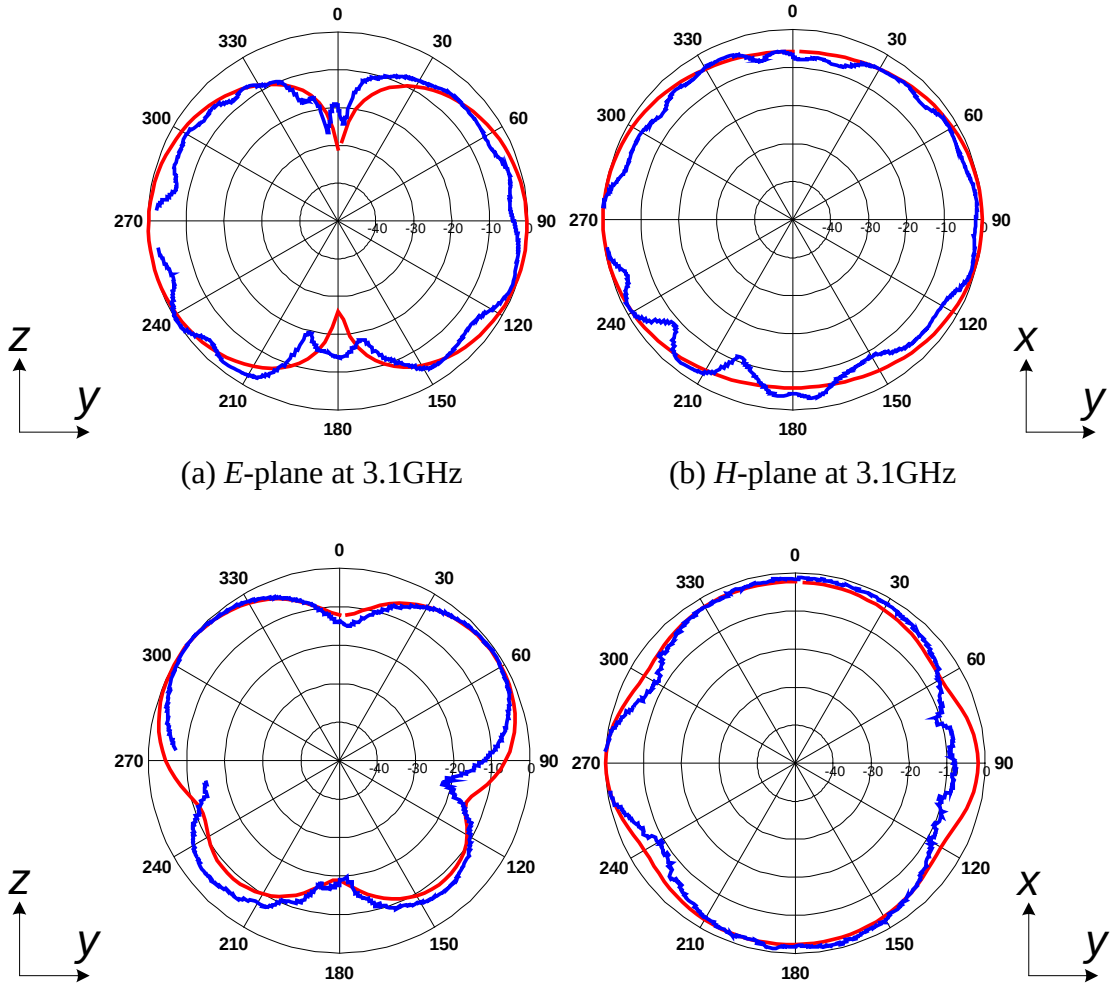


(a) E-plane at 3.1GHz (b) H-plane at 3.1GHz



(c) E-plane at 10GHz (d) H-plane at 10GHz

Figure 5.13: Measured (blue line) and simulated (red line) radiation patterns of microstrip line fed elliptical slot antenna



(a) *E*-plane at 3.1GHz (b) *H*-plane at 3.1GHz

(c) *E*-plane at 10GHz (d) *H*-plane at 10GHz

Figure 5.14: Measured (blue line) and simulated (red line) radiation patterns of CPW fed elliptical slot antenna

5.3.3 Antenna Gain

The measured gains of the four antennas are presented in Figure 5.15. It is seen that the measured gains fluctuate within the range from 2dBi to 7dBi and reach the maximum values at 10GHz for all of the four slot antennas. Generally speaking, the measured gains are similar to those presented in [13–16] over most parts of the bandwidth. However, due to the wider operational bandwidth compared with those in [13–16], the gains have more variations, as shown in Figure 5.15.

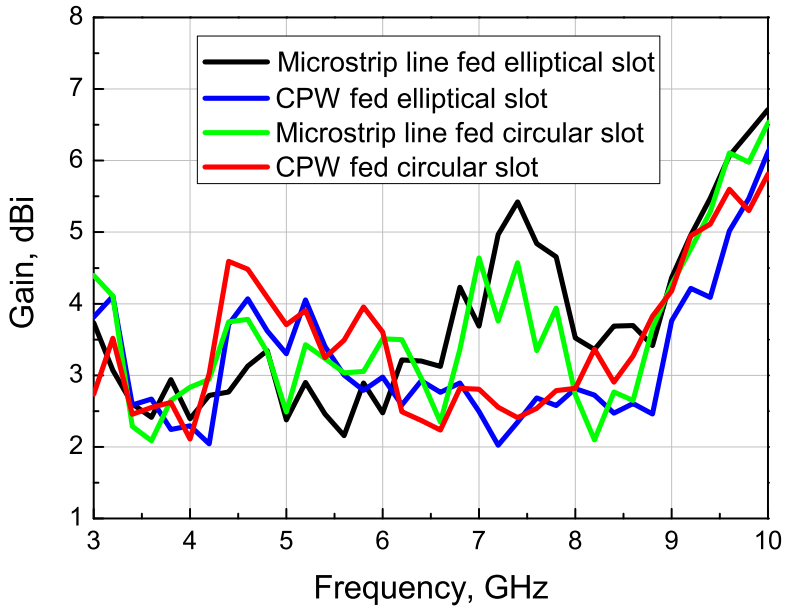
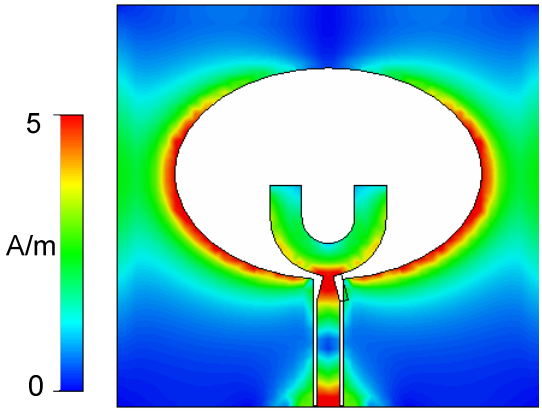


Figure 5.15: The measured gains of the four slot antennas

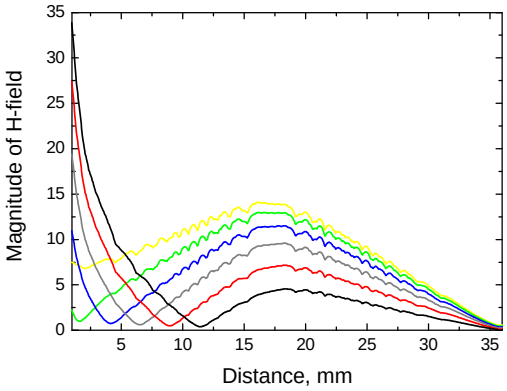
5.3.4 Current Distributions

The simulated current distributions of CPW fed elliptical slot antenna at three frequencies are presented in Figure 5.16, as a typical example. On the ground plane, the current is mainly distributed along the edge of the slot for all of the three different frequencies. The current patterns indicate the existence of different resonance modes, i.e. the first harmonic at 3.3GHz in Figure 5.16(a), the second harmonic around 5GHz in Figure 5.16(b) and the 4th harmonic around 10GHz in Figure 5.16(c). This confirms that the elliptical/circular slot is capable of supporting multiple resonant modes, and the overlapping of these multiple modes leads to the UWB characteristic, as analysed in the disc type of monopoles.

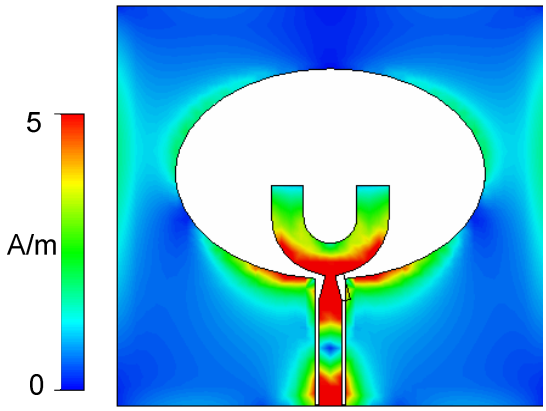
Again, the current variations have been observed in Figures 5.16 (d) - (f). The feature is similar to that of the microstrip line fed disc monopole. The travelling wave at high frequency is not well supported in this enclosed structure. Hence, the impedance matching becomes worse at the high frequency and the -10 dB bandwidth is limited at the high end.



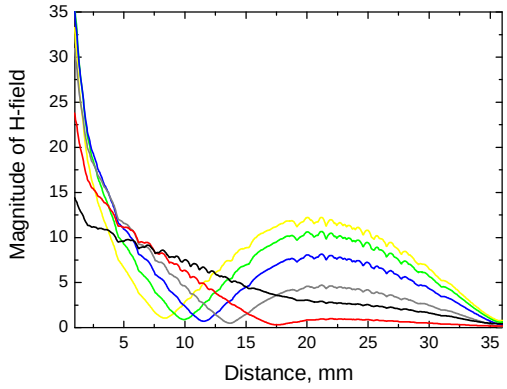
(a) Current distribution at 3.3GHz



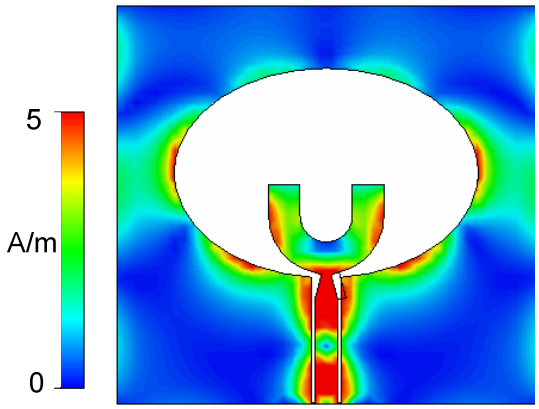
(d) Magnetic field variation at 3.3GHz



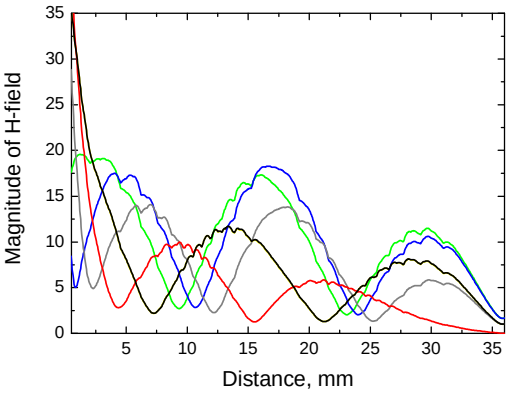
(b) Current distribution at 5GHz



(e) Magnetic field variation at 5GHz



(c)Current distribution at 10GHz



(f) Magnetic field variation at 10GHz

Figure 5.16: Simulated current distributions (a-c) and magnetic field distributions along the edge of the half slot L ($L = 0$ - 36mm: bottom to top) at different phases (d-f) of CPW fed elliptical slot antenna

5.4 Design Considerations

Studies in the previous sections have indicated that the ultra wide bandwidth of the slot antenna results from the overlapping of the multiple resonances introduced by the combination of the elliptical slot and the feeding line with U-shaped tuning stub. Thus, the slot dimension, the distance S and the slant angle θ are the most important design parameters which affect the antenna performance and need to be further investigated.

5.4.1 Dimension of Elliptical Slot

It is noticed that the dimension of the slot antenna is directly related to the lower edge of the impedance bandwidth. In the case of elliptical disc monopoles [17], an empirical formula for estimating the lower edge frequency of the -10dB bandwidth f_l is derived based on the equivalence of a planar configuration to a cylindrical wire, as shown:

$$f_l = \frac{30 \times 0.32}{L + r} \tag{5.1}$$

where f_l in GHz, L and r in cm. L is the disc height, r is equivalent radius of the cylinder given by $2\pi r\ L{=}\pi AB$.

In this study, the elliptical slot can be regarded as an equivalent magnetic surface. Equation 5.1 is modified empirically as:

$$f_l = \frac{30 \times C}{L + r} \tag{5.2}$$

where L, r in cm, and f_l in GHz; $L{=}2A$, $r{=}0.25B$. C is the element factor which equals to 0.32 for elliptical slot and 0.35 for circular slot, respectively. The comparison between the calculated f_l and the measured one for different printed slot antennas are tabulated in Table 5-C. It is shown that the measured f_l matches the calculated one quite well.

Table 5-C: The calculated and measured lower edge of -10dB bandwidth

	Microstrip line fed elliptical slot	Microstrip line fed circular slot	CPW fed elliptical slot	CPW fed circular slot
A(mm)	16	13.3	14.5	13.3
B(mm)	11.5	13.3	10	13.3
Measured f_l (GHz)	2.6	3.46	3.1	3.75
Calculated f_l (GHz)	2.75	3.51	3.17	3.51

5.4.2 Distance *S*

The simulated return loss curves of CPW fed elliptical slot antenna for various S (S=0.15mm, 0.4mm, 0.65mm) with A=14.5mm, B=10mm and θ=15 degrees are illustrated in Figure 5.17. It is seen that the curves for different S have similar shape and variation trend, but the optimal distance is S=0.4mm because it can provide the widest -10dB bandwidth.

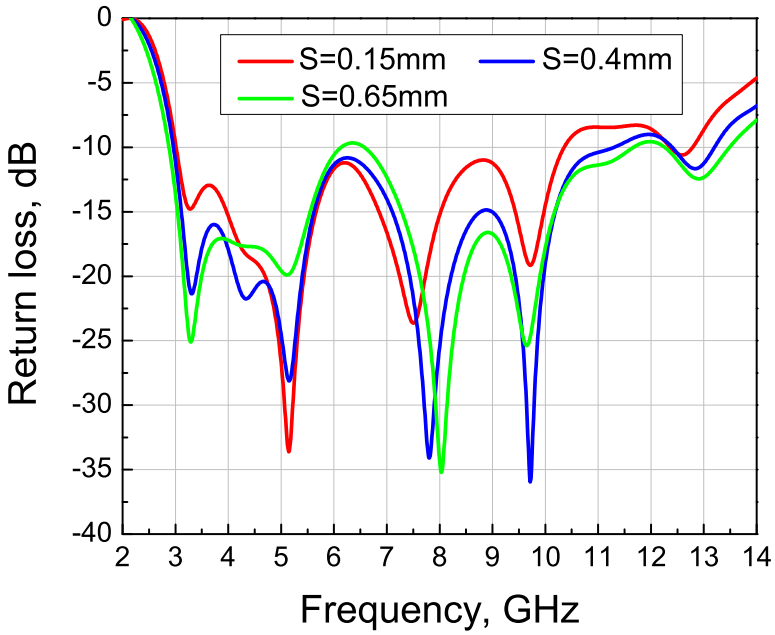


Figure 5.17: Simulated return loss curves of CPW fed elliptical slot antenna for different S with A=14.5mm, B=10mm and θ=15 degrees

5.4.3 Slant Angle θ

In Figure 5.18, the return loss curves of CPW fed elliptical slot antenna for different slant angles with A=14.5mm, B=10mm and S=0.4mm are plotted. It is observed that the lower edge of the -10dB bandwidth is independent of θ, but the upper edge is very sensitive to the variation of θ. The optimal slant angle is found to be at θ=15 degrees, with a bandwidth of 8.4GHz (from 3.0GHz to 11.4GHz).

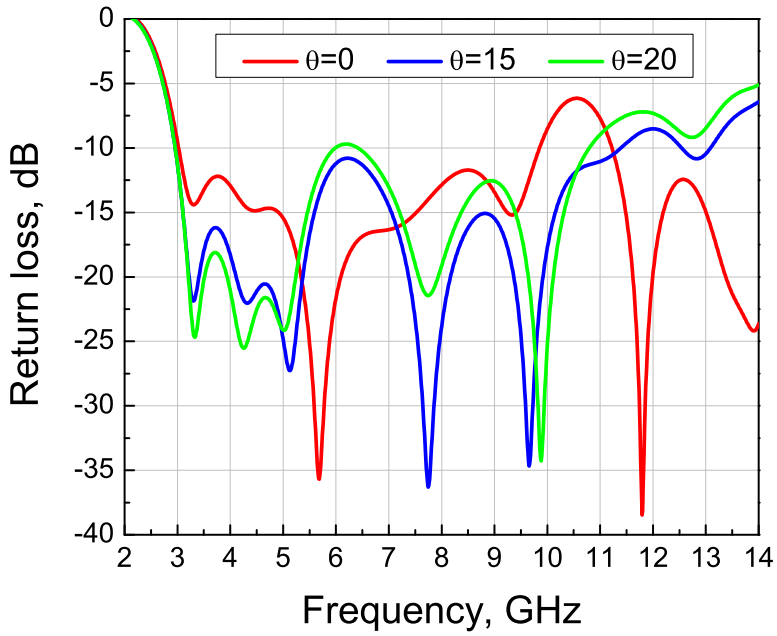


Figure 5.18: Simulated return loss curves of CPW fed elliptical slot antenna for different θ with A=14.5mm, B=10mm and S=0.4mm

Figure 5.17 and Figure 5.18 also confirm that the first resonant frequency (hence the lower edge of the -10dB bandwidth f_l is mainly dependent on the slot dimension. A variation of either S or θ does not cause any shift of f_l once both A and B are fixed.

It is clear that tuning of the design parameters of distance S and slant angle θ can affect the wave transmission at the feed point to the slot. A proper adjustment of these parameters can either enhance or shift the antenna resonances to overlap more across the spectrum, hence, a more wide operating bandwidth.

5.5 Summary

The frequency domain performances of printed elliptical/circular slot antennas have been addressed in this chapter.

The frequency bandwidth can be widened by using tapered microstrip or CPW feeding line with U-shaped tuning stub to enhance the coupling between the slot and the feed line. Thus, the slot dimension, the distance S and the slant angle θ are the most important design parameters that determine the antenna performance.

It has been shown that ultra wideband characteristic was achieved due to the overlapping of the closely spaced resonances over the frequency band. Similar to UWB disc monopole antennas, printed elliptical/circular slot antennas operate in a hybrid mode of standing and travelling waves. However, the travelling wave at high frequency is not well supported in the enclosed structure, so the frequency bandwidth is limited at the high end.

Experimental results have confirmed UWB characteristics of the proposed antennas as well as nearly omni-directional radiation properties over a majority fraction of the bandwidth. These features and their small sizes make them attractive for future UWB applications

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Chapter 6

Time Domain Characteristics of UWB Antennas

In Chapter 4 and Chapter 5, UWB disc monopoles and slot antennas have been investigated with an emphasis on their frequency domain performances. However, because UWB systems directly transmit narrow pulses rather than employing a continuous wave carrier to convey information, the effect of the antenna on the transmitted pulse becomes a crucial issue. In such a system, the antenna behaves like a bandpass filter and reshapes the spectra of the pulses [1]. The signal waveforms arriving at the receiver usually do not resemble the waveforms of the source pulses at the transmitter. The antenna, hence, should be designed with care to avoid undesired distortions. In other words, a good time domain performance is a primary requirement of UWB antenna, as mentioned in Chapter 3.

In addition, due to the potential interference with other wireless systems, the FCC regulated the emission mask which defines the maximum allowable radiated power for UWB systems. Thus, the source pulses also play an important role on the performance of the UWB system. Properly selecting the source pulse can maximize the radiated power within the UWB band and meet the required emission limit [2].

In this Chapter, time domain characteristics of UWB disc monopoles and slot antennas will be analysed. Firstly, the performances of these UWB antennas are evaluated from a system point of view. Secondly, the transmit and receive responses of UWB antennas are investigated. Thirdly, the radiated power spectral density of the antenna is studied in comparison with the FCC emission mask. And finally, a convolution approach is used to obtain the measured received pulses and the transmitting/receiving antenna systems are assessed by the pulse fidelity.

6.1 Performances of UWB Antenna System

6.1.1 Description of UWB Antenna System

Consider a typical UWB antenna system as a two-port network [2], as shown in Figure 6.1.

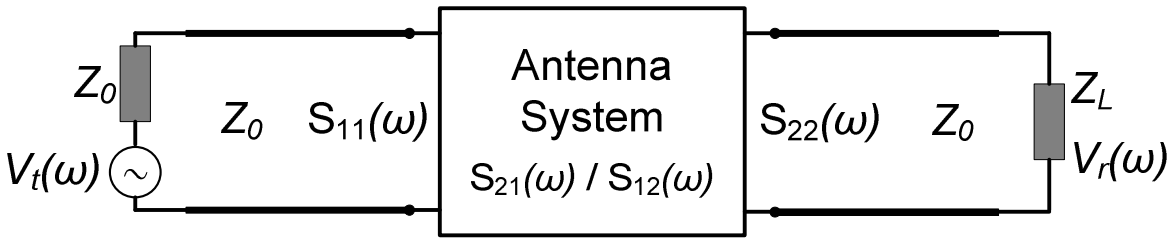


Figure 6.1: Configuration of UWB antenna system

The Friis Transmission Equation relates the power received to the power transmitted between the two antennas [3], as given in Equation 6.1.

$$\frac{P_r}{P_t} = (1 - |\Gamma_t|^2)(1 - |\Gamma_r|^2) G_r G_t |\hat{\rho}_t \cdot \hat{\rho}_r|^2 (\frac{\lambda}{4\pi d})^2 \tag{6.1}$$

where P_t, P_r: time average input power of the transmitting antenna and time average output power of the receiving antenna; Γ_t, Γ_r: return loss at the input of the transmitting antenna and the output of the receiving antenna; G_t, G_r: gain of the transmitting antenna and the receiving antenna; $|\hat{\rho}_t \cdot \hat{\rho}_r|^2$: polarization matching factor between the

transmitting and receiving antennas; λ: operating wavelength; and d: distance between the two antennas.

Since UWB systems operate over a large frequency range, all of the parameters in Equation 6.1 are frequency-dependent. The Friis Equation can be rewritten as follows:

$$\frac{P_r(\omega)}{P_t(\omega)} = (1 - |\Gamma_t(\omega)|^2)(1 - |\Gamma_r(\omega)|^2)G_r(\omega)G_t(\omega)|\hat{\rho}_t(\omega) \cdot \hat{\rho}_r(\omega)|^2(\frac{\lambda}{4\pi d})^2 \tag{6.2}$$

For reflection and polarization-matched antennas aligned for maximum directional radiation and reception, Equation 6.2 reduces to:

$$\frac{P_r(\omega)}{P_t(\omega)} = (\frac{\lambda}{4\pi d})^2 G_r(\omega)G_t(\omega) \tag{6.3}$$

The system transfer function, i.e. S_{21}, is defined as the ratio of the output signal over the source signal. According to Figure 6.1, $[V_t(\omega)/2]^2 = P_t(\omega)Z_0$, ${V_r}^2(\omega)/2 = P_r Z_L$, so S_{21} is given by:

$$\begin{aligned} S_{21}(\omega) &= \frac{V_r(\omega)}{V_t(\omega)} = |\sqrt{\frac{P_r(\omega)}{P_t(\omega)}\frac{Z_L}{4Z_0}}|e^{-j\phi(\omega)} = |S_{21}(\omega)|e^{-j\phi(\omega)} \\ \phi(\omega) &= \phi_t(\omega) + \phi_r(\omega) + \frac{\omega d}{c} \end{aligned} \tag{6.4}$$

in which c is the velocity of light, $\phi_t(\omega)$ and $\phi_r(\omega)$ are the phase variations related to the transmitting and receiving antennas, respectively.

It is manifest in Equation 6.2 and Equation 6.4 that the transfer function is determined by the characteristics of both transmitting and receiving antennas, such as impedance matching, gain, polarization matching and the distance between the antennas. S_{21} also integrates the important system performances, such as path loss and phase delay. Therefore, it can be used to evaluate the performance of the antenna systems.

Furthermore, the system response can be completely determined when the transfer function is known. To minimize the distortions in the received signal waveform, the

transfer function is required to have flat magnitude and linear phase response over the operational band. Usually, the group delay is used to evaluate the phase response of the transfer function because it is defined as the rate of change of the total phase shift with respect to angular frequency. Ideally, when the phase response is strictly linear, the group delay is constant.

6.1.2 Measured Results of UWB Antenna System

The measurements of the antenna system were carried out inside an anechoic chamber by using an HP8720ES vector network analyzer. The antenna system is comprised of two identical UWB antennas, as shown in Figure 6.2. Since UWB technology will be mainly employed in WPAN systems, in the measurements, the transmitter and receiver are vertically placed with a separation of d=1.2m. Further, to investigate the system performances in different directions, the two antennas are measured in two different orientations, namely face to face and side by side, respectively, as shown in Figure 6.3.

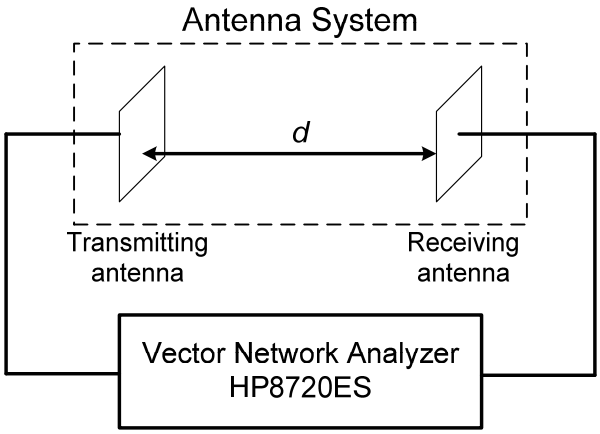


Figure 6.2: System set-up

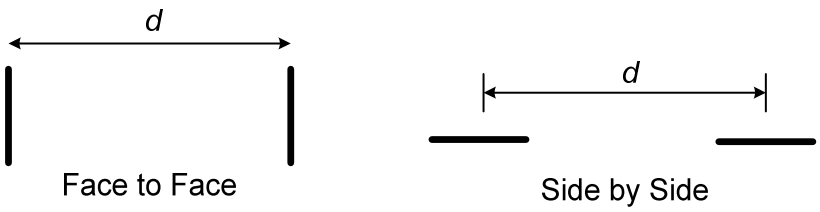


Figure 6.3: Antenna orientation (top view)

For UWB disc monopoles, the two discs are facing to each other in "face to face" orientation; in "side by side" case, the two monopoles are aligned along the same line with the two discs pointing to the same direction. For UWB slot antennas, the two slots are facing to each other in "face to face" orientation and pointing to the same direction in "side by side" case.

6.1.2.1 Vertical Disc Monopole Antennas

The measured transfer functions and group delays of vertical disc monopole pair are plotted in Figure 6.4, 6.6 and 6.7.

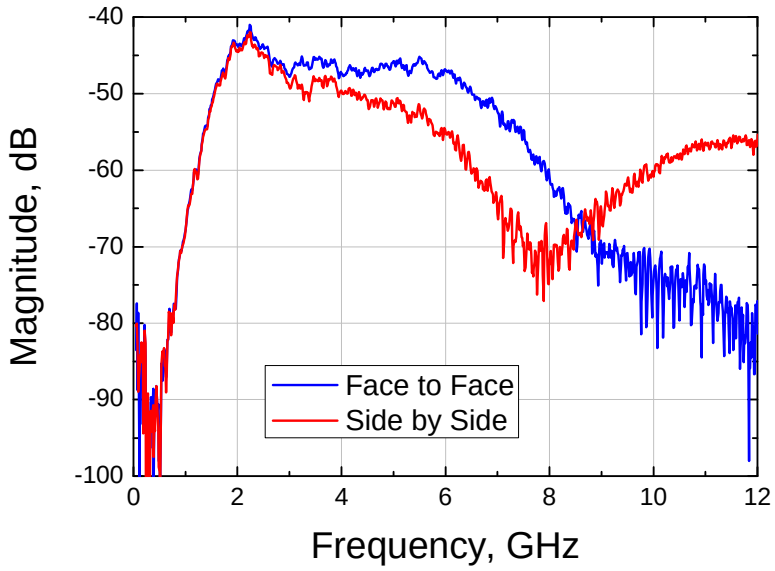


Figure 6.4: Magnitude of measured transfer function of vertical disc monopole pair

It is shown in Figure 6.4 that the magnitude curves of the transfer functions correspond well to the antenna gain (as illustrated in Figure 6.5). At lower frequency range (less than 4GHz), the y-direction (the direction which is parallel to the disc radiator) gain is quite close to that of the x-direction (the direction which is normal to the disc radiator), as a result, the two magnitude curves for both scenarios, i.e. face to face and side by side, are almost identical; Within the frequency range from 4GHz to 8GHz, the y-direction gain becomes smaller than that of the x-direction, so the magnitude

curve of face to face case is slightly higher than that of side by side case; However, at higher frequencies (more than 8GHz), the x-direction gain decreases substantially with the increase of frequency, at the same time, the y-direction gain rises remarkably. Thus, the magnitude of side by side case becomes bigger than the face to face case.

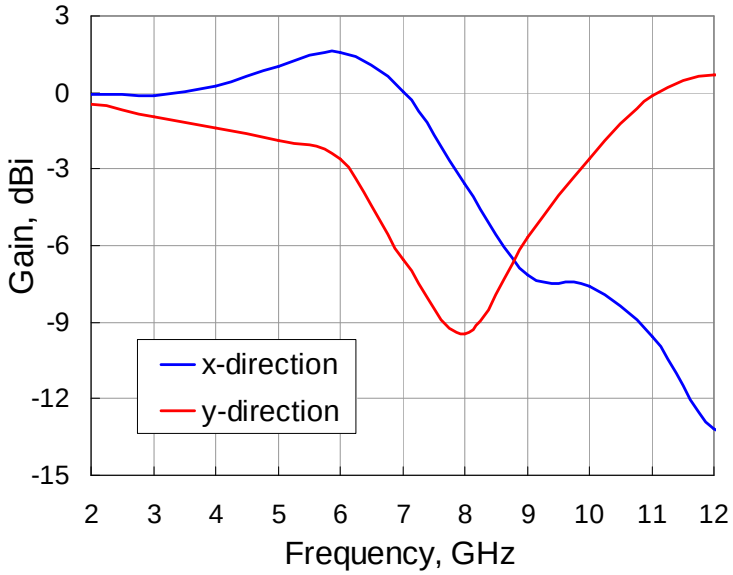


Figure 6.5: Simulated gain of vertical disc monopole in the x-direction and the y-direction

It is also noticed that the operating band (-10dB below the peak) of transfer function for the face to face case spans from 1.5GHz to 7GHz. Outside this band, the magnitude curve decays very sharply. This means the signal within the frequency range of 1.5GHz to 7GHz will be almost equally received. For frequencies higher than 7GHz the frequency components would be further attenuated, which will result in distortion of the total received signal. The operating band of the side by side case is narrower than the face to face case, from 1.5GHz to 5.5GHz. And the magnitude curve undergoes a deep null at around 8GHz.

As shown in Figure 6.6, the phase curves of the transfer functions for both face to face and side by side cases are quite similar. They are nearly linear over an ultra wide frequency range from about 1GHz to 7GHz. The non-linear properties outside this range will definitely cause distortions of the received signal waveforms.

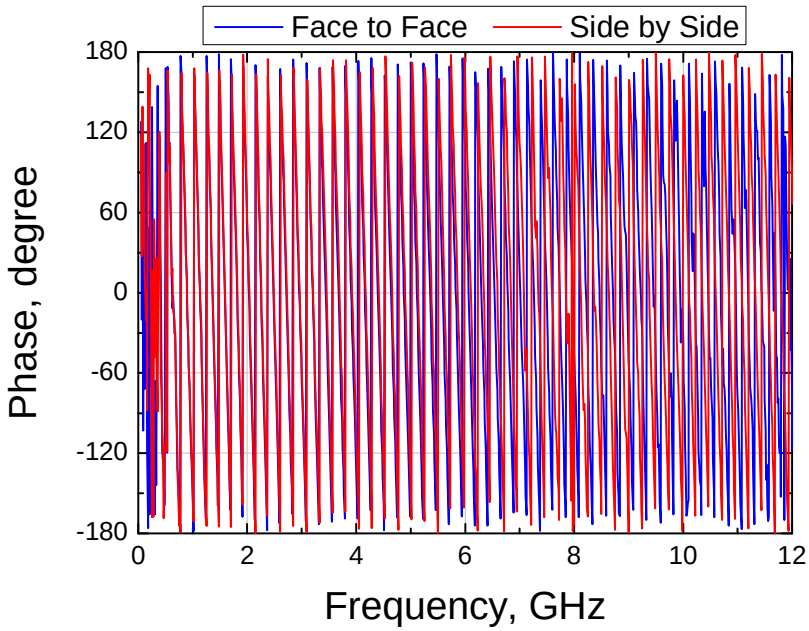


Figure 6.6: Phase of measured transfer function of vertical disc monopole pair

The group delays of the two cases, as given in Figure 6.7, are quite stable with variation less than 3ns within the frequency range from 1GHz to 7GHz which corresponds well to the phase curves of the transfer functions.

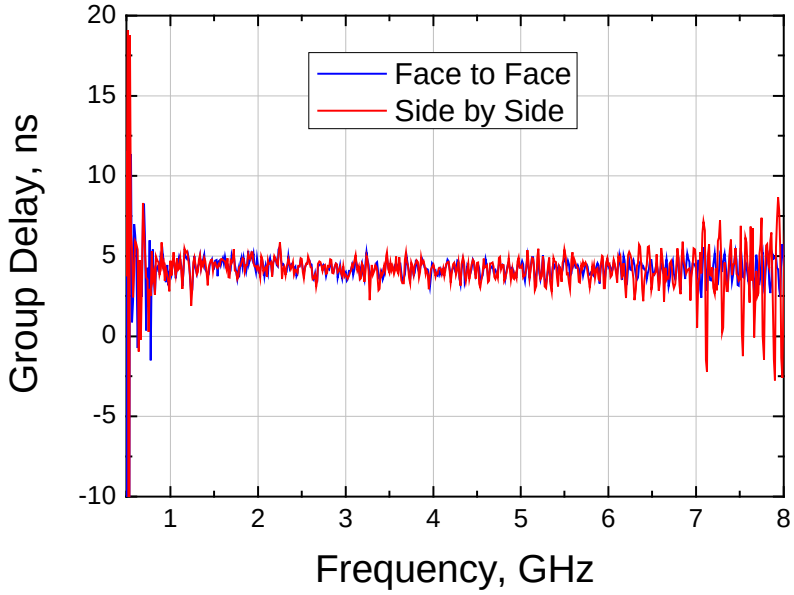


Figure 6.7: Group delay of measured transfer function of vertical disc monopole pair

6.1.2.2 CPW Fed Disc Monopole Antennas

Figure 6.8-6.10 present the measured transfer functions and group delays of CPW fed disc monopole pair.

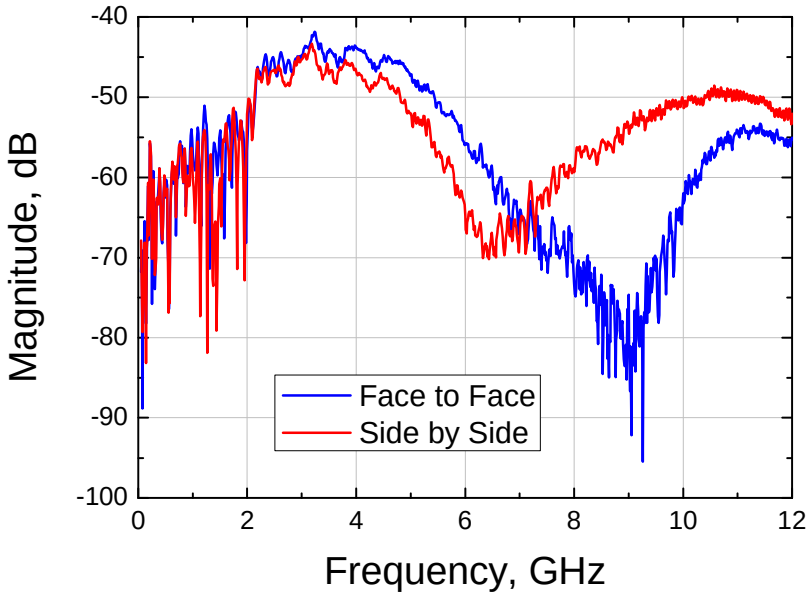


Figure 6.8: Magnitude of measured transfer function of CPW fed disc monopole pair

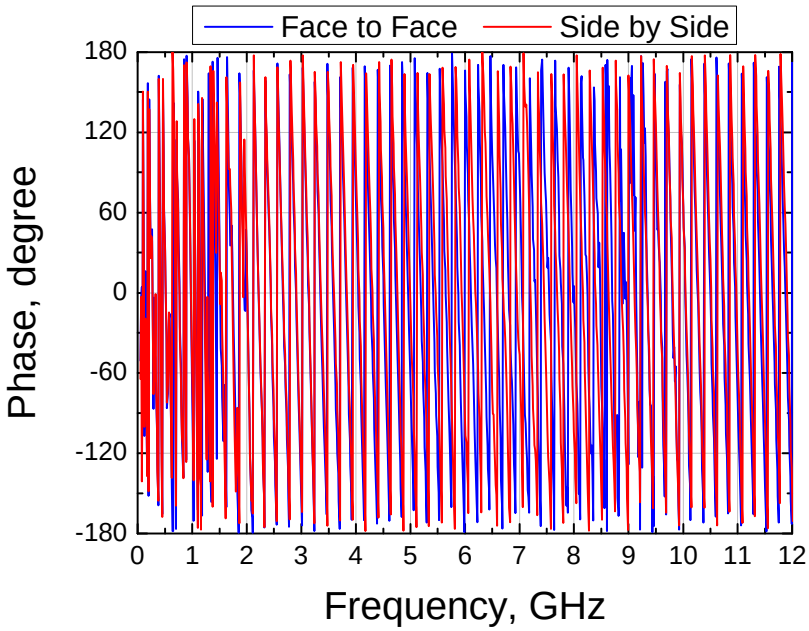


Figure 6.9: Phase of measured transfer function of CPW fed disc monopole pair

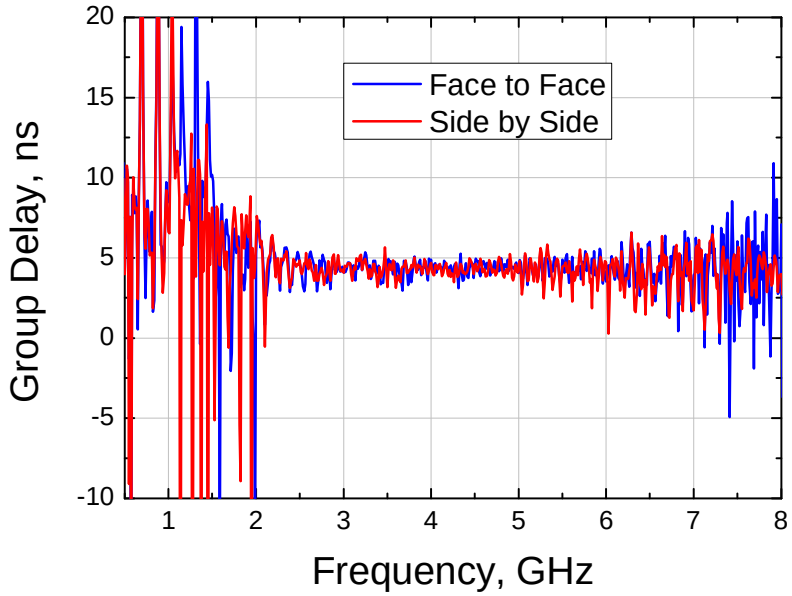


Figure 6.10: Group delay of measured transfer function of CPW fed disc monopole pair

It is noticed in Figure 6.8 that the operating bands are from 2GHz to 5.8GHz for face to face case and 2GHz to 5.2GHz for side by side case, respectively. Besides, linear phase responses and less variable group delays are also observed over the frequency range from 2GHz to 6GHz, as shown in Figure 6.9 and 6.10.

6.1.2.3 Microstrip Line Fed Circular Slot Antennas

The measured transfer functions of microstrip line fed circular slot antenna pair are illustrated in Figure 6.11–6.13.

As shown in Figure 6.11, the magnitude curves of face to face and side by side cases are quite similar over the entire frequency range from 0 to 12GHz, except that the side by side curve has more ripples due to the noise. The operating band ranges from 2.68GHz to 6.93GHz in face to face case and from 2.85GHz to 6.35GHz in side by side case.

For face to face case, linear phase response with group delay variation less than 3ns is obtained within the frequency range from 2GHz to 8GHz; for side by side case, the

variation of group delay is more notable within the operating band and it becomes bigger than 20ns at 7GHz, as shown in Figure 6.12 and 6.13.

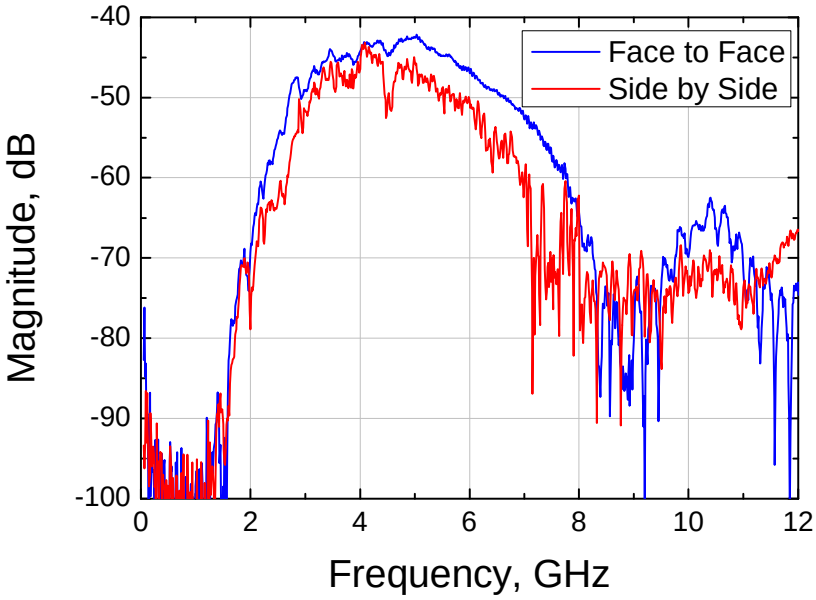


Figure 6.11: Magnitude of measured transfer function of Microstrip line fed circular slot antenna pair

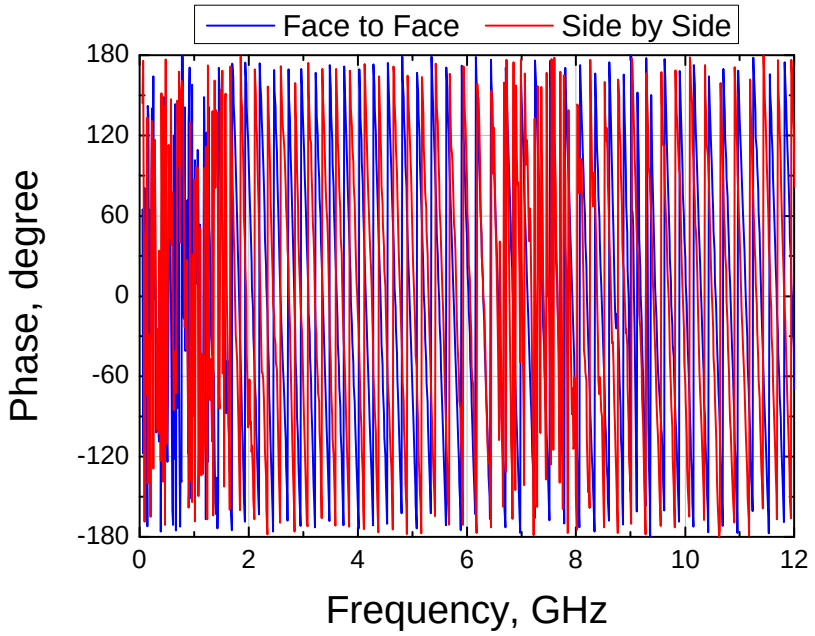


Figure 6.12: Phase of measured transfer function of Microstrip line fed circular slot antenna pair

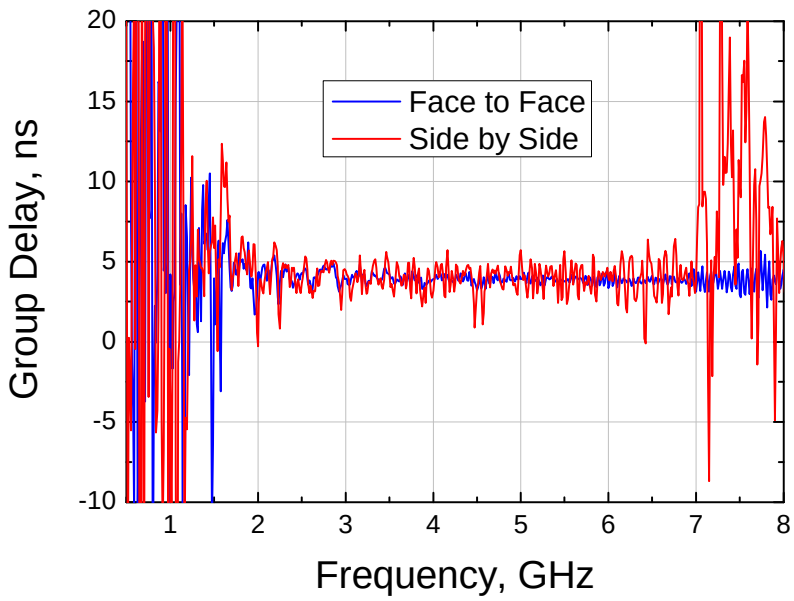


Figure 6.13: Group delay of measured transfer function of Microstrip line fed circular slot antenna pair

6.2 Impulse Responses of UWB Antennas

In Section 6.1, the performances of UWB antennas are evaluated from a system point of view. The system transfer function S_{21} has taken into account the effects of both transmitting and receiving antennas. This section will investigate the impulse responses of UWB antennas when they are used as transmitters and receivers, respectively.

6.2.1 Transmitting and Receiving Responses

Now consider a practical antenna whose performances are frequency-dependent. The transmitting and receiving characteristics are represented by $A(\omega)$ $(a(t))$ and $H(\omega)$ $(h(t))$, respectively, as shown in Figure 6.14. Furthermore, using the principle of reciprocity, it can be shown that the transmitting and receiving characteristics are related by [4, 5]:

$$A(\omega) \propto j\omega H(\omega) \tag{6.5}$$

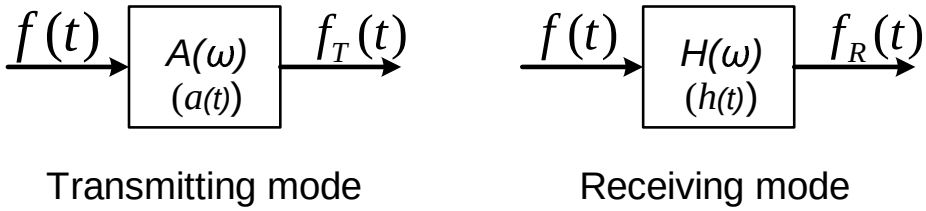


Figure 6.14: Antenna operating in transmitting and receiving modes

For the same stimulus $f(t)$ $(F(\omega))$, the transmitting response $f_T(t)$ and receiving response $f_R(t)$ can be expressed as:

$$f_T(t) \triangleq \int F(\omega)A(\omega)e^{j\omega t}d\omega \tag{6.6}$$

$$f_R(t) \triangleq \int F(\omega)H(\omega)e^{j\omega t}d\omega \tag{6.7}$$

Using Equation 6.5, Equation 6.6 can be rewritten as:

$$f_T(t) \propto \int F(\omega)j\omega H(\omega)e^{j\omega t}d\omega = \frac{\partial}{\partial t}f_R(t) \tag{6.8}$$

Equation 6.8 indicates that the transmitting response is proportional to the temporal derivative of the receiving response. Consequently, $f_T(t)$ and $f_R(t)$ will exhibit different shapes to the same stimulus $f(t)$ $(F(\omega))$.

If the bandwidth $B(\Omega, \omega_c)$ of the stimulus $F(\omega)$ is defined as [4]:

$$B(\Omega, \omega_c) \triangleq \{\omega : ||\omega| - \omega_c| \leqq \Omega/2\}$$

where $\omega_c, \Omega > 0$, ω_c is the center frequency of the bandwidth.

Then Equation 6.8 may be modified as:

$$f_T(t) \propto \frac{\partial}{\partial t}f_R(t) = \int F(\omega)j\omega H(\omega)e^{j\omega t}d\omega = \int |\omega|F(\omega)H(\omega)e^{j\omega(t+\frac{\pi}{2|\omega|})}d\omega \tag{6.9}$$

In narrowband case, $\Omega \ll \omega_c$, $|\omega| \approx \omega_c$, hence Equation 6.9 can be approximated to:

$$f_T(t) \propto \frac{\partial}{\partial t} f_R(t) \approx \omega_c \int F(\omega) H(\omega) e^{j\omega(t+\frac{\pi}{2\omega_c})} d\omega \propto f_R(t + \frac{\pi}{2\omega_c}) \tag{6.10}$$

In the extreme case of a CW signal $f_R(t) = \cos(\omega_c + \phi)$,

$$\frac{\partial}{\partial t} f_R(t) \propto f_R(t + \frac{\pi}{2\omega_c}) \tag{6.11}$$

which is not an approximation, but exact.

Therefore, the antenna transmitting and receiving responses are approximately equal up to scaling and shifting when the signal bandwidth is sufficiently narrow.

For ultra wideband case, the approximation ($\Omega \ll \omega_c$) leading to Equation 6.10 is not valid. The transmitting response is proportional to the temporal derivative of the receiving response, as shown in Equation 6.8. But this derivative relationship can not be represented by scaling and shifting.

6.2.2 Transmitting and Receiving Responses of UWB Antennas

The transmitting and receiving responses of UWB antennas can be obtained based on the frequency domain measurement results.

We still consider the antenna system given in Figure 6.2 and 6.3. Suppose the two antennas are placed in ideal free space, then the system transfer function $S_{21}(\omega)$ with face to face orientation can be expressed as [4]:

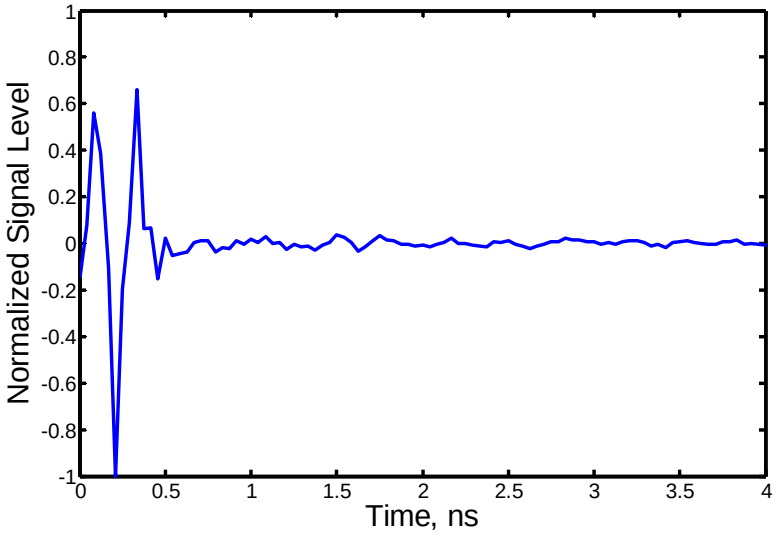
$$S_{21}(\omega) = A(\omega) H(\omega) \frac{e^{-jkd}}{d} \tag{6.12}$$

where $A(\omega)$ and $H(\omega)$ denote the transmitting and receiving characteristics of the antenna, respectively; d is the spacing between the two antennas.

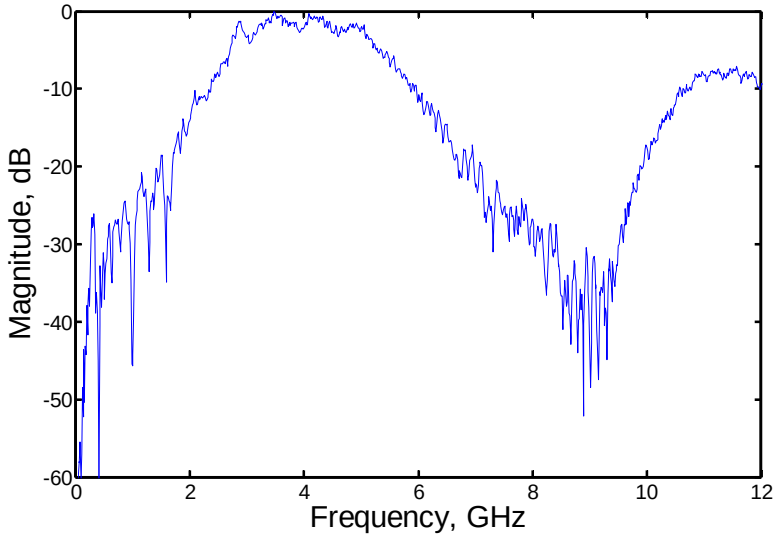
Using the derivative relationship between $A(\omega)$ and $H(\omega)$, Equation 6.12 may be modified as:

$$S_{21}(\omega) = A(\omega)H(\omega)\frac{e^{-jkd}}{d} \propto \frac{1}{j\omega}A(\omega)A(\omega)\frac{e^{-jkd}}{d} \propto j\omega H(\omega)H(\omega)\frac{e^{-jkd}}{d} \tag{6.13}$$

Hence, the transmitting characteristic $A(\omega)$ $(a(t))$ and receiving characteristic $H(\omega)$ $(h(t))$ can be calculated when the measured system transfer function $S_{21}(\omega)$ is available.



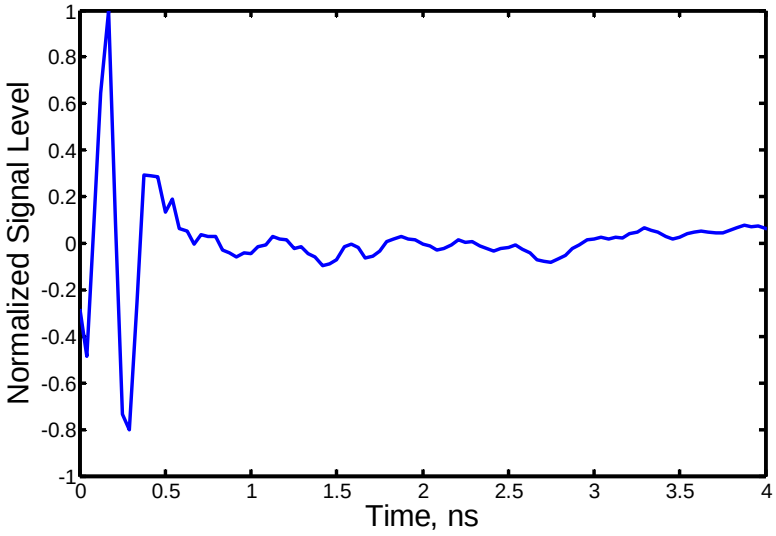
(a) $a(t)$



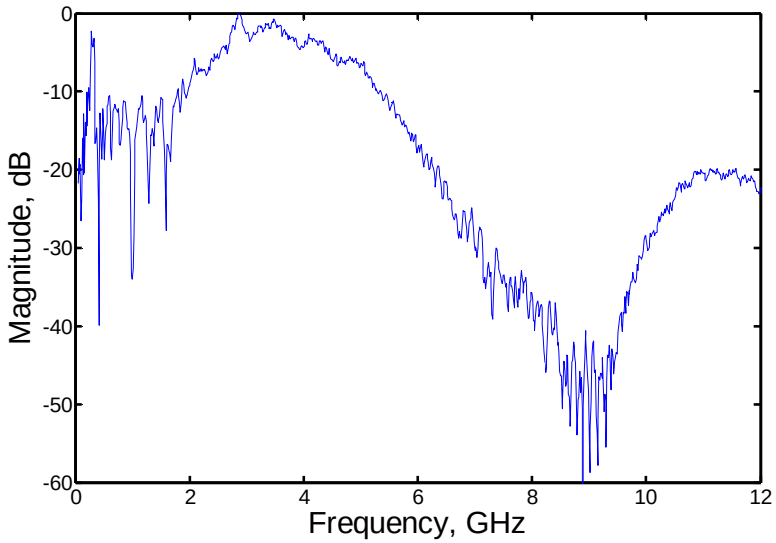
(b) $A(\omega)$

Figure 6.15: Transmitting characteristic of CPW fed disc monopole

The transmitting and receiving characteristics of CPW fed disc monopole are illustrated in Figure 6.15 and Figure 6.16, respectively. The curves have already been normalized to their respective maximum value. The results confirm the disc monopole exhibits different characteristics when operating in transmitting and receiving modes.



(a) $h(t)$



(b) $H(\omega)$

Figure 6.16: Receiving characteristic of CPW fed disc monopole

In Figure 6.15 (b) and Figure 6.16 (b), it is also observed that the CPW disc monopole acts as a band-pass filter with pass band from about 2GHz to 6GHz in transmitting mode and from about 2GHz to 5.4GHz in receiving mode.

If a Gaussian pulse with pulse parameter of a=45ps, as given in Equation 6.14 and Figure 6.17, is used as the stimulus, the resultant transmitting and receiving responses are illustrated in Figure 6.18.

$$f(t) = e^{-(\frac{t-1}{a})^2} \tag{6.14}$$

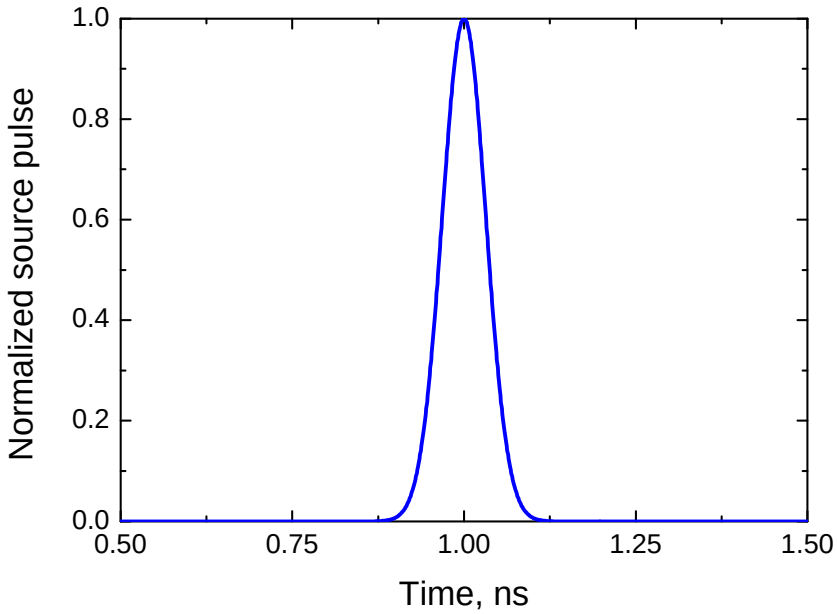


Figure 6.17: Gaussian pulse with a=45ps

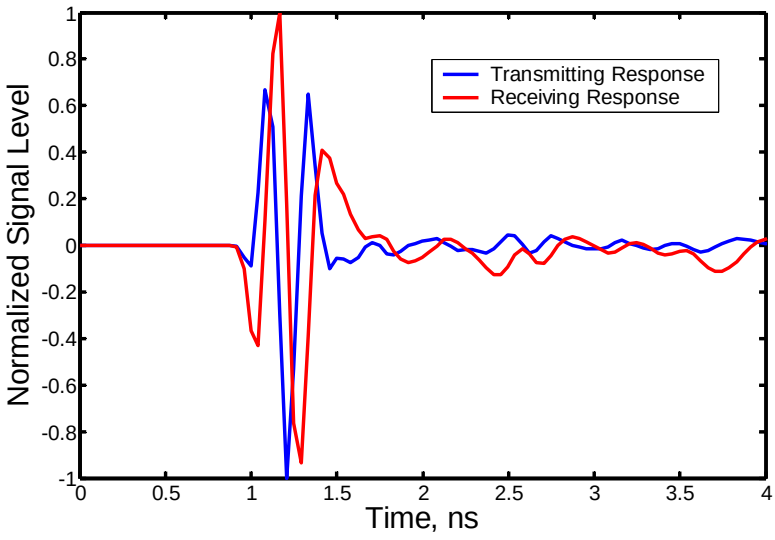


Figure 6.18: Transmitting and receiving responses of CPW fed disc monopole to Gaussian pulse with a=45ps

It is evident in Figure 6.18 that the two responses feature different waveforms.

Fundamentally, this is due to the derivative relationship between the transmitting and receiving characteristics of the antenna.

6.3 Radiated Power Spectral Density

As mentioned in Chapter 2, UWB systems may cause interferences to other wireless systems since they operate over a large frequency range, which covers many bands being used. Thus, the emission limit is a crucial consideration for the design of both source pulses and UWB antennas.

6.3.1 Design of Source Pulses

The first order Rayleigh pulse, as given in Equation 6.15, is widely used as the source signal to drive the transmitter in a UWB system.

$$f(t) = \frac{-2(t-1)}{a^2} e^{-(\frac{t-1}{a})^2} \tag{6.15}$$

where the pulse parameter a stands for the characteristic time. Large a corresponds to wide waveform in the time domain but narrow bandwidth in frequency domain. Figure 6.19 presents the normalized source pulses with three different pulse parameters. Their respective power spectral densities (PSD) against FCC mask are given in Figure 6.20.

As shown in Figure 6.20, the peak value position of the PSD as well as the 10dB bandwidth, i.e. the frequency band bounded by the points that are 10dB below the highest emission, increases with the decrease of a. When a=30ps, the PSD curve peaks at 7.7GHz, and its 10dB bandwidth ranges from 1.5GHz to 16.6GHz; when a rises to 45ps, the peak value of PSD occurs at 5GHz, and the 10dB bandwidth is reduced to 10.1GHz, from 1GHz to 11.1GHz; with the further increase of a to 80ps, the 10dB bandwidth only spans from 0.55GHz to 6.2GHz, with the maximum PSD at 2.8GHz.

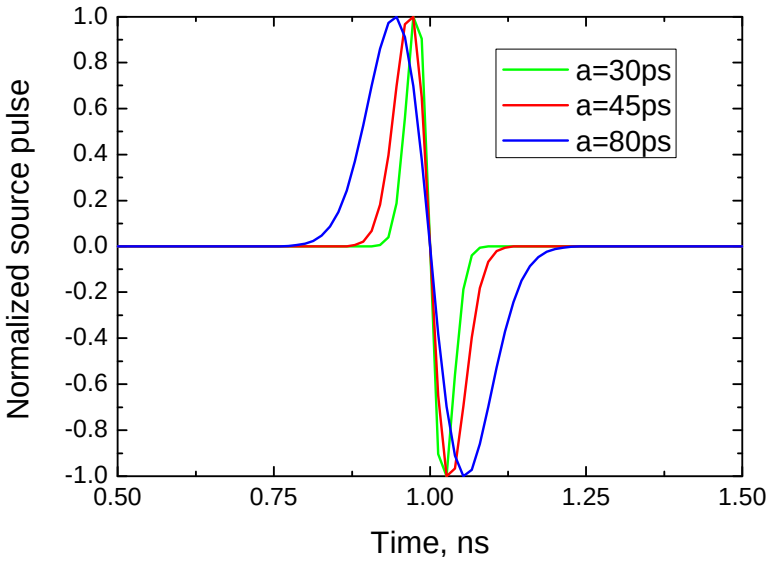


Figure 6.19: First order Rayleigh pulses with different a

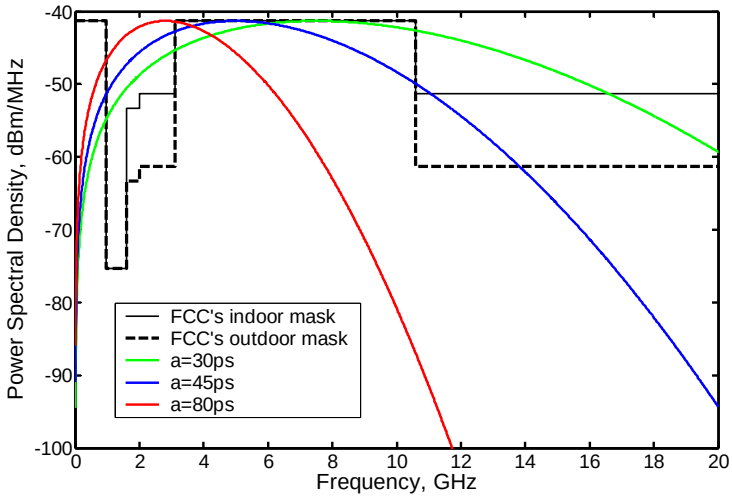


Figure 6.20: Power spectral densities of first order Rayleigh pulses with different a

Figure 6.20 also indicates that none of the three pulses can fully comply with the FCC's emission mask because of the high PSD at the frequencies lower than 3.1GHz. In addition, the pulse with a=30ps has high PSD level within the frequency range between 10.6GHz and 16.4GHz. Comparatively, the pulse with a=45ps is more suitable for UWB systems because its bandwidth matches the UWB band better, while the bandwidth of a=80ps is a bit narrow.

It is feasible to move the spectra of first order Rayleigh pulse into UWB band to completely meet the FCC's emission limits using continuous sine wave carrier if the carrier frequency and the pulse parameter are properly selected [6].

Besides, it is shown that some higher order Rayleigh pulses [2] can comply with the FCC's indoor emission mask directly such as the fourth order Rayleigh pulse with pulse parameter of a=67ps, as presented in Equation 6.16, Figure 6.21 and Figure 6.22.

$$f(t) = \left[\frac{12}{a^4} - \frac{48}{a^6}(t-1)^2 + \frac{16}{a^8}(t-1)^4\right] e^{-(\frac{t-1}{a})^2} \tag{6.16}$$

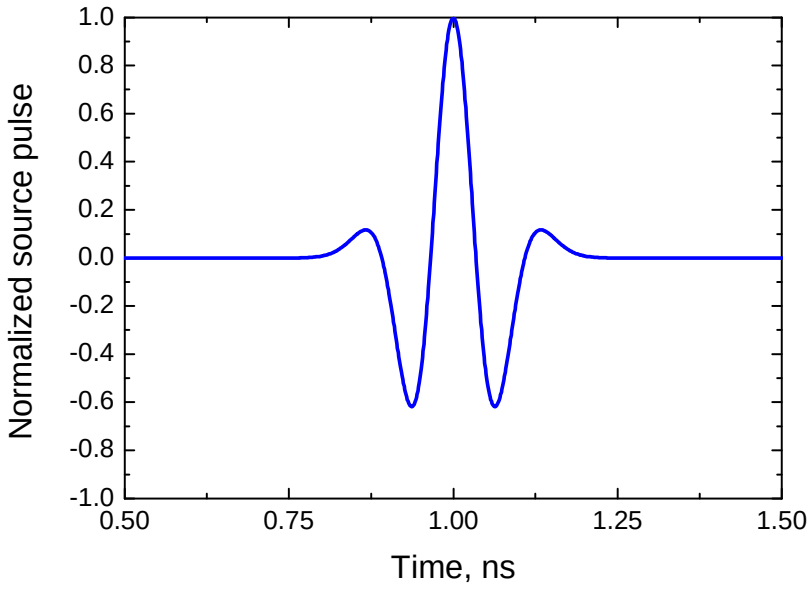


Figure 6.21: Fourth order Rayleigh pulse with a=67ps

Although the power spectral density of first order Rayleigh pulse can not fully meet the FCC's emission mask, it is widely exploited in UWB systems due to its simple monocycle waveform which can be easily generated by RF circuits. So in this chapter, we still use this type of signal as source pulse to study the time domain performances of UWB antennas.

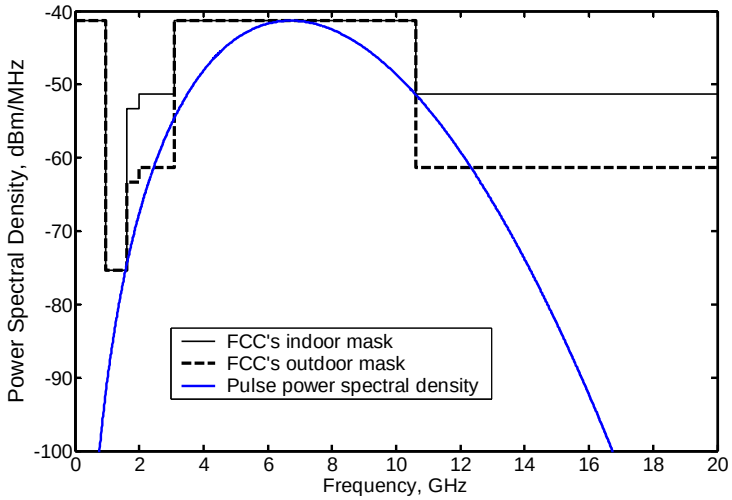


Figure 6.22: Power spectral density of fourth order Rayleigh pulse with a=67ps

6.3.2 Radiated Power Spectral Density of UWB antennas

In previous section, it has been shown that the antenna transmitting characteristic can be calculated based on the measured system transfer function. Thus, the radiated pulse can be easily obtained through the convolution of input signal and the antenna transmitting response.

6.3.2.1 Results of CPW Fed Disc Monopole

The power spectral densities of the radiated pulses against the FCC emission mask are illustrated in Figure 6.23 and 6.24. Here, first order Rayleigh pulse with a =45ps and fourth order Rayleigh pulse with a =67ps, as discussed in previous section, are used as the source signals to drive CPW fed disc monopole. Their power spectral densities are also presented in the Figures to compare with the radiated ones accordingly.

As shown in Figure 6.23, when the source signal is first order Rayleigh pulse with a=45ps, the radiated PSD complies with the FCC emission mask at most part of the frequency band except at the frequencies lower than 3.1GHz, which is mainly due to the high spectra level of the source pulse within this frequency range. At around 9GHz, the

radiated PSD curve undergoes a deep null, which agrees well to the antenna transmitting characteristic $A(\omega)$, as given in Figure 6.15 (b).

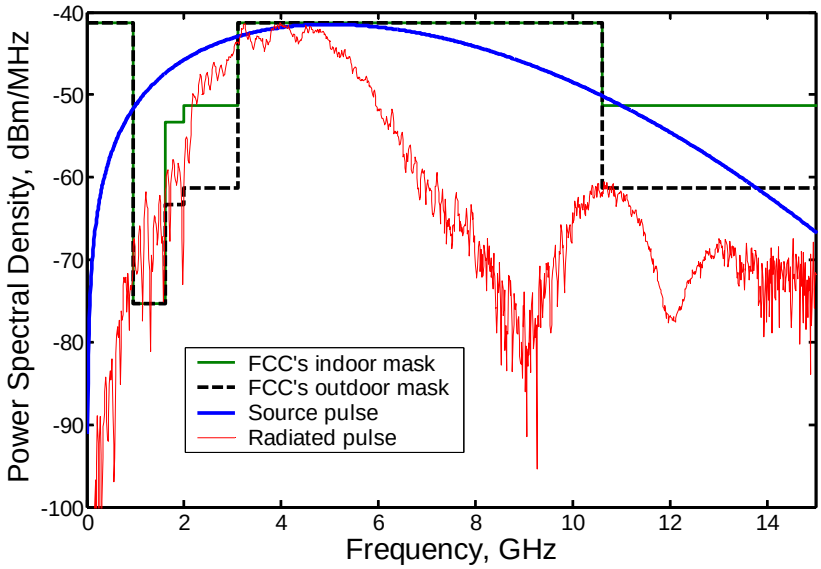


Figure 6.23: Radiated power spectral density with first order Rayleigh pulse of a=45ps

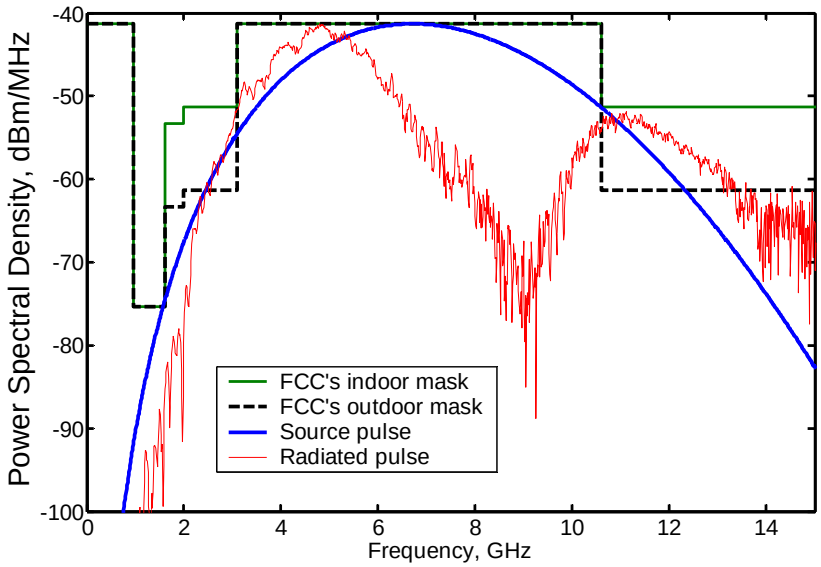


Figure 6.24: Radiated power spectral density with fourth order Rayleigh pulse of a=67ps

It is understandable that the fourth order Rayleigh pulse with a=67ps can lead to a radiated PSD totally compliant with the FCC indoor mask since the source spectra completely meet this emission limit, as shown in Figure 6.24.

6.3.2.2 Results of Other UWB Antennas

The radiated power spectral densities of vertical disc monopole and microstrip line fed circular slot antenna are presented in Figure 6.25 and 6.26, respectively.

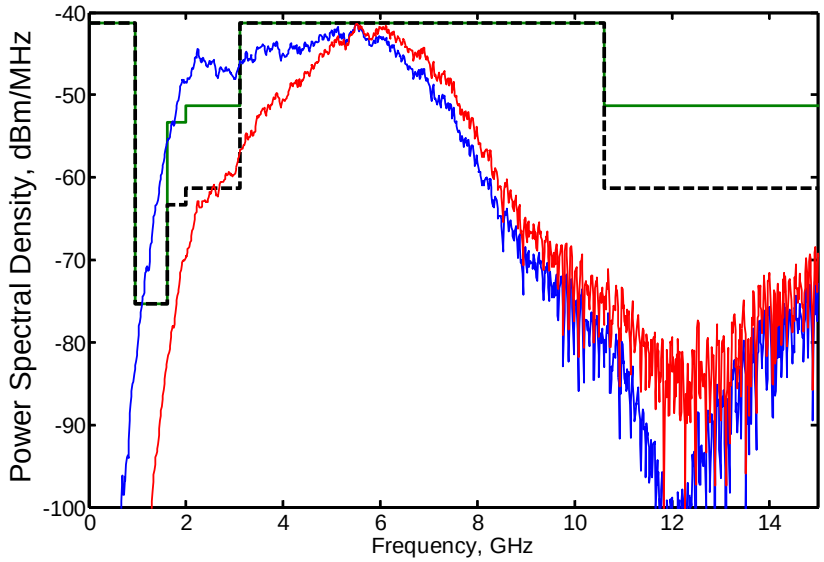


Figure 6.25: Radiated power spectral densities of vertical disc monopole for different source signals (blue curve: first order Rayleigh pulse with a=45ps; red curve: fourth order Rayleigh pulse with a=67ps)

For vertical disc monopole, the results are similar to those of CPW fed disc monopole, i.e. the radiated PSD is fully compliant with the indoor emission mask only when the source signal is fourth order Rayleigh pulse with a=67ps, as shown in Figure 6.25.

In contrast, the radiated power spectral densities of microstrip line fed circular slot antenna can completely match the indoor emission mask for both of the two different signals, i.e. first order Rayleigh pulse with a=45ps and fourth order Rayleigh pulse with a=67ps, as illustrated in Figure 6.26.

The results indicate that the radiated power spectral density shaping can be controlled to conform to the mandated emission limit by properly selecting the source pulse and the antenna.

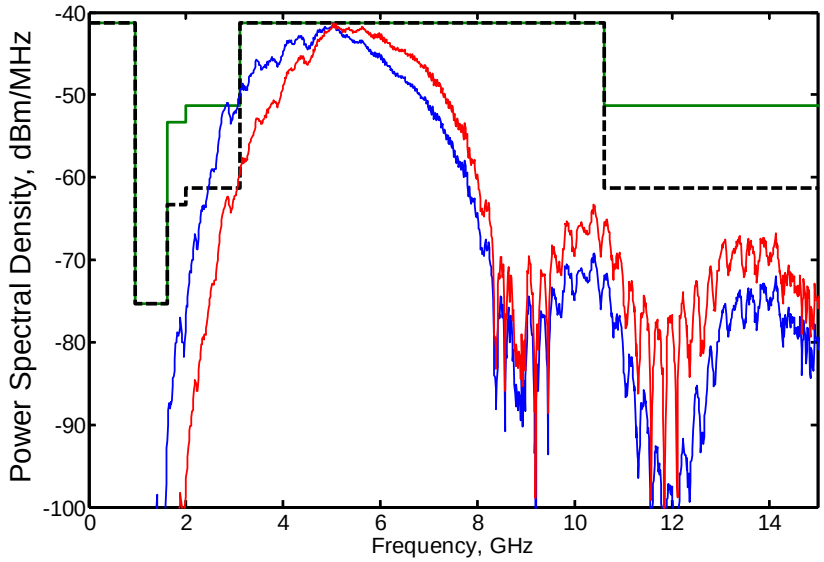


Figure 6.26: Radiated power spectral densities of microstrip line fed circular slot antenna for different source signals (blue curve: first order Rayleigh pulse with a=45ps; red curve: fourth order Rayleigh pulse with a=67ps)

6.4 Received Signal Waveforms

For a UWB system, as depicted in Figure 6.2, the received signal is required to match the source pulse with minimum distortions because the signal is the carrier of useful information. The received waveform is determined by both source pulse and the system transfer function which has already taken into account the effects from the whole system including the transmitting/receiving antennas.

The transfer function measured by vector network analyzer is a frequency response of the system. However, the frequency domain raw data can be transformed to the time domain. Here, Hermitian processing is used for the data conversion [7], as illustrated in Figure 6.27. Firstly, the pass-band signal is obtained with zero padding from the lowest frequency down to DC (direct current); Secondly, the conjugate of the signal is taken and reflected to the negative frequencies. The resultant double-sided spectrum corresponds to a real signal, i.e. the system impulse response. It is then transformed to the time domain using inverse fast Fourier transform (IFFT); Finally, the system impulse

response is convolved with the input pulse to achieve the received signal.

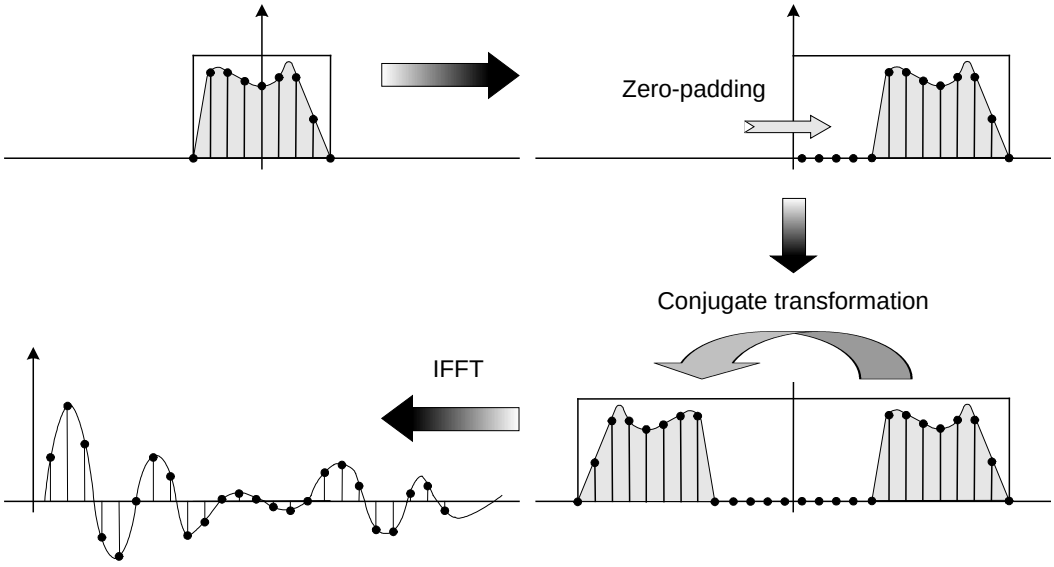


Figure 6.27: Hermitian processing (Reproduced from [7])

A well-defined parameter named fidelity [1] is proposed to assess the quality of a received signal waveform, as given in Equation 6.17.

$$F = \max_{\tau} \left[\frac{\int_{-\infty}^{+\infty} f(t) s_R(t+\tau) dt}{\sqrt{\int_{-\infty}^{+\infty} f^2(t) dt \int_{-\infty}^{+\infty} s_R^2(t) dt}} \right] \tag{6.17}$$

where the source pulse $f(t)$ and the received signal $s_R(t)$ are normalized by their energy. The fidelity F is the maximum correlation coefficient of the two signals by varying the time delay τ. It reflects the similarity between the source pulse and the received pulse. When the two signal waveforms are identical to each other, the fidelity reaches its peak, i.e. unity, which means the antenna system does not distort the input signal at all. In the extreme case that the two pulses are totally different in shape, the fidelity decreases to the minimum value of zero. In practice, a UWB system normally provides a fidelity between 0 and 1. Undoubtedly, a big fidelity is always desirable.

When first order Rayleigh pulse with a=45ps is chosen as the input signal, the received signals of vertical disc monopole are given in Figure 6.28.

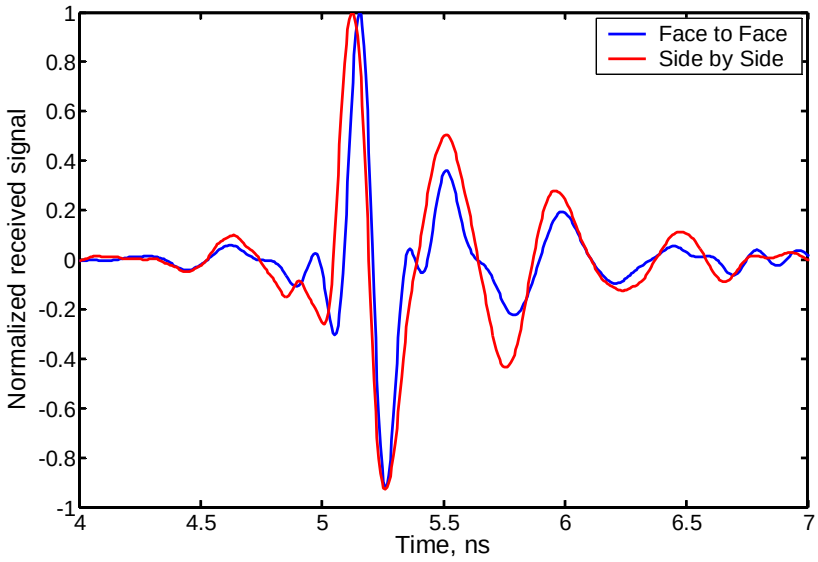


Figure 6.28: Received signal waveforms by vertical disc monopole with first order Rayleigh pulse of a=45ps as input signal (as shown in Figure 6.19)

The received waveforms of the two scenarios, i.e. face to face and side by side, match with each other very well, which corresponds to the nearly omni-directional radiation patterns of vertical disc monopole. The signal waveforms generally follow the shape of source pulse and only have slight distortions. The calculated fidelity F is 0.8318 for face to face case and 0.7160 for side by side case.

The distortions of the received signal waveforms can be explained by comparing the bandwidths between the transfer function S_{21} and the spectrum of the source pulse, as illustrated in Figure 6.29.

The input signal has a power spectrum across 1GHz to 11.1GHz at -10dB points. However, the operating band of the transfer functions, as shown in Figure 6.4, are much less than that of the source pulse spectrum. That means the input frequency components outside the transfer function bandwidth can not be efficiently transmitted by the disc monopole. Furthermore, there are some non-linear parts in the phase curves of the transfer functions, as plotted in Figure 6.6. Consequently, the received waveforms undergo some distortions.

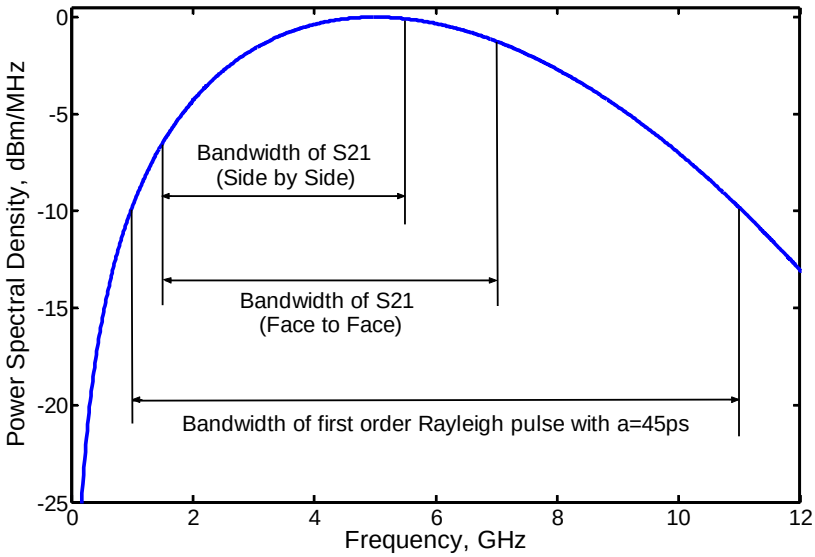


Figure 6.29: Spectrum of first order Rayleigh pulse with a=45ps

Furthermore, the transfer function curves (both magnitude and phase) of the two scenarios are quite similar within their respective operating bands, leading to similar received signal waveforms. However, face to face case achieves flatter magnitude curve, which indicates the signal frequency components are received more equally, leading to a bigger fidelity than side by side case.

The further analysis shows that the distortions of the received waveforms may be minimized if the source signal bandwidth falls into the band of the system transfer function.

Now we consider a Gaussian pulse modulated by a continuous sine wave carrier, as given in Equation 6.18.

$$f(t) = \sin\left[2\pi f_c(t-1)\right] e^{-(\frac{t-1}{a})^2} \tag{6.18}$$

where the carrier frequency f_c is set at 4GHz and pulse parameter a at 350ps.

The signal waveform of the modulated Gaussian pulse and its spectrum are plotted in Figure 6.30 and Figure 6.31, respectively. It is shown that the spectrum peaks at

4GHz with a -10dB bandwidth from 3GHz to 5GHz, which means the main energy of the signal is totally moved into the operating band of the system transfer function.

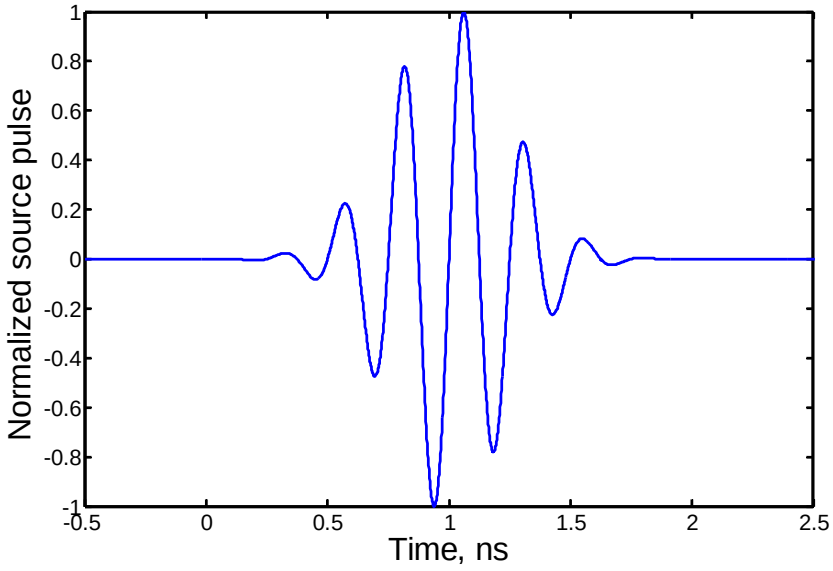


Figure 6.30: Gaussian pulse modulated by sine signal with f_c=4GHz and a=350ps

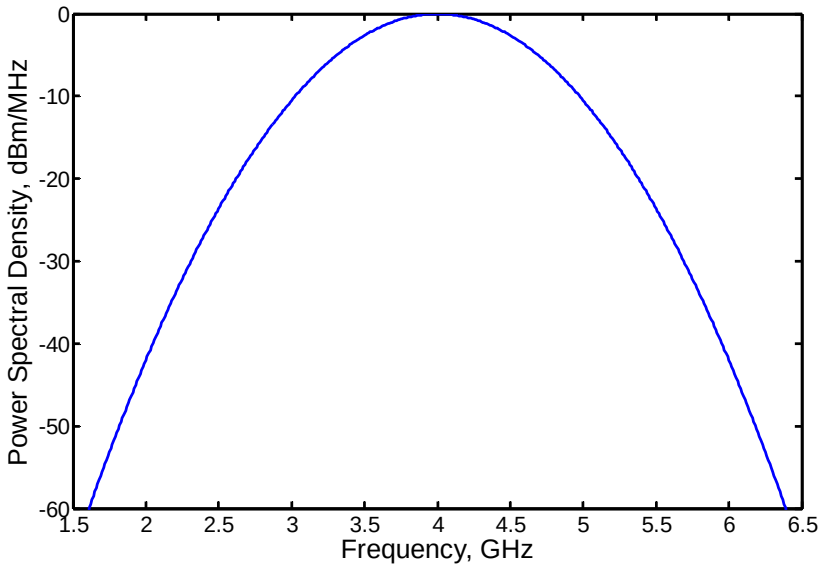


Figure 6.31: Spectrum of modulated Gaussian pulse with a=350ps and f_c=4GHz

When this modulated Gaussian pulse is excited to the vertical disc monopole pair, the received signal waveforms are quite similar to the source pulse in both orientations, as illustrated in Figure 6.32.

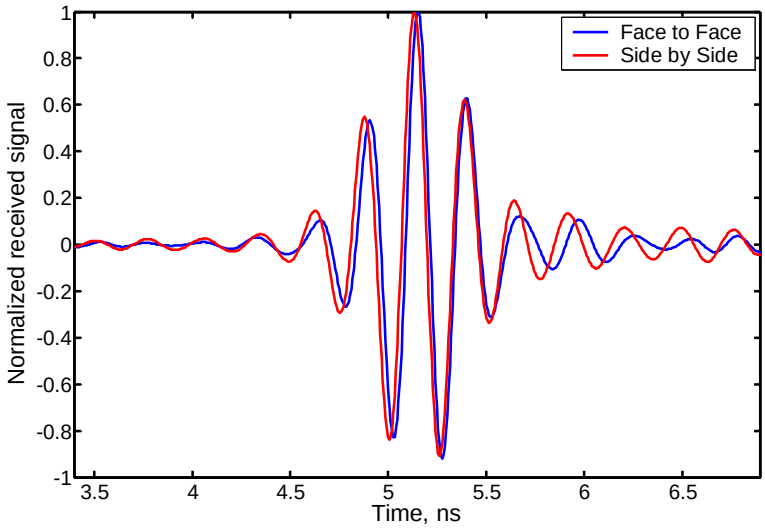


Figure 6.32: Received signal waveforms by vertical disc monopole with modulated Gaussian pulse (a=350ps, f_c=4GHz) as input signal

The calculated fidelity reaches as high as 0.9914 for face to face case and 0.9913 for side by side case, much bigger than those when first order Rayleigh pulse is used as the source signal. A significant improvement of received signal quality has been achieved by using modulated Gaussian pulse.

The received pulses by other UWB antennas are plotted in Figure 6.33 – Figure 6.36.

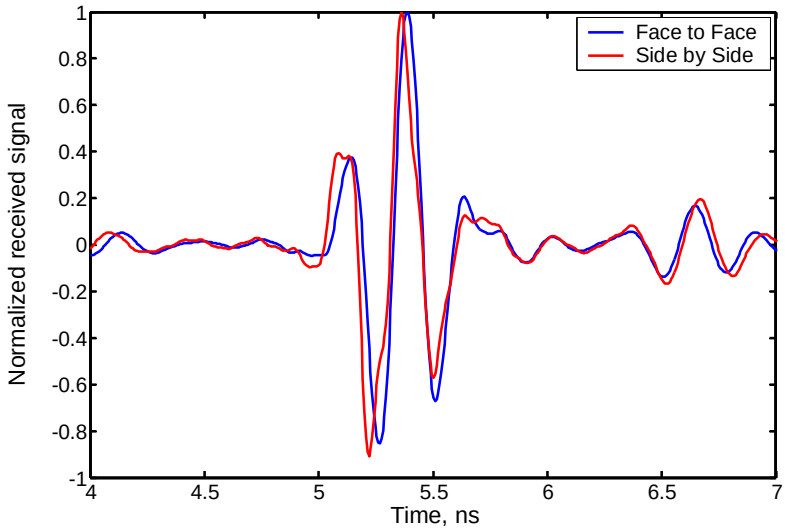


Figure 6.33: Received signal waveforms by CPW fed disc monopole with first order Rayleigh pulse of a=45ps as input signal

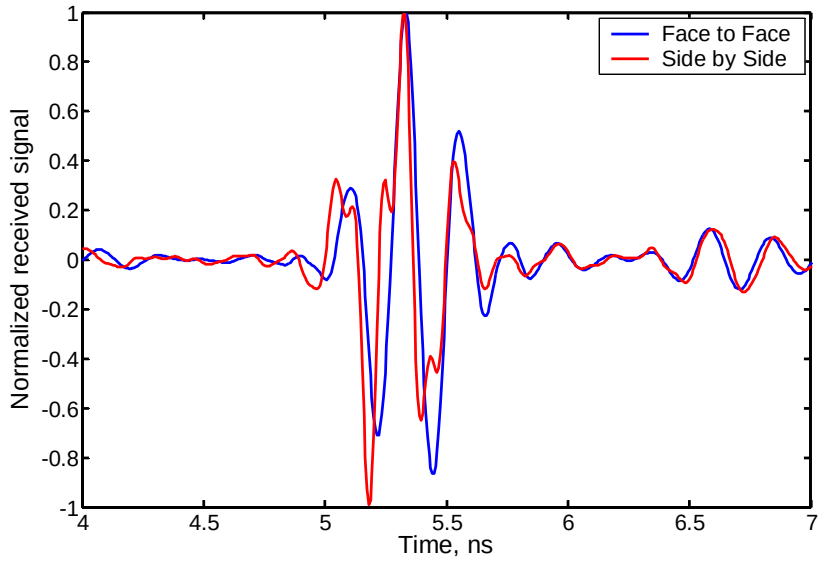


Figure 6.34: Received signal waveforms by CPW fed disc monopole with fourth order Rayleigh pulse of a=67ps as input signal

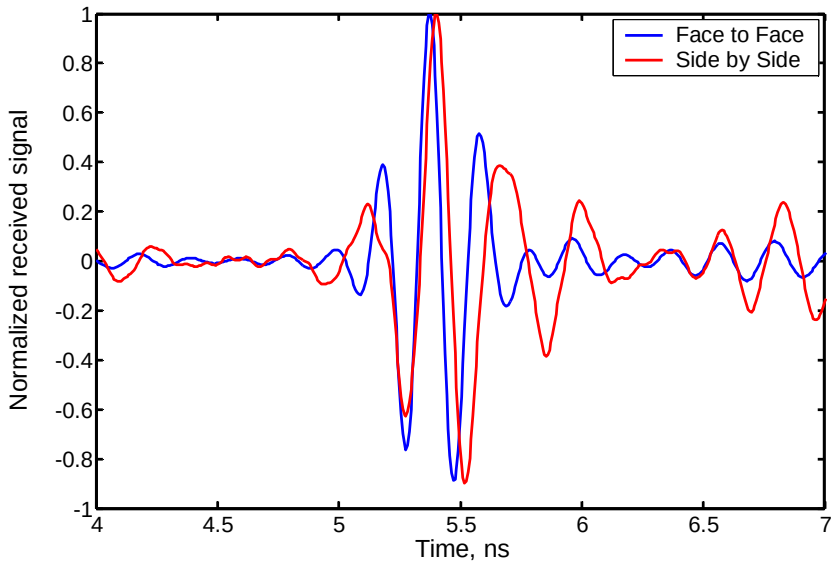


Figure 6.35: Received signal waveforms by microstrip line fed circular slot antenna with first order Rayleigh pulse of a=30ps as input signal

The calculated fidelities for various source pulses in different UWB antenna systems are tabulated in Table 6-A– 6-C.

According to the Tables, a fidelity greater than 0.95 is always achieved for different antenna pairs when the modulated Gaussian pulse is used as the source pulse. It is even better than 0.99 for disc monopole antennas in both of the two orientations. This is well

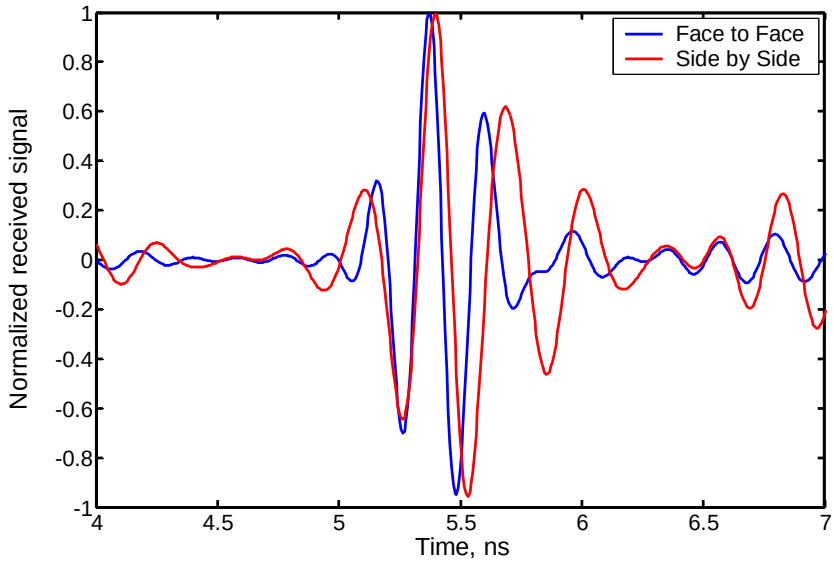


Figure 6.36: Received signal waveforms by microstrip line fed circular slot antenna with first order Rayleigh pulse of a=80ps as input signal

understood since the pulse spectrum is full within the band of transfer function. Most of the frequency components can be received effectively and equally. Thus, the antenna system does not cause distortions to the signal.

For fourth order Rayleigh pulse with a=67ps, its spectrum coincide well with the 3.1GHz–10.6GHz UWB frequency band, as shown in Figure 6.22. The upper end of the transfer function operating band for each antenna pair is typically at around 6GHz, less than 10.6GHz. Signal frequency components between 6GHz and 10.6GHz are attenuated substantially, leading to some distortions in the received signal waveforms. The calculated fidelity fluctuates around 0.9 for all of the three antenna pairs in different orientations.

The spectrum of first order Rayleigh pulse is critically dependent on the pulse parameter a. A larger a corresponds to a narrower spectrum bandwidth, as illustrated in Equation 6.20. For a=80ps, the spectrum bandwidth is from 0.55GHz to 6.2GHz. The upper frequency end of the pulse spectrum is close to those of the transfer functions, but the lower end is smaller than those of the transfer functions. That means some low frequency components of the signal are filtered by the antenna system. With the decrease

of a, the spectrum band is getting broader, and some high frequency components are also filtered. As a result, the value of fidelity becomes smaller and it is even lower than 0.6 for microstrip line fed circular slot antenna in both orientations when a=30ps.

Table 6-A: Fidelity for vertical disc monopole antenna pair

	1st order Rayleigh a=30ps	1st order Rayleigh a=45ps	1st order Rayleigh a=80ps	4th order Rayleigh a=67ps	modulated Gaussian a=350ps
Face to Face	0.6231	0.8318	0.8751	0.9300	0.9914
Side by Side	0.6031	0.7160	0.8156	0.9293	0.9913

Table 6-B: Fidelity for CPW fed disc monopole antenna pair

	1st order Rayleigh a=30ps	1st order Rayleigh a=45ps	1st order Rayleigh a=80ps	4th order Rayleigh a=67ps	modulated Gaussian a=350ps
Face to Face	0.6315	0.7474	0.8369	0.9000	0.9933
Side by Side	0.6229	0.6303	0.8599	0.9250	0.9970

Table 6-C: Fidelity for microstrip line fed circular slot antenna pair

	1st order Rayleigh a=30ps	1st order Rayleigh a=45ps	1st order Rayleigh a=80ps	4th order Rayleigh a=67ps	modulated Gaussian a=350ps
Face to Face	0.5787	0.7502	0.7419	0.9076	0.9609
Side by Side	0.4666	0.7230	0.8108	0.8876	0.9583

It is also noticed that for a given input signal, face to face case always produces a higher fidelity than side by side case in various antenna pairs. This is because the system function in face to face case normally has a flatter magnitude within the operating band.

6.5 Summary

In a UWB system, the antenna behaves like a bandpass filter and reshapes the pulse spectrum. The antenna transmitting response is related to its receiving response by a temporal derivative. Consequently, the signal waveform arriving at the receiver usually does not resemble the input pulse.

To obtain a high fidelity, which describes the similarity between the input signal and the received one, the system transfer function is required to have flat magnitude with linear phase within the operating band. Moreover, the spectrum of source pulse needs to match the transfer function. Thus, the received signal waveform is determined by both the antenna system and the source pulse.

First order Rayleigh pulse features simple monocycle shape. However, its power spectral density can not fully comply with the FCC emission mask. Besides, the obtained fidelity may be very low due to the mismatch between the pulse spectrum and the transfer function. Some higher order Rayleigh pulses (for example, the fourth order pulse with a=67ps) can completely conform to the FCC emission mask, which makes it qualified for DS-UWB systems. Using modulated Gaussian pulse, a very high fidelity can be achieved and the received signal does not have many distortions. This relatively narrow band pulse with carrier is suitable for MB-OFDM UWB systems.

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Chapter 7

Conclusions and Future Work

7.1 Summary

The UWB technology will be the key solution for the future WPAN systems. This is due to its ability to achieve very high data rate which results from the large frequency spectrum occupied. Besides, extremely low power emission level will prevent UWB systems from causing severe interference with other wireless systems. As the only non-digital part of a UWB system, antenna remains as a particular challenging topic because there are more stringent requirements for a suitable UWB antenna compared with a narrowband antenna. Therefore, the antenna design and analysis for UWB systems were carried out in this thesis.

Circular disc monopole antenna originates from a conventional monopole by replacing the wire element of with a circular disc element. The antenna configuration has also evolved from a vertical disc to a planar version by using microstrip line and CPW feeding structure for the ease of integration with printed circuit board.

Studies indicate that the disc element is capable of supporting multiple resonant modes and these modes are closely spaced. It is the overlapping of these resonances that

leads to the UWB characteristic. The disc monopole operates in a pure standing wave mode at the first resonance, and in a hybrid mode of standing and travelling waves at higher order resonances. Besides, travelling wave becomes more dominant in the antenna operation with the increase of frequency. Therefore, it is essential to design a smooth transition between the feeding line and the antenna for good impedance matching over the entire operational bandwidth.

Investigations have also been carried out in this thesis to analyse the design parameters of circular disc monopole. In a broad sense, the ground plane serves as an impedance matching circuit, and it tunes the input impedance and hence changes the operating bandwidth when the feed gap is varied. The dimension of the disc also has an impact on the antenna performance because the current is mainly distributed along the edge on the disc. Thus, the multiple resonances are directly (first resonance) or indirectly (other resonances) associated with the dimension of the disc. In addition, the current distributions also imply that cutting the central part of the disc will not degrade the antenna performance. Consequently, circular ring monopole exhibits the similar characteristics as its disc counterpart. Both of them can provide UWB characteristics with nearly omni-directional radiation patterns over the entire bandwidth.

Elliptical/circular slot antennas are studied in this thesis as a type of antenna candidates for the future UWB applications. Similar to circular disc monopole antennas, the UWB characteristics of elliptical/circular slot antennas are also due to the overlapping of the closely spaced resonances over the frequency band. The antennas operate in a hybrid mode of standing and travelling waves. However, the travelling wave at high frequency is not well supported in the enclosed structure, so the frequency bandwidth is limited at the high end. It has been shown that the operating bandwidth can be enhanced significantly by using tapered microstrip or CPW feeding line with U-shaped tuning stub. Therefore, the slot dimension, the slant angle and the distance between the tuning stub and the slot are the most important design parameters that determine the antenna performance.

The time domain behaviors of circular disc monopole and elliptical/circular slot antennas are also evaluated in this thesis. The received signal waveform is affected by both the transmit/receive antennas and the source pulse. In order to achieve high signal fidelity, the antenna system transfer function is required to have flat magnitude and linear phase response over the operational band. Also, the spectrum of the source pulse is required to match the operating band of the system transfer function. Three different pulses, i.e. the first order Rayleigh pulse, the fourth order Raleigh pulse and the carrier-modulated Gaussian pulse, are investigated and selected as source pulse for circular disc monopole and elliptical/circular slot antenna systems. Studies have shown that a very high signal fidelity was always achieved by using the carrier-modulated Gaussian pulse because the pulse spectrum matches the antenna system transfer function well.

7.2 Key Contributions

The major contributions in the thesis are detailed below.

Firstly, the mechanism which leads to UWB characteristics was proposed based on the further insight of the operations of UWB disc monopoles and elliptical/circular slot antennas. The overlapping of multiple resonances which are evenly and closely spaced accounts for the UWB characteristics. In addition, the antenna operates in a hybrid mode of standing and travelling waves, and the travelling wave becomes dominant with the increase of frequency.

Secondly, the miniaturization of vertical type disc monopole was realized by narrowing the ground plane while retaining the UWB characteristics. The proposed disc monopole antenna is simple in design, small in size and easy to manufacture.

Thirdly, two planar versions of UWB disc monopole antennas were proposed. One is fed by microstrip line, and the other by CPW feeding structure. Both of them feature small size, ease of fabrication, low profile and compatibility with printed circuit board.

Fourthly, printed UWB circular ring monopoles were realized by replacing the disc elements of planar circular disc monopole antennas with circular ring elements. The proposed circular ring monopole antennas can exhibit nearly same characteristics as their disc counterparts.

Lastly, four printed UWB elliptical/circular slot antennas fed by microstrip line or CPW were also proposed in the thesis.

All of these antennas proposed in the thesis can provide ultra wide bandwidth with nearly omni-directional radiation patterns and satisfactory time domain perofrmances which make them very suitable for the future UWB applications.

7.3 Future Work

Based on the conclusions drawn and the limitations of the work presented, future work can be carried out in the following areas:

Firstly, it has been shown that both UWB disc monopole and elliptical/circular slot antennas operate in a hybrid mode of standing and travelling waves. A more detailed understanding of the travelling wave mechanism and the impedance variations could lead to improved design of UWB antennas.

Secondly, in this thesis, all of the antenna measurements are carried out inside an anechoic chamber. However, in the future UWB systems, antenna might be embedded inside a laptop or other devices. Thus, the devices' effects on the antenna performances need to be investigated. When the antenna is built on a portable device, the impact from human body should also be considered.

Thirdly, UWB antenna with small size is always desirable for the WPAN applications, especially for mobile and portable devices. Future research may focus on finding out new methods to further reducing the sizes of UWB disc monopole and elliptical/circular slot

antennas.

Fourthly, due to the collocation of UWB system with frequency bands reserved for other wireless systems, sometimes the UWB device may be required to provide filtering in those bands to avoid potential interference. UWB disc monopole and slot antennas with band rejection properties can be an objective of future work.

Fifthly, UWB systems operate at extremely low power level which limits its transmission range. In order to enhance the quality of the communication link and improve channel capacity and range, directional systems with high gain are required for some applications. Therefore, research on UWB directional antenna and antenna array could be carried out.

Lastly, good time domain performance is a primary requirement for UWB antennas. Studies can be carried out to investigate the antenna effect on the transmitted signal and improve the time domain behaviors by optimizing the antenna configuration.

Appendix A

Author's Publications

Journal Papers

1. **J. Liang**, C Chiau, X. Chen and C.G. Parini, "Study of a Printed Circular Disc Monopole Antenna for UWB Systems", *IEEE Transactions on Antennas and Propagation*, vol. 53, no. 11, November 2005, pp.3500-3504.

2. **J. Liang**, L.Guo, C.C.Chiau, X. Chen and C.G.Parini, "Study of CPW-Fed circular disc monopole antenna", *IEE Proceedings Microwaves, Antennas & Propagation*, vol. 152, no. 6, December 2005, pp. 520-526.

3. **J. Liang**, C Chiau and X. Chen, "Printed circular ring monopole antennas", *Microwave and Optical Technology Letters*, vol. 45, no. 5, June 5, 2005, pp. 372-375.

4. **J. Liang**, C Chiau, X. Chen and C.G. Parini, "Printed circular disc monopole antenna for ultra wideband applications", *IEE Electronic Letters*, vol. 40, no. 20, September 30th, 2004, pp.1246-1248.

5. P. Li, **J. Liang** and X. Chen, "Ultra-wideband elliptical slot antenna fed by tapered micro-strip line with U-shaped tuning stub", *Microwave and Optical Technology*

Letters, vol. 47, no. 2, October 20, 2005, pp. 140-143.

6. P. Li, **J. Liang** and X. Chen, "Study of Printed Elliptical/Circular Slot Antennas for Ultra Wideband Applications", *IEEE Transactions on Antennas and Propagation*, vol. 54, no. 6, June 2006, pp.1670-1675.

7. Xiaodong Chen, **Jianxin Liang**, Pengcheng Li and Choo C. Chiau, "UWB Electric and Magnetic Monopole Antennas", *African Journal of Information& Communication Technology*, vol. 2, no. 1, 2006.

8. L.Guo, **J. Liang**, C.C.Chiau, X. Chen, C.G.Parini and J.Yu, "Performances of UWB disc monopoles in time domain", *IEE Proceedings Microwaves, Antennas & Propagation*, **(Submitted)**

Conference Papers

1. **J. Liang**, K. Wu, C. C. Chiau, X. Chen, Clive Parini, "PRINTED UWB ELLIPTICAL DISC MONOPOLE", *2006 Loughborough Antennas and Propagation Conference*, Loughborough, UK, April 11-12, 2006.

2. **J. Liang**, C Chiau and X. Chen, "Time domain characteristics of UWB disc monopole antennas", *35th European Microwave Conference*, Paris, France, October 3-7, 2005.

3. **J. Liang**, C Chiau, X. Chen and C.G. Parini, "CPW-Fed Circular Ring Monopole Antenna", *2005 IEEE AP-S International Symposium on Antennas and Propagation*, Washington, DC, USA, July 3-8, 2005.

4. **J. Liang**, C Chiau and X. Chen, "Design analysis in a planar UWB circular ring monopole", *2005 Loughborough Antennas and Propagation Conference*, Loughborough, UK, April 4-6, 2005.

5. **J. Liang**, L. Guo, C. Chiau and X. Chen, "CPW-Fed Circular Disc Monopole

Antenna for UWB Applications", *IEEE International Workshop on Antenna Technologies (iWAT 2005)*, Singapore, 7-9 March 2005.

6. **J. Liang**, C. C. Chiau, X. Chen and J. Yu, "Effect of the ground plane on the operation of a UWB monopole", *2004 Progress in Electromagnetics Research Symposium*, 28 - 31 August, 2004, Nanjing, China.

7. **J. Liang**, C. C. Chiau, X. Chen and J. Yu, "Study of a circular disc monopole antenna for ultra wideband applications", *2004 International Symposium on Antennas and Propagation*, August 17-21, 2004, Sendai, Japan.

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19. Xiaodong Chen, **Jianxin Liang**, Pengcheng Li and Choo C. Chiau, "electric and magnetic monopole antennas for UWB applications", *ICT 2005–12th International Conference on Telecommunications*, Cape Town, South Africa, May 3-6, 2005.

20. P. Li, **J. Liang** and X. Chen, "a CPW-fed UWB hexagonal monopole antenna", *2005 Loughborough Antennas and Propagation Conference*, Loughborough, UK, April 4-6, 2005.

21. L. Guo, **J. Liang**, X. Chen and C.G. Parini, "Study of tapering effects on the CPW-Fed of a circular disc monopole antenna", *2005 Loughborough Antennas and Propagation Conference*, Loughborough, UK, April 4-6, 2005.

Appendix B

Electromagnetic (EM) Numerical Modelling Technique

The technology of wireless communications is established on the principles of electromagnetic (EM) fields and waves. Thus, numerical techniques are playing an important role in solving EM field problems especially when the problems' complexity increases.

Currently, several numerical techniques are available to solve the EM problems, such as Finite Element (FE) method , the Method of Moments (MoM), Finite-Difference Time-Domain (FDTD) method and Finite Integration Technique (FIT). FE and MoM solve the EM problems in frequency domain whilst FDTD and FIT solve the EM problems in time domain instead. A particular numerical technique is well suited for the analysis of a particular type of problem. Analyses have shown that FDTD/FIT is fast in computation and the resolution is better than other available numerical software package [1]. Therefore the CST Microwave Studio® which is based on the FIT numerical method has been used as the modelling tool in this thesis.

B.1 Maxwell's Equations

The basis of EM theory is based on the relationship between the electric and magnetic fields, charges and currents. In 1886, James C. Maxwell assembled the Faraday's Law, Ampere's Law, Gauss's Law and magnetic field law into a set of equations which form the basis of EM theory [2, 3].

The Maxwell's equations can be written in the differential form:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{Faraday's Law} \tag{B.1}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{Magnetic Field Law} \tag{B.2}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{Ampere's Law} \tag{B.3}$$

$$\nabla \cdot \vec{D} = \rho \quad \text{Gauss's Law} \tag{B.4}$$

or in the equivalent integral form:

$$\oint \vec{E} \cdot d\vec{s} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A} \quad \text{Faraday's Law} \tag{B.5}$$

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \text{Magnetic Field Law} \tag{B.6}$$

$$\oint \vec{H} \cdot d\vec{s} = \int (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{A} \quad \text{Ampere's Law} \tag{B.7}$$

$$\oint \vec{D} \cdot d\vec{A} = \int \rho dV \quad \text{Gauss's Law} \tag{B.8}$$

In addition to the above four Maxwell's equations, there are three material equations:

$$\vec{D} = \varepsilon \vec{E} \tag{B.9}$$

$$\vec{B} = \mu \vec{H} \tag{B.10}$$

$$\vec{J} = \sigma \vec{E} \tag{B.11}$$

where E is the electric field intensity (v/m); H is the magnetic field intensity (A/m); D is the electric flux density; B is the magnetic flux density; J is the electric current density (A/m^2); σ is the electric conductivity (S/m); $\varepsilon = \varepsilon_0 \varepsilon_r$, is the electrical permittivity (F/m); $\mu = \mu_0 \mu_r$ is the magnetic permeability (H/m).

B.2 Finite Integral Technique (FIT)

Finite Integration Technique (FIT) was first proposed by Weiland in 1977 [4]. Equivalent to FDTD, FIT is a time-domain numerical technique for solving Maxwell's equations. However, it discretises the integral form rather than the differential form of Maxwell's equations.

The first step of the FIT discretisation is to define the computation domain which contains the space region of interest. The computation domain is enclosed by the restriction of the electromagnetic field problem, which normally represents an open boundary problem to a bounded space region.

The next step is to decompose the computation domain into a finite number of the simplicial cell complex G, which serves as a computational grid. The primary grid G can be visualised in the CST Microwave Studio®, whilst internally a dual grid $\widetilde{G}$ is set up orthogonally to the primary one. In the Cartesian system, the dual grid $\widetilde{G}$ is defined by taking the foci of the cells of G as grid points for the mesh cells of as shown in Figure B.1.

The electric voltages e and magnetic fluxes b are allocated on the primary grid G whilst the dielectric fluxes d and the magnetic voltages h are allocated on the dual grid $\widetilde{G}$. A voltage is defined as the integral of a field strength value (electric or magnetic) along a (dual) mesh edge whilst a flux is defined as the integral of a flux density value (electric or magnetic) across a (dual) mesh cell facette.

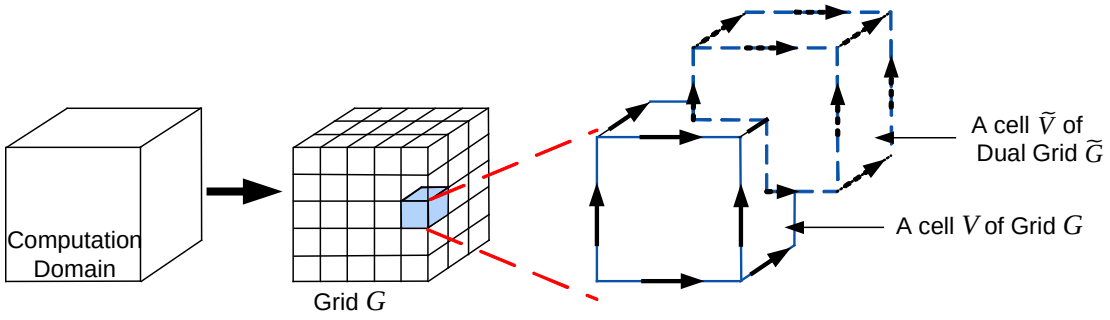


Figure B.1: FIT discretization

B.3 Faraday's Law

Consider a single cell V of the grid G as shown in Figure B.2, the integration form of Faraday's Law (Equation B.5) can be rewritten for a facet A_n as a sum of four grid voltages:

$$e_i + e_j - e_k - e_l = -\frac{d}{dt} b_n \tag{B.12}$$

where the scalar value e is the electric voltage along one edge of the surface A_n, whilst the scalar b_n represents the magnetic flux though the cell facet A_n.

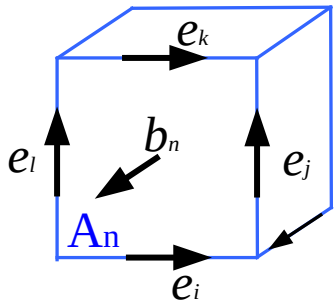


Figure B.2: A cell V of the grid G with the electric grid voltage e on the edges of A_n and the magnetic facet flux b_n through this surface

Therefore, the discrete form of Faraday's Law can be expressed in the general form:

$$Ce = -\frac{d}{dt} b_n \tag{B.13}$$

where C=(1,1,-1,-1), is a matrix coefficient which contains the incident relation of the cell edges within G and on their orientation.

B.4 Magnetic Field Law

For a cell V of the grid G as shown in Figure B.3, the integration form of Magnetic Field Law (Equation B.6) can be represented as:

$$-b_1 + b_2 - b_3 + b_4 - b_5 + b_6 = 0 \tag{B.14}$$

Again, the relation in Equation B.14 can be expanded to all the available cells and expressed in a general form as:

$$Sb = 0 \tag{B.15}$$

where S is a matrix which contains the incident relation of the cell facet, representing the discrete divergence-operator for grid G.

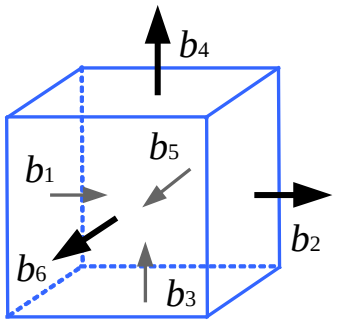


Figure B.3: A cell V of the grid G with six magnetic facet fluxes which have to be considered in the evaluation of the closed surface integral for the non-existance of magnetic charges within the cell volume

B.5 Ampere's Law

The discretisation of Ampere's Law (Equation B.7) using the FIT requires the dual grid $\widetilde{G}$, as given in Figure B.1.

As shown in Figure B.4, on a facet $\widetilde{A_n}$ of a dual grid cell $\widetilde{G}$, the summing of the magnetic grid voltages to obtain the displacement current and the conductive current

through the facet can be expressed as follows:

$$h_1 + h_2 - h_3 - h_4 = \frac{d}{dt} d_n + j \tag{B.16}$$

i.e.:

$$\widetilde{C} h = \frac{d}{dt} d + j \tag{B.17}$$

where the matrix $\widetilde{C}$ contains the incident relation of the cell edges within $\widetilde{G}$ and their orientation.

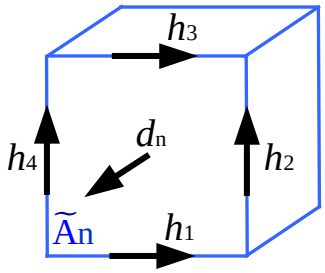


Figure B.4: A cell $\widetilde{V}$ of the grid $\widetilde{G}$ with the magnetic grid voltage h on the edges of $\widetilde{A_n}$ and the electric facet flux d_n through this surface

B.6 Gauss's Law

The integral form of Gauss's Law (Equation B.8) can be discretised for the dual grid cells and its discrete matrix form is:

$$\widetilde{S} d = q \tag{B.18}$$

where the matrix $\widetilde{S}$ contains the incident relation of the cell facet, representing the discrete divergence-operator for the dual grid $\widetilde{G}$.

B.7 Maxwell's Grid Equations (MGE's)

It has been shown in previous sections that in the FIT discretisation, the integral form of Maxwell's equations (Equation B.5– B.8) is transformed into a complete set of discrete matrix equations, (i.e. Equation B.13, B.15, B.17 and B.18), termed the Maxwell Grid Equations (MGE's). Furthermore, the curl (C, $\widetilde{C}$) and divergence (S, $\widetilde{S}$) matrices from the MGE's have the following properties:

$$SC = 0 \tag{B.19}$$

$$\widetilde{S}\widetilde{C} = 0 \tag{B.20}$$

$$C = \widetilde{C}^T \tag{B.21}$$

The relations in Equation B.19 and B.20 have ensured that there is no electric or magnetic charges arising during the computation due to the numerical algorithm.

Finally, the material equations (Equation B.9– B.11) can also be rewritten in terms of material matrices M_ε, M_μ and M_σ, as follows:

$$d = M_\varepsilon e \tag{B.22}$$

$$b = M_\mu h \tag{B.23}$$

$$j = M_\sigma e \tag{B.24}$$

B.8 Advanced techniques in CST Microwave Studio®

The most common disadvantage of the FIT in three-dimensional modelling is the usage of Yee-type Cartesian grids [5]. The standard gridding scheme introduces errors to the geometry representation of the curved structure surface due to the staircase approxima-

tion, as shown in Figure B.5(a).

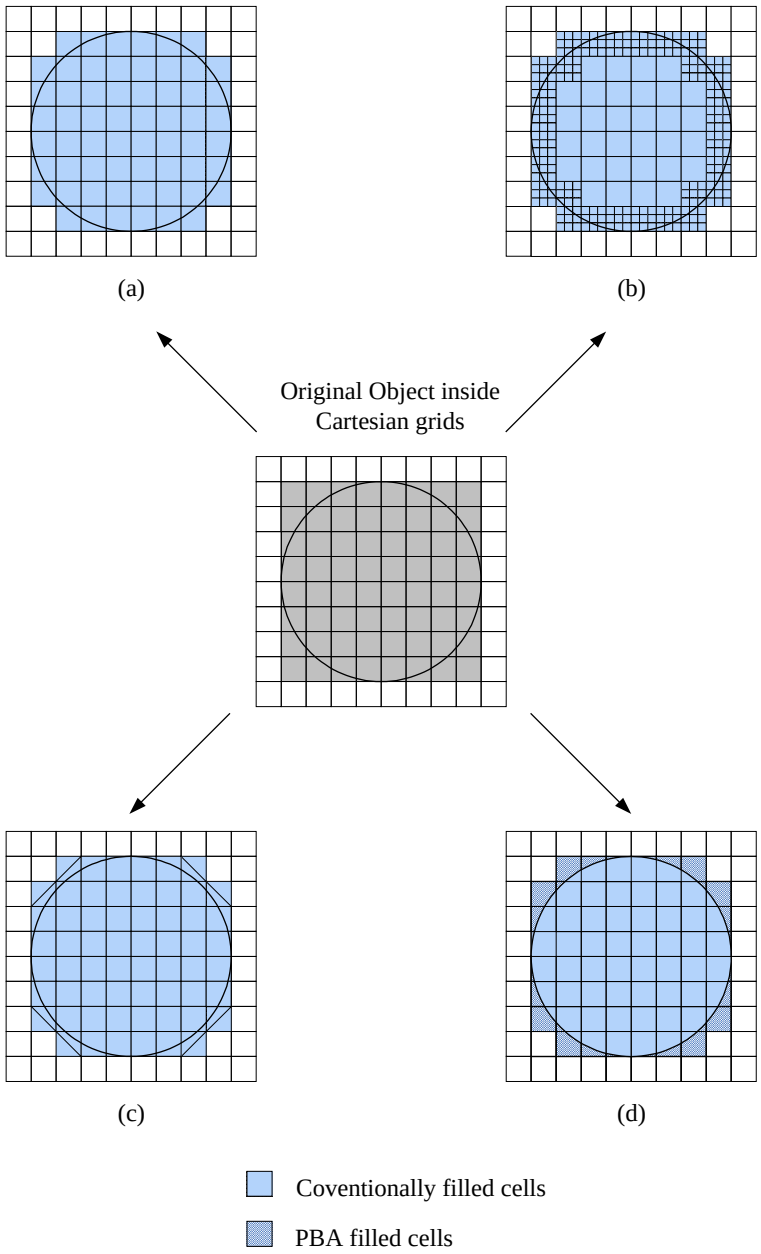


Figure B.5: Grid approximation of rounded boundaries: (a) standard (stair case), (b) sub-gridding, (c) triangular and (d) Perfect Boundary Approximation (PBA)

In order to reduce the errors, a fine mesh is usually introduced around the curved surface. However, it leads to an overall fine mesh in the whole structure. Therefore, sub-gridding technique has been introduced. This technique is more efficient because it

refines the mesh density only within the desired area (e.g. curve surface) instead of the whole structure, as illustrated in Figure B.5(b). Figure B.5(c) shows the triangular filling, which is another approach introduced to overcome the geometry approximation problem. However, most of these techniques have stability problems or low efficiency.

A more accurate and efficient technique termed Perfect Boundary Approximation (PBA®), as shown in Figure B.5(d) has been implemented in the commercial EM modelling package, CST Microwave Studio® [6]. Using this technique, the computational grid does not have to conform to the curved surface/boundaries. Instead, the sub-cellular information is taken into consideration resulting in an algorithm with second order accuracy for arbitrary shaped boundaries. Unlike other techniques, PBA only requires slightly higher numerical cost during the iteration. The algorithm of PBA has never been published by CST due to commercial reasons.

However, PBA can only define one field value within PEC partially filled cells. There is still fine mesh to be defined in the thin PEC region of the structure. As such, Thin Sheet Technology^{TM} (TST) has been introduced in the CST Microwave Studio® to solve the problem. It is possible for TST to handle two different field values within one cell, as shown in Figure B.6.

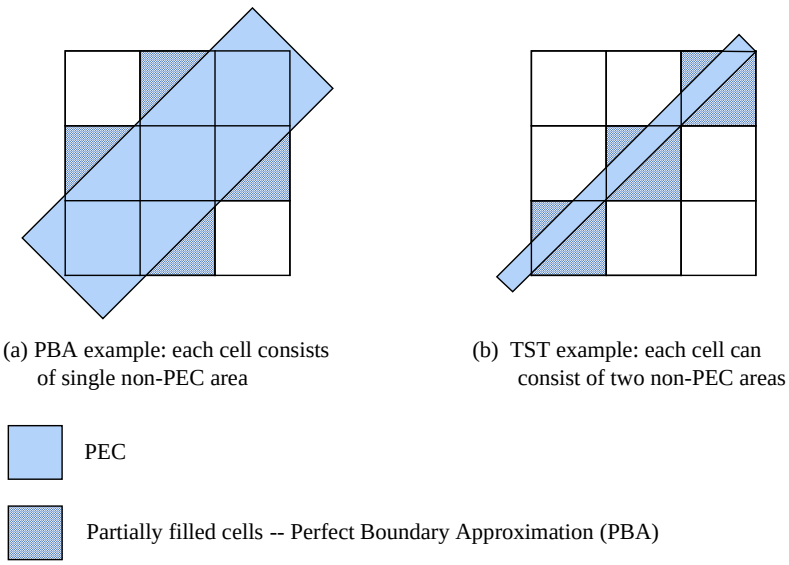


Figure B.6: TST technique

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